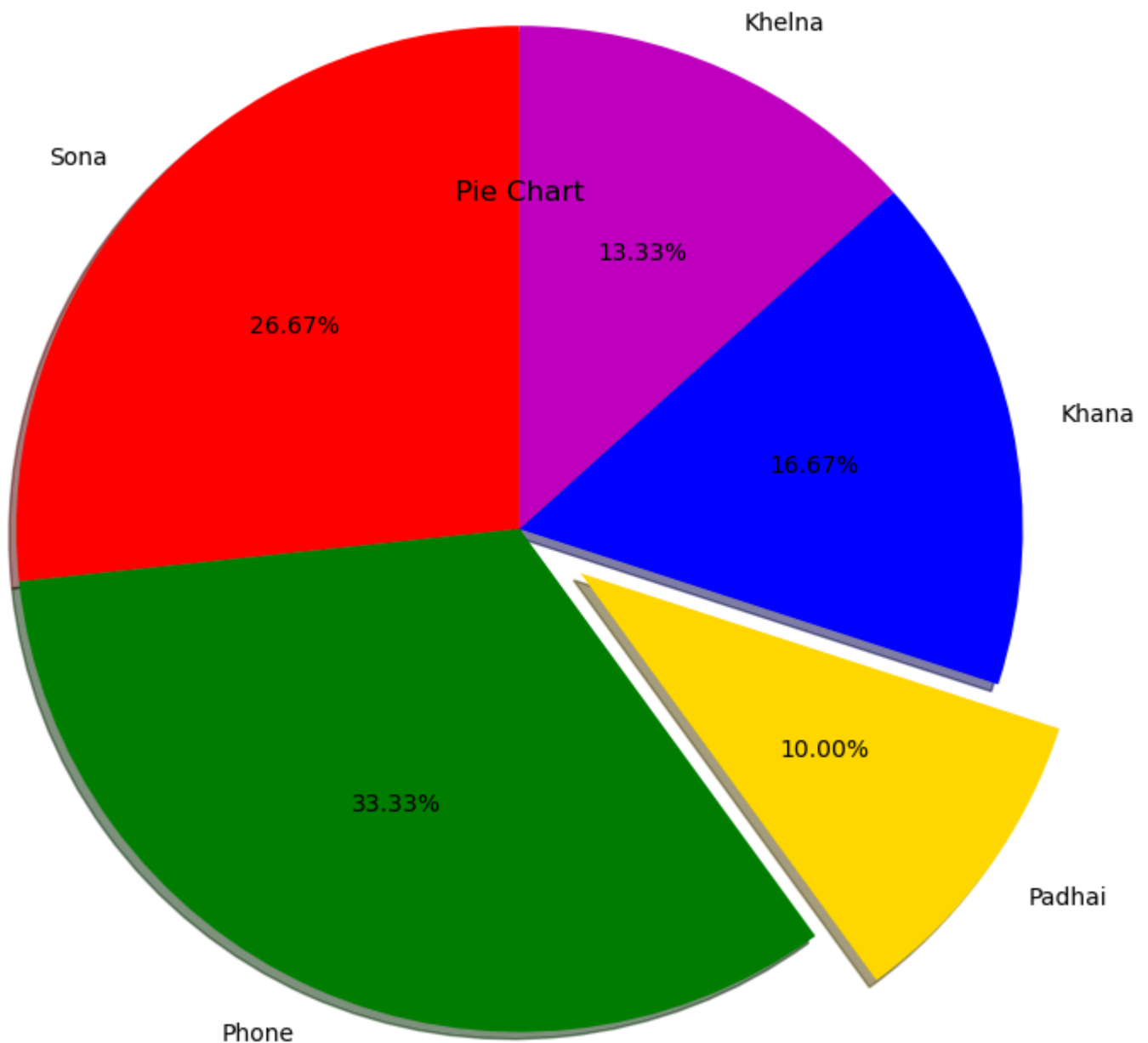


```
In [1]: import matplotlib.pyplot as plt
```

```
In [141... #Pie Chart
slices = [8,10,3,5,4]
activities = ['Sona','Phone','Padhai','Khana','Khelna']
cols = ['r','g','gold','b','m']

plt.pie(slices,labels=activities,colors=cols,
        startangle=90,shadow=True,explode=(0,0,0.3,0,0),
        autopct='%1.2f%%',radius=2)

plt.title('Pie Chart')
plt.show()
```



Seaborn

Statistical Data Visualization

Nature of Data :

--> Continuous (continuous numbers)
(Ex : 2.5,2.7,2.9,3.2,3.8,4.0.....)

--> Categorical (string or integer)
(Ex : M/F, T/F, Y/N, 0/1, 0/1/2)

In [142...

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [143...

```
iris = pd.read_csv(r"C:\Users\CTTC\Downloads\iris (4)\iris.data",header=None)
iris
```

Out[143...

	0	1	2	3	4
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Iris-virginica
146	6.3	2.5	5.0	1.9	Iris-virginica
147	6.5	3.0	5.2	2.0	Iris-virginica
148	6.2	3.4	5.4	2.3	Iris-virginica
149	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 5 columns

In [144...

```
iris.columns = ['SL','SW','PL','PW','Flower']
iris.head(3)
```

Out[144...

	SL	SW	PL	PW	Flower
0	5.1	3.5	1.4	0.2	Iris-setosa
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa

In [145...

```
#Data Cleaning
iris.isnull().sum()
```

Out[145...

```
SL      0
SW      0
PL      0
PW      0
Flower  0
dtype: int64
```

In [146...

```
for i in iris.columns:
    print(i,':','\n',iris[i].unique(),'\n')
```

```
SL :  
[5.1 4.9 4.7 4.6 5.  5.4 4.4 4.8 4.3 5.8 5.7 5.2 5.5 4.5 5.3 7.  6.4 6.9  
6.5 6.3 6.6 5.9 6.  6.1 5.6 6.7 6.2 6.8 7.1 7.6 7.3 7.2 7.7 7.4 7.9]
```

```
SW :  
[3.5 3.  3.2 3.1 3.6 3.9 3.4 2.9 3.7 4.  4.4 3.8 3.3 4.1 4.2 2.3 2.8 2.4  
2.7 2.  2.2 2.5 2.6]
```

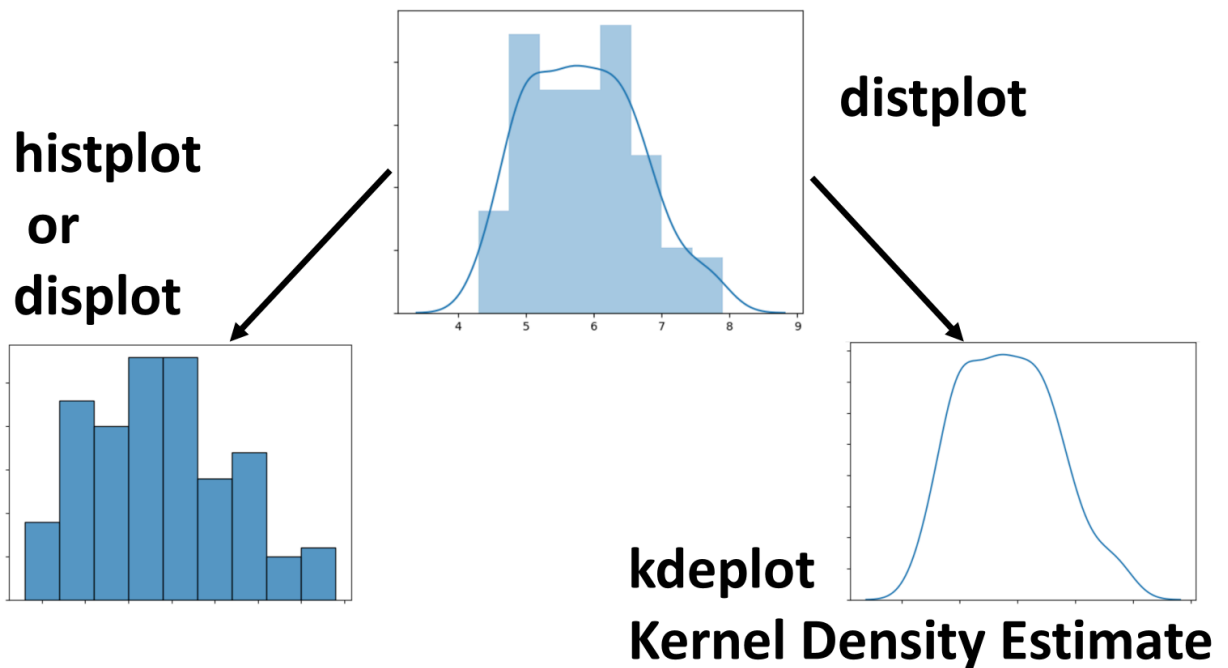
```
PL :  
[1.4 1.3 1.5 1.7 1.6 1.1 1.2 1.  1.9 4.7 4.5 4.9 4.  4.6 3.3 3.9 3.5 4.2  
3.6 4.4 4.1 4.8 4.3 5.  3.8 3.7 5.1 3.  6.  5.9 5.6 5.8 6.6 6.3 6.1 5.3  
5.5 6.7 6.9 5.7 6.4 5.4 5.2]
```

```
PW :  
[0.2 0.4 0.3 0.1 0.5 0.6 1.4 1.5 1.3 1.6 1.  1.1 1.8 1.2 1.7 2.5 1.9 2.1  
2.2 2.  2.4 2.3]
```

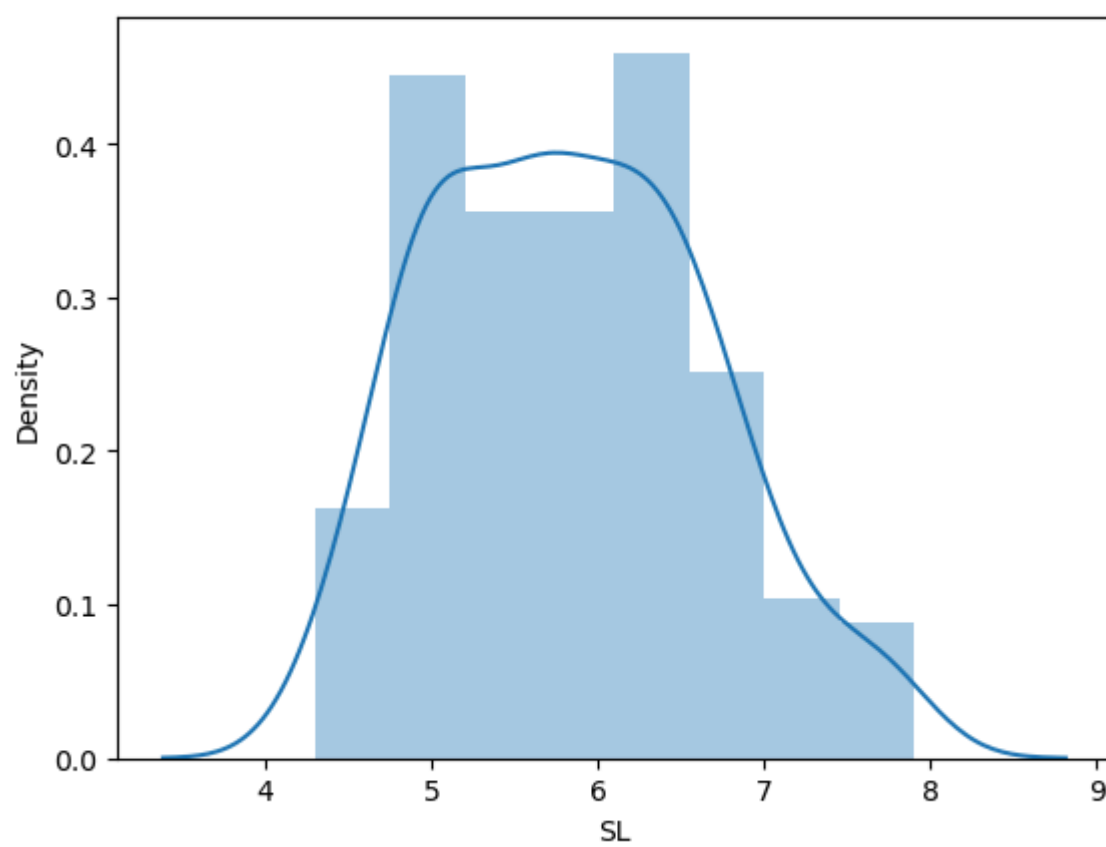
```
Flower :  
['Iris-setosa' 'Iris-versicolor' 'Iris-virginica']
```

```
In [11]: import warnings  
warnings.filterwarnings('ignore')
```

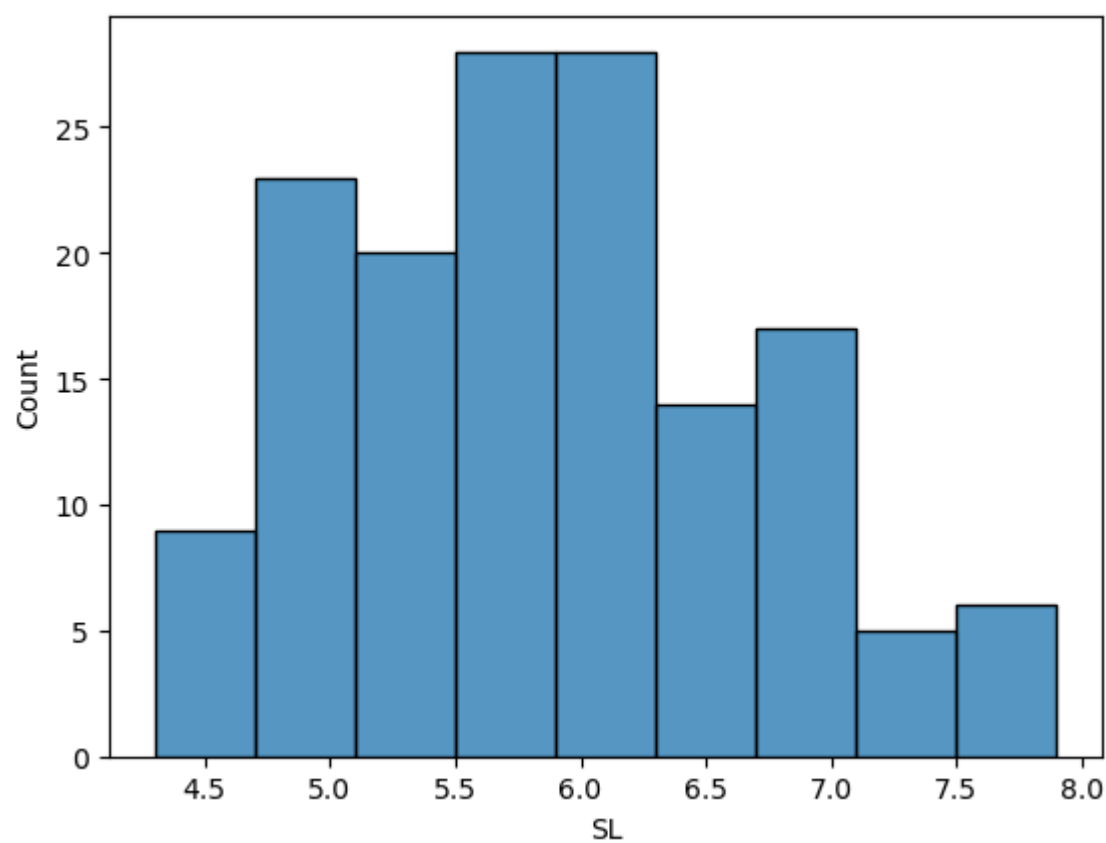
Distribution Plot



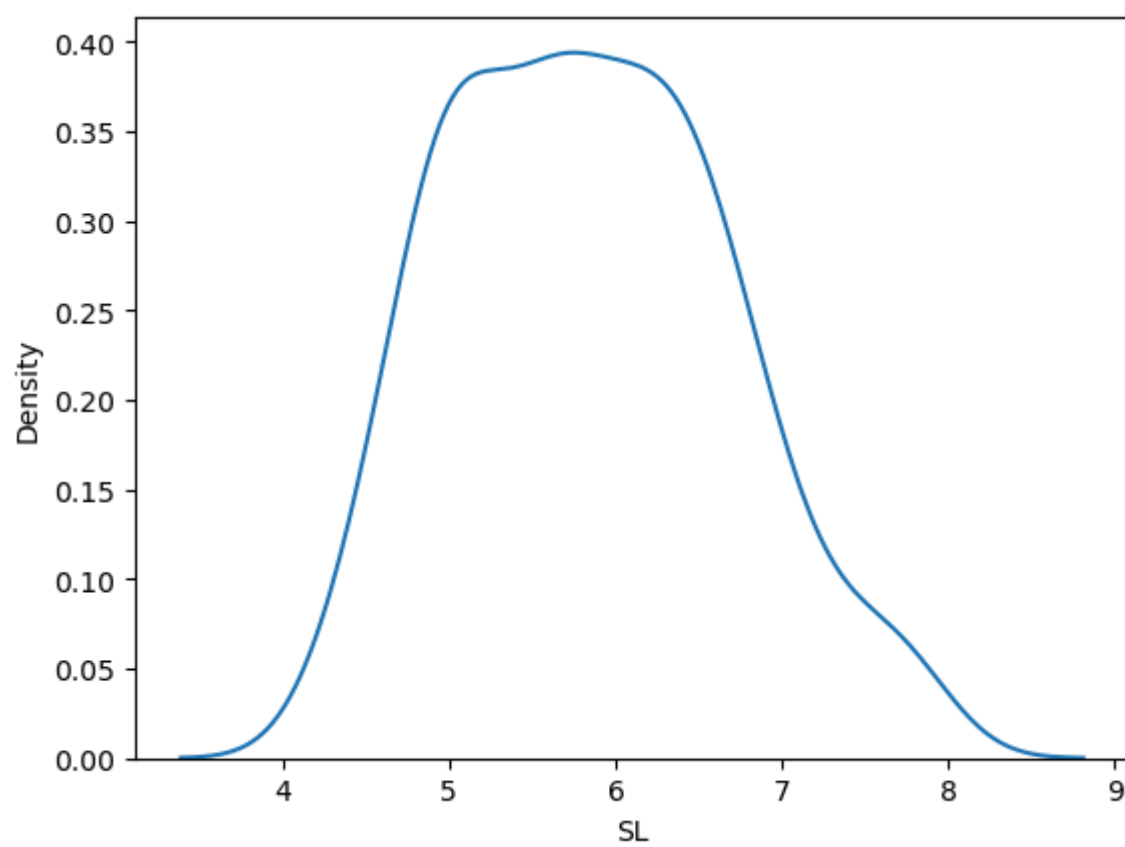
```
In [12]: #Continuous and Categorical  
#Distribution Plot  
sns.distplot(iris.SL)  
plt.show()
```



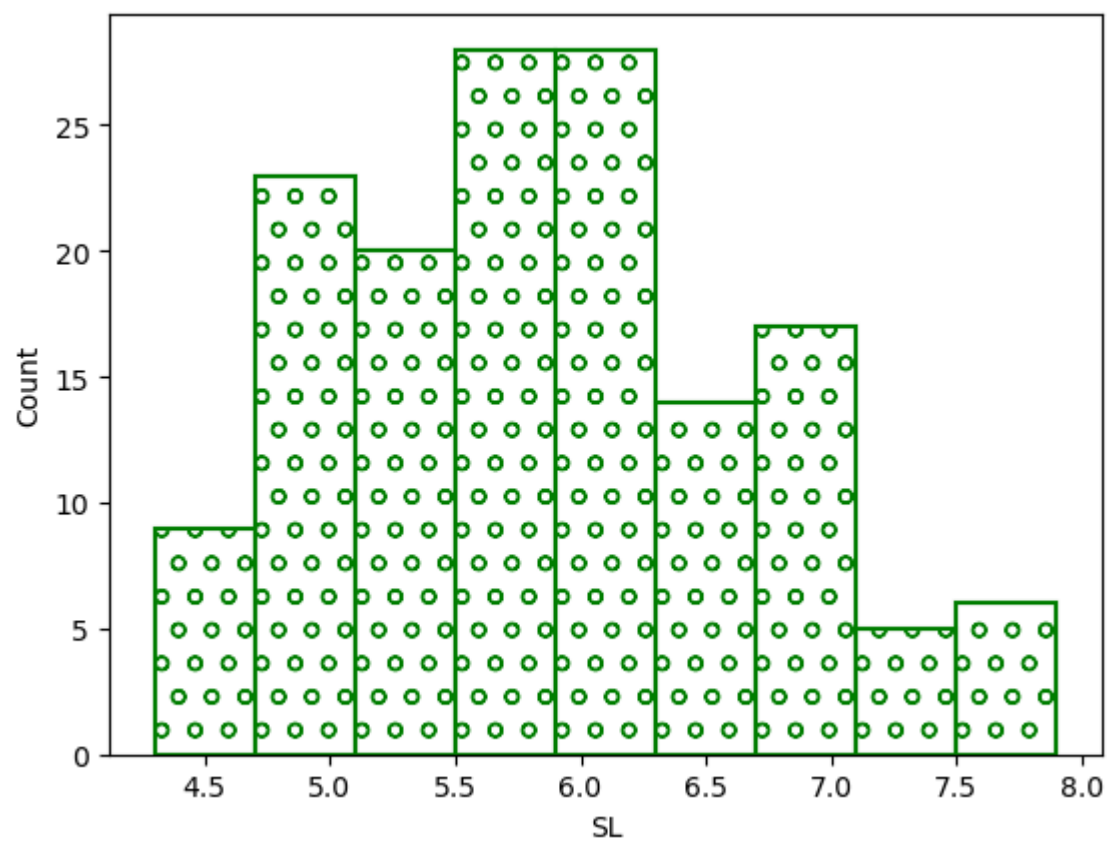
```
In [13]: sns.histplot(iris.SL)
plt.show()
```



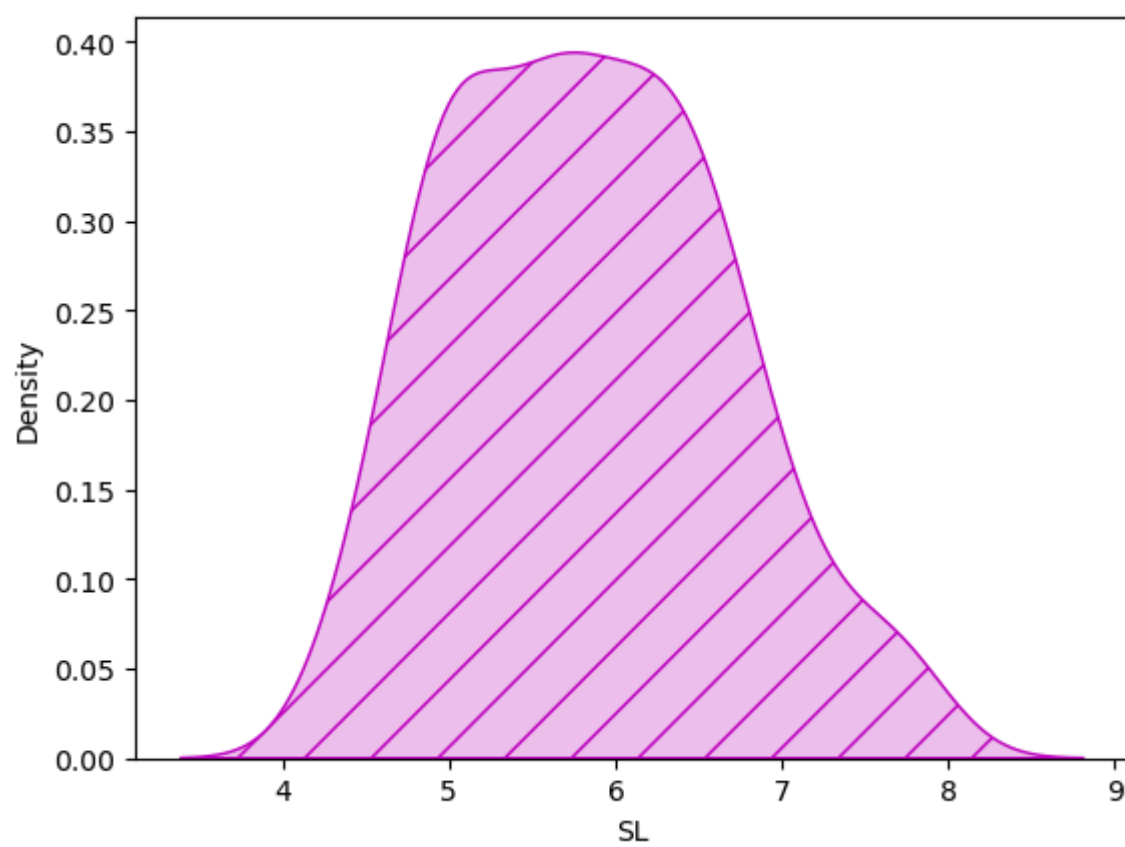
```
In [14]: sns.kdeplot(iris.SL)
plt.show()
```



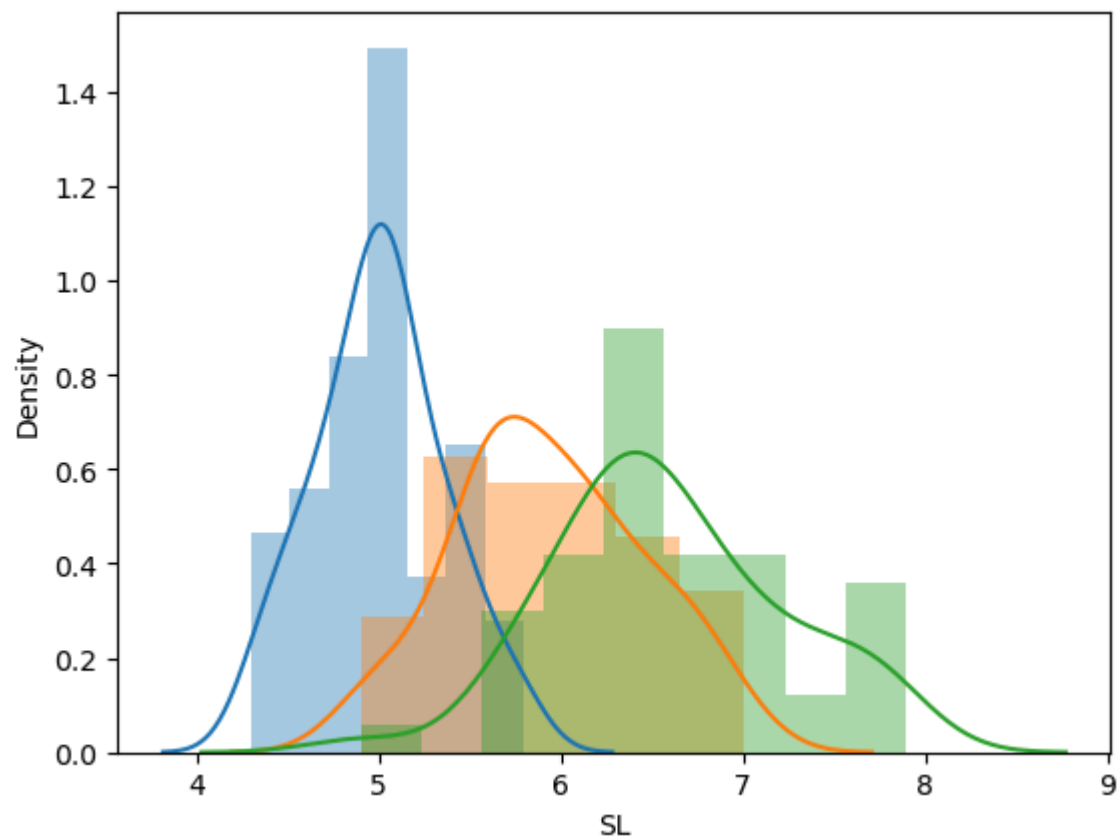
```
In [154]: sns.histplot(iris.SL, fill=False, color='g', hatch='o')  
plt.show()
```



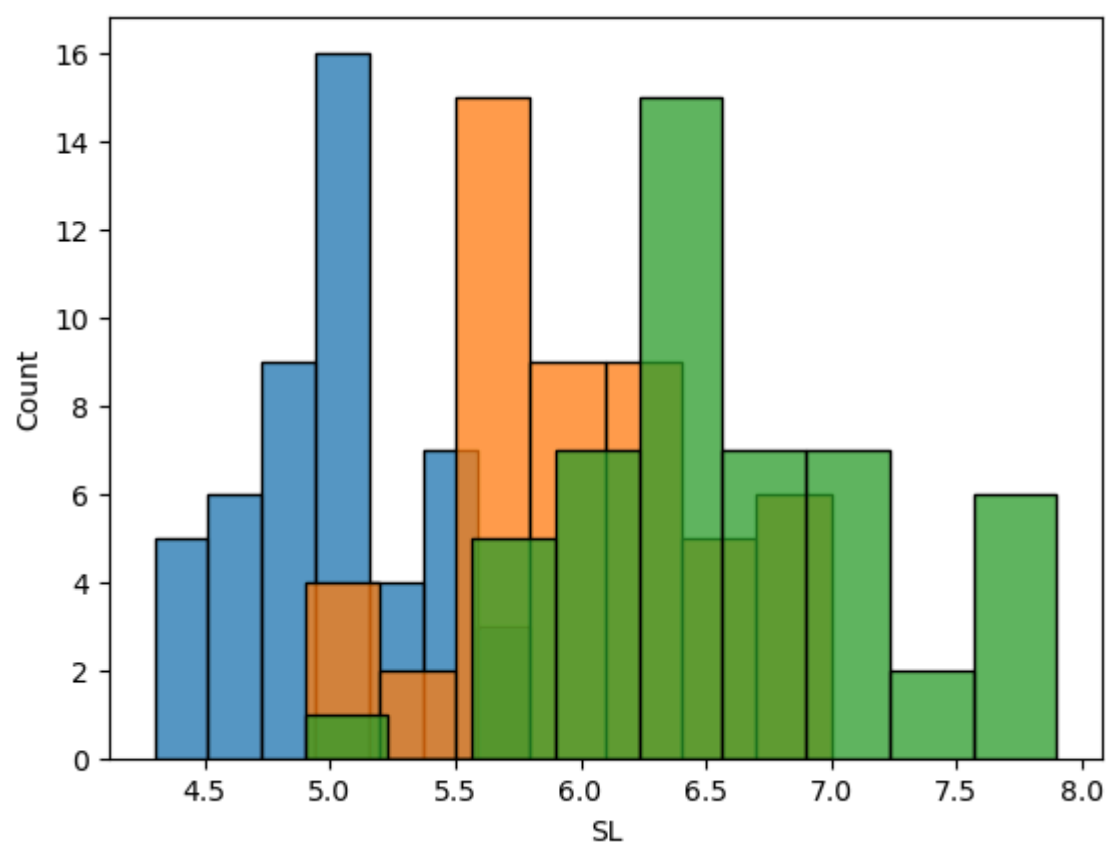
```
In [16]: sns.kdeplot(iris.SL, fill=True, color='m', hatch='/')  
plt.show()
```



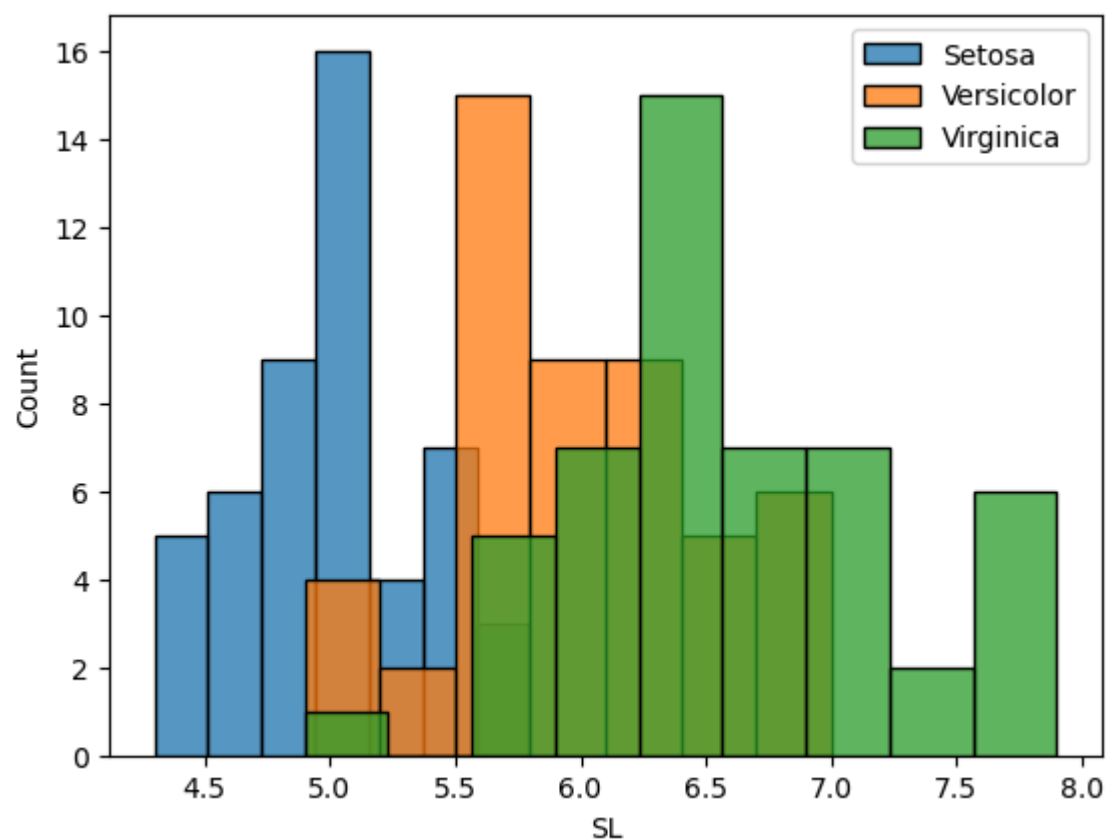
```
In [155... sns.distplot(iris.SL[iris['Flower']=='Iris-setosa'])
sns.distplot(iris.SL[iris.Flower=='Iris-versicolor'])
sns.distplot(iris.SL[iris.Flower=='Iris-virginica'])
plt.show()
```



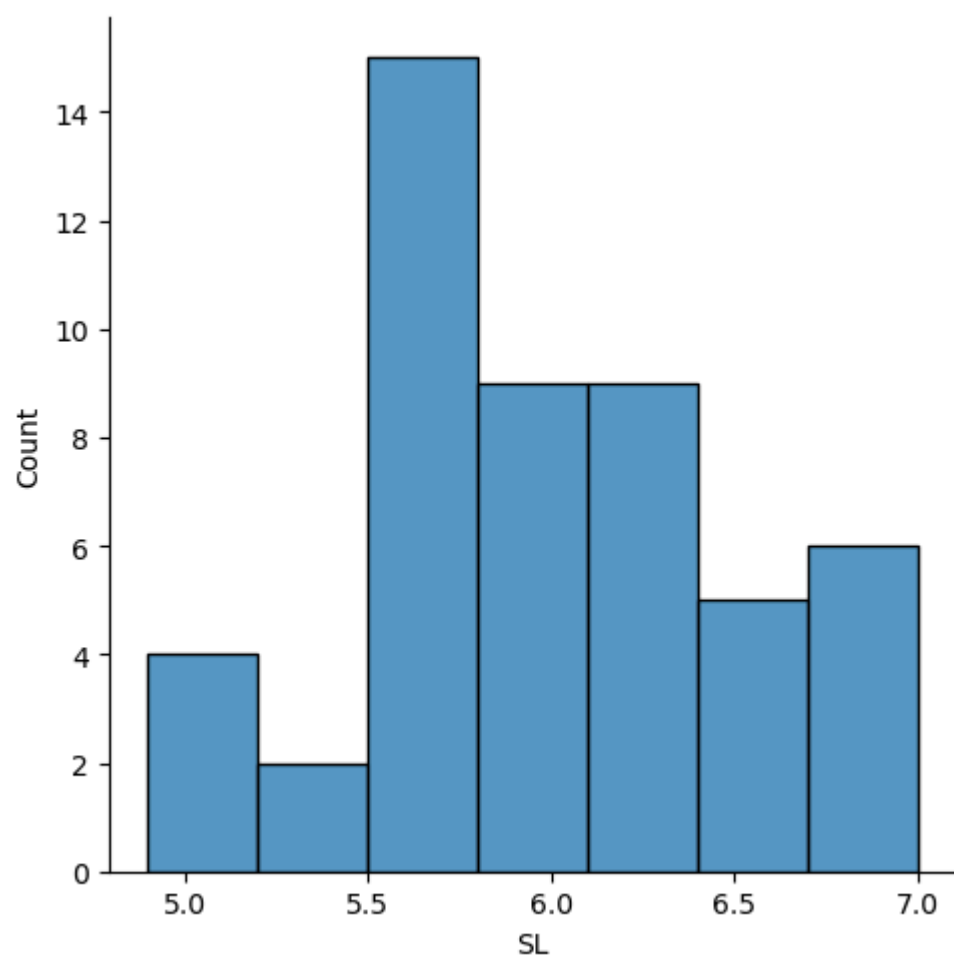
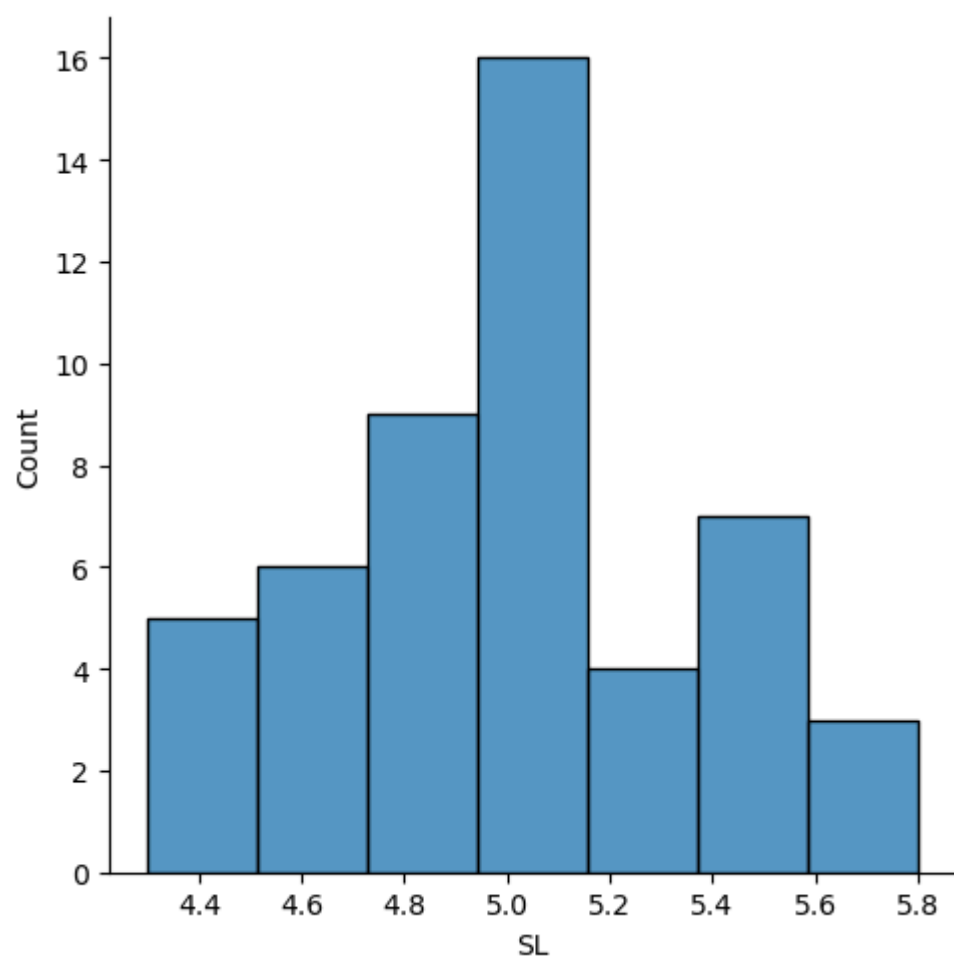
```
In [156... sns.histplot(iris.SL[iris['Flower']=='Iris-setosa'])
sns.histplot(iris.SL[iris.Flower=='Iris-versicolor'])
sns.histplot(iris.SL[iris.Flower=='Iris-virginica'])
plt.show()
```

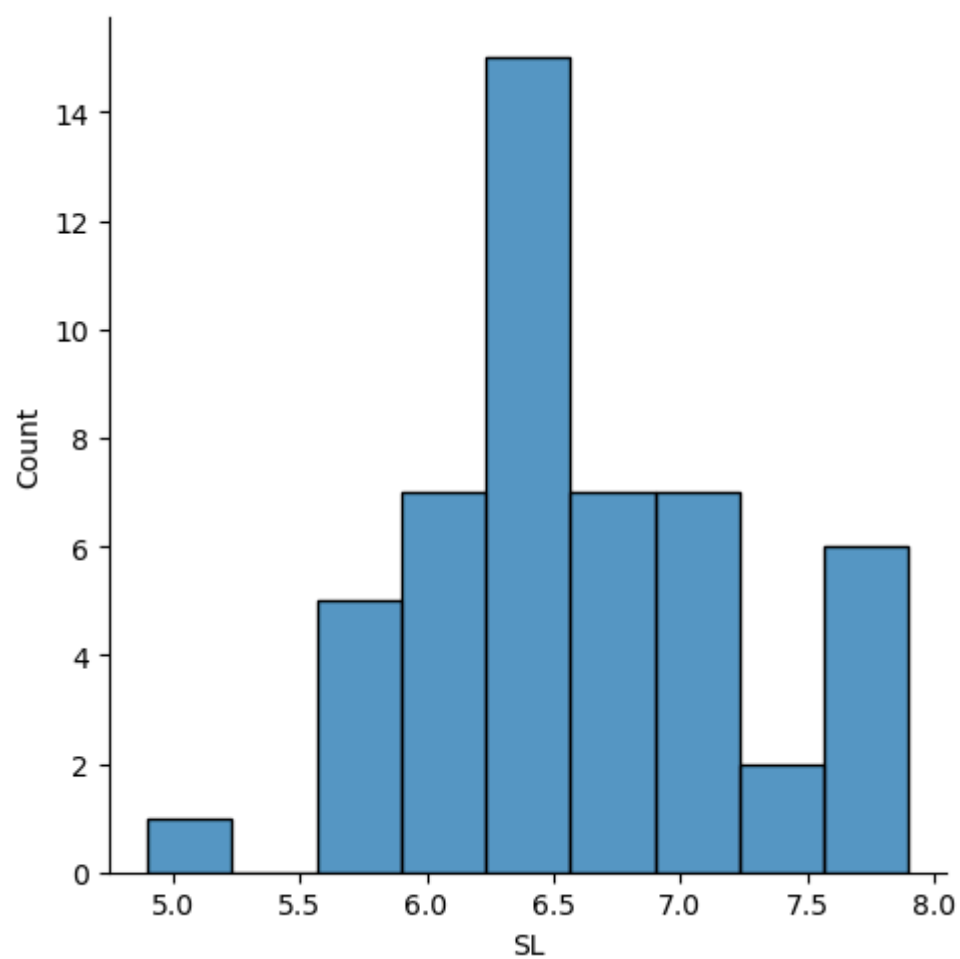


```
In [18]: sns.histplot(iris.SL[iris['Flower']=='Iris-setosa'],label='Setosa')
sns.histplot(iris.SL[iris.Flower=='Iris-versicolor'],label='Versicolor')
sns.histplot(iris.SL[iris.Flower=='Iris-virginica'],label='Virginica')
plt.legend()
plt.show()
```



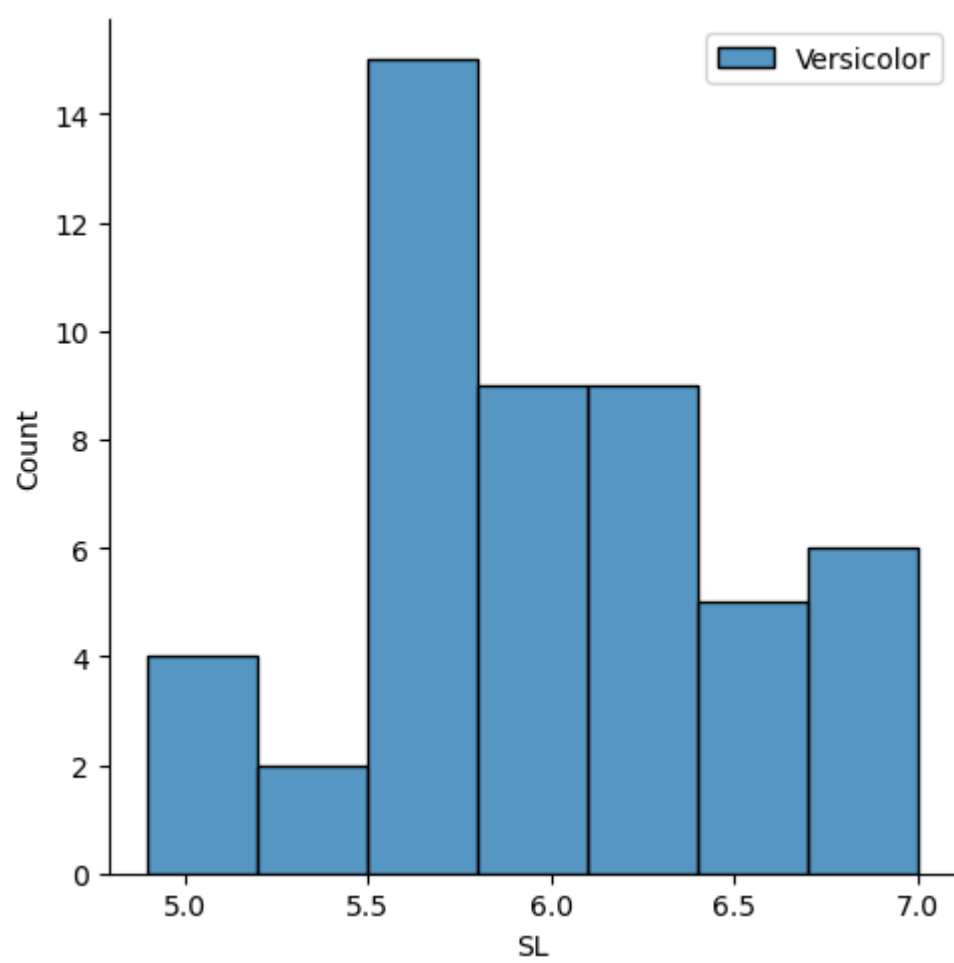
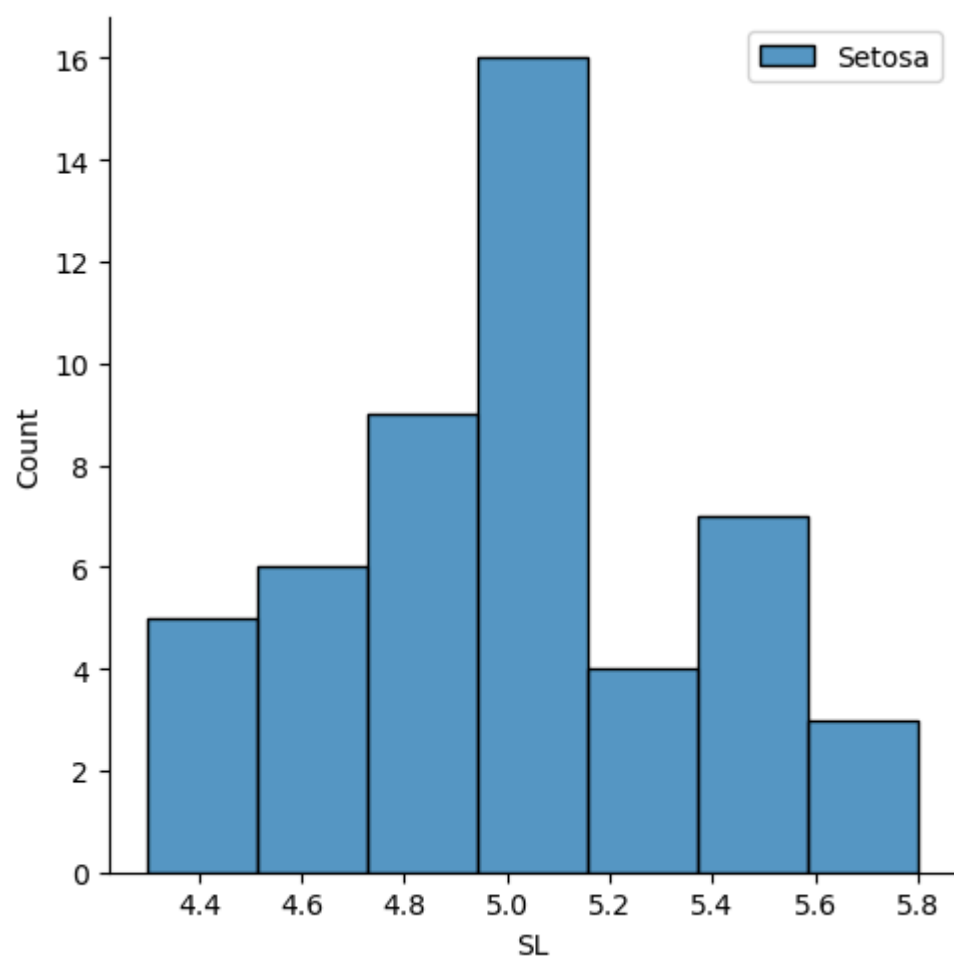
```
In [157... sns.displot(iris.SL[iris['Flower']=='Iris-setosa'])
sns.displot(iris.SL[iris.Flower=='Iris-versicolor'])
sns.displot(iris.SL[iris.Flower=='Iris-virginica'])
plt.show()
```

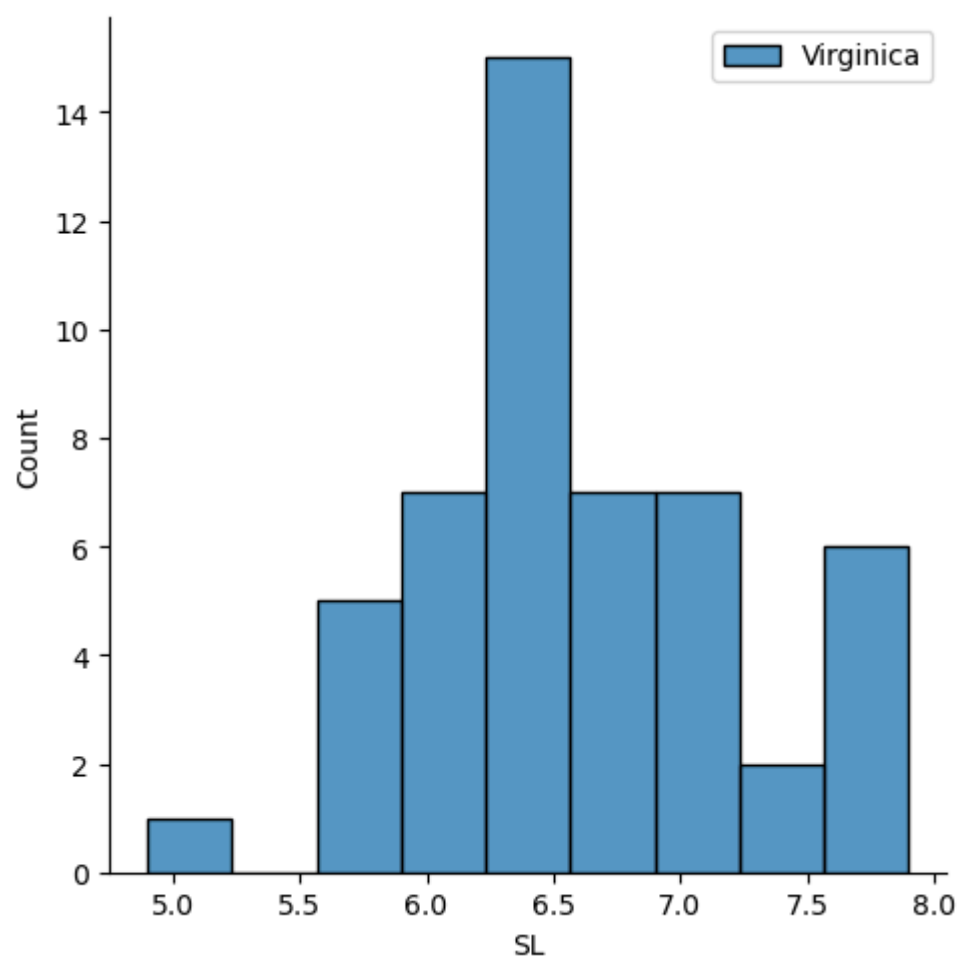




In [158...

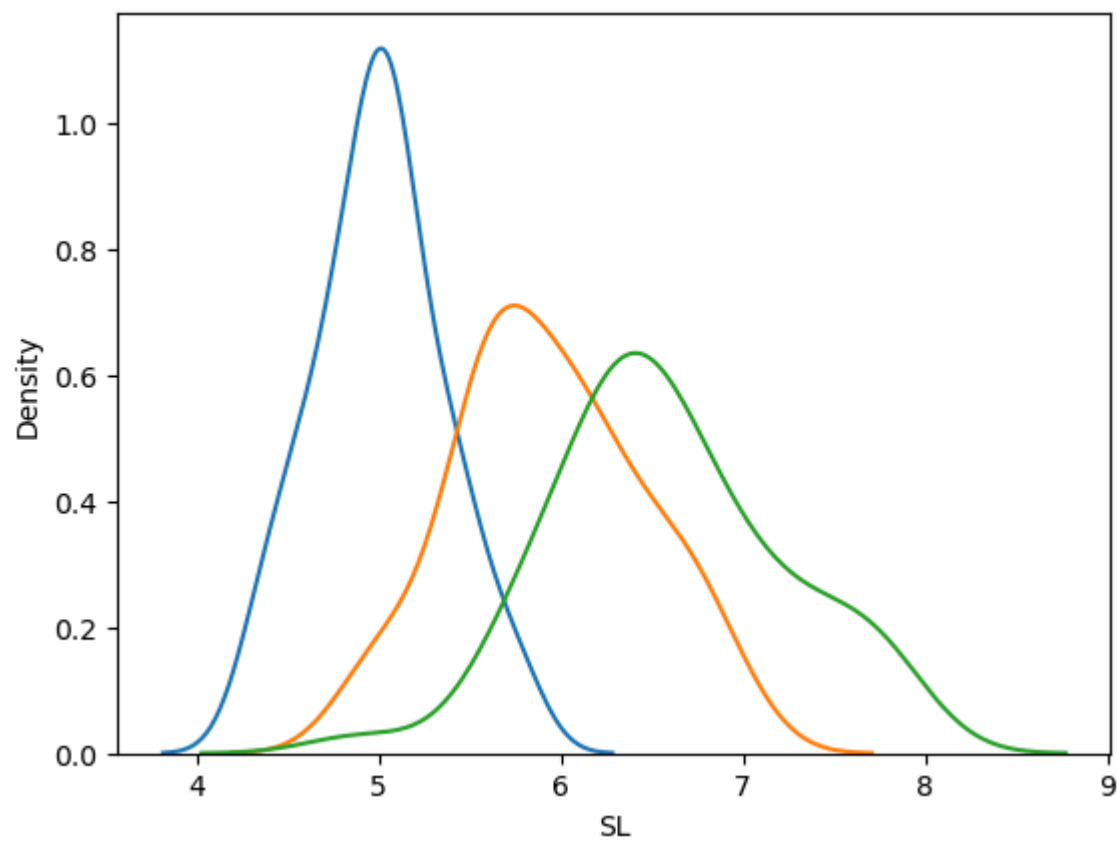
```
sns.displot(iris.SL[iris['Flower']=='Iris-setosa'],label='Setosa')
plt.legend()
sns.displot(iris.SL[iris.Flower=='Iris-versicolor'],label='Versicolor')
plt.legend()
sns.displot(iris.SL[iris.Flower=='Iris-virginica'],label='Virginica')
plt.legend()
plt.show()
```





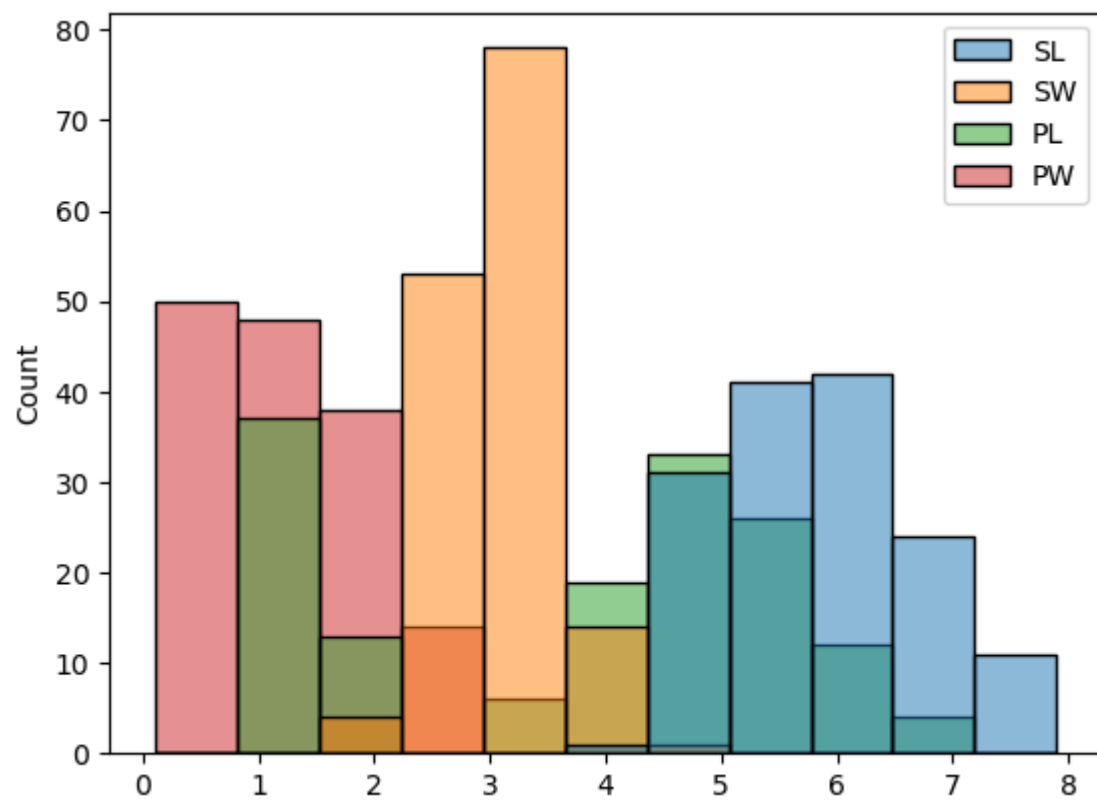
In [159...

```
sns.kdeplot(iris.SL[iris['Flower']=='Iris-setosa'])  
sns.kdeplot(iris.SL[iris.Flower=='Iris-versicolor'])  
sns.kdeplot(iris.SL[iris.Flower=='Iris-virginica'])  
plt.show()
```

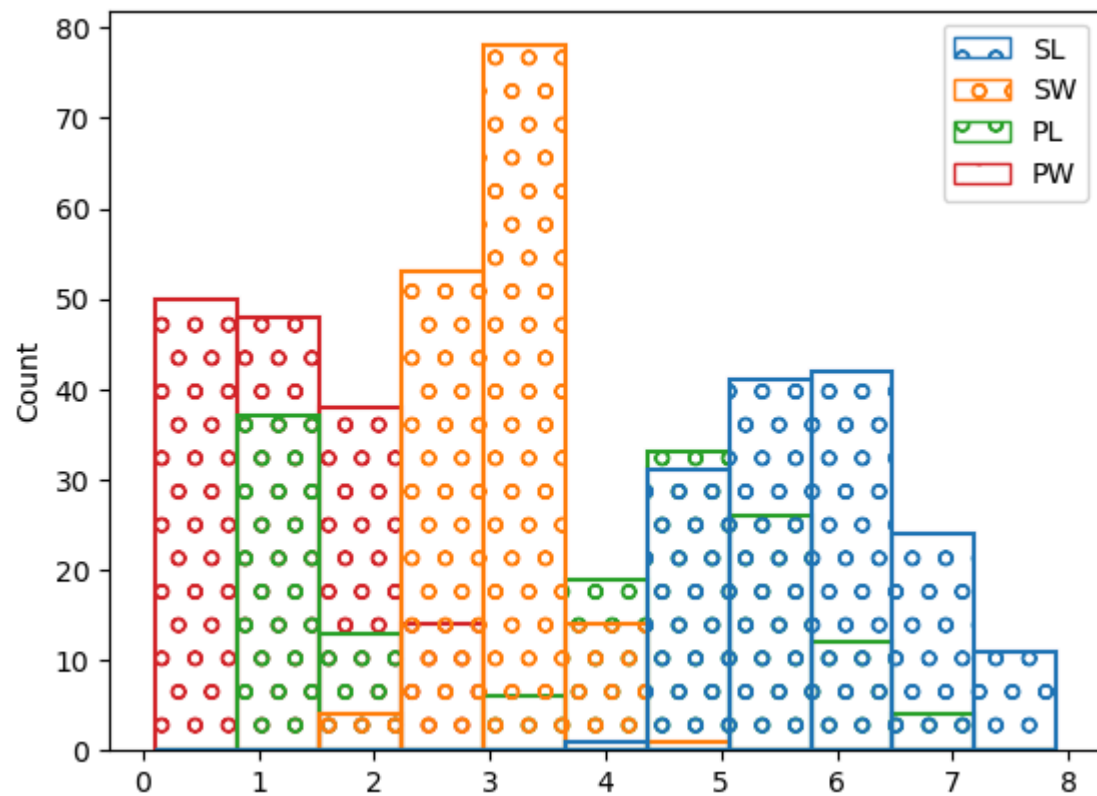


In [160...

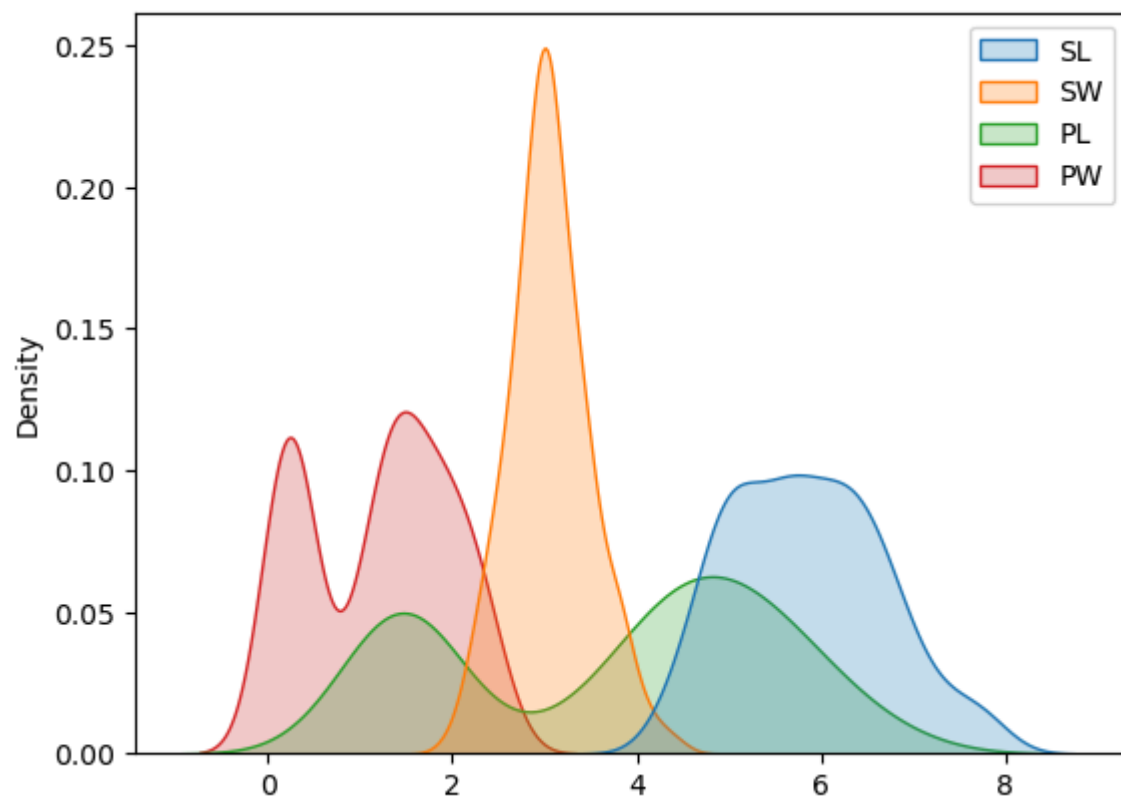
```
sns.histplot(data=iris)  
plt.show()
```



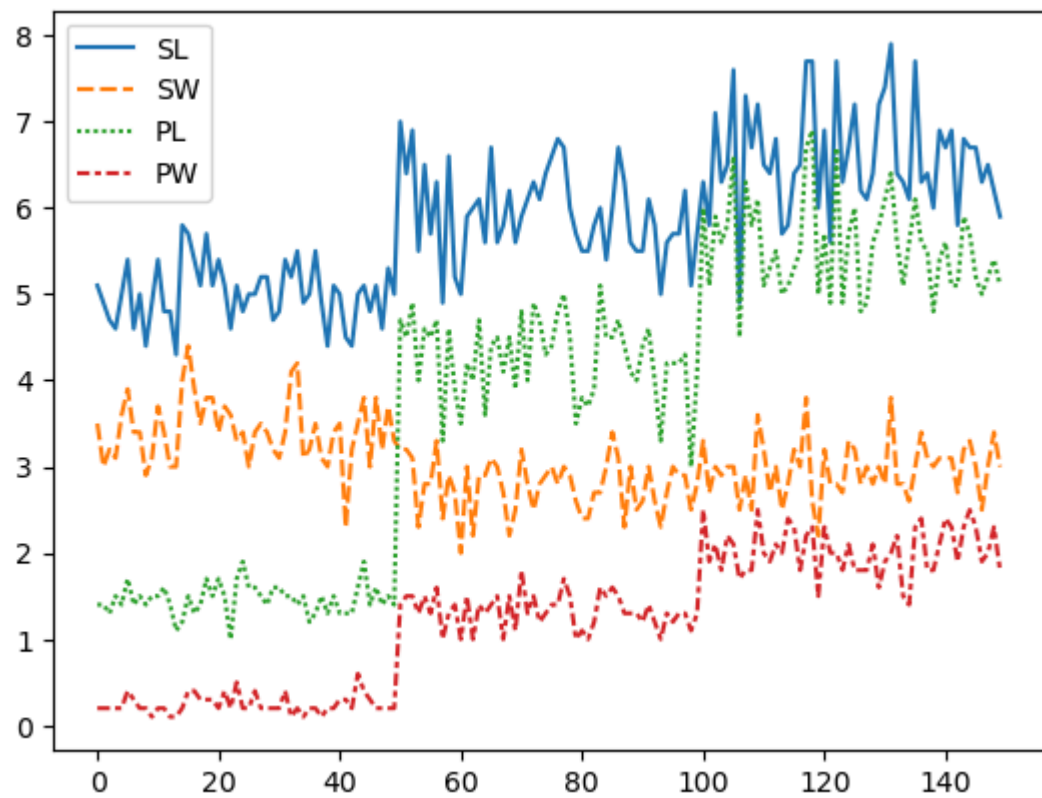
```
In [22]: sns.histplot(data=iris,fill=False,hatch='o')
plt.show()
```



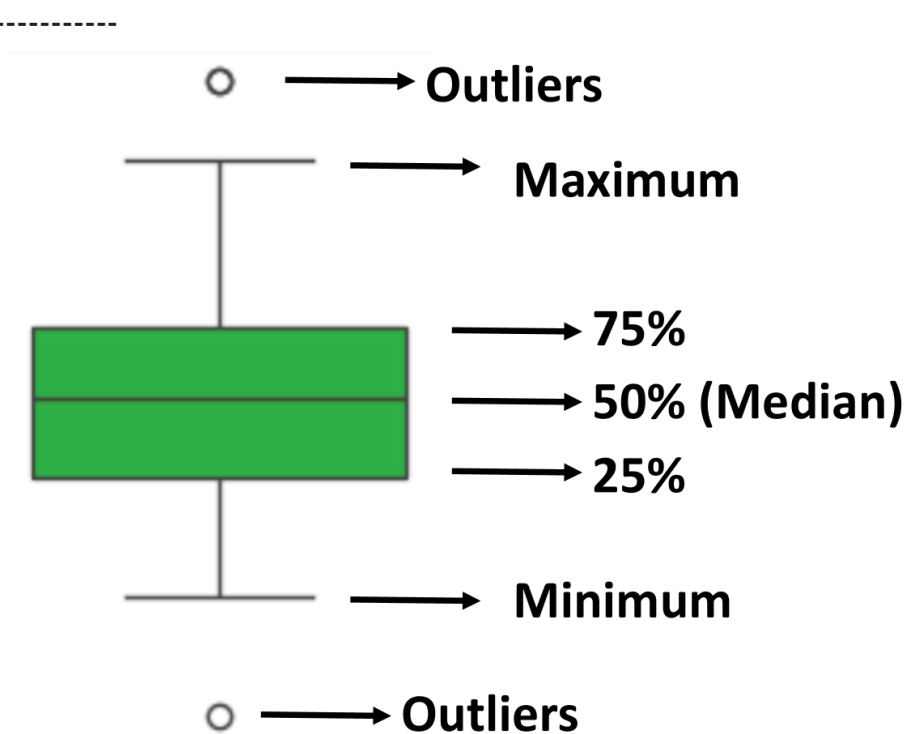
```
In [23]: sns.kdeplot(data=iris,fill=True)
plt.show()
```



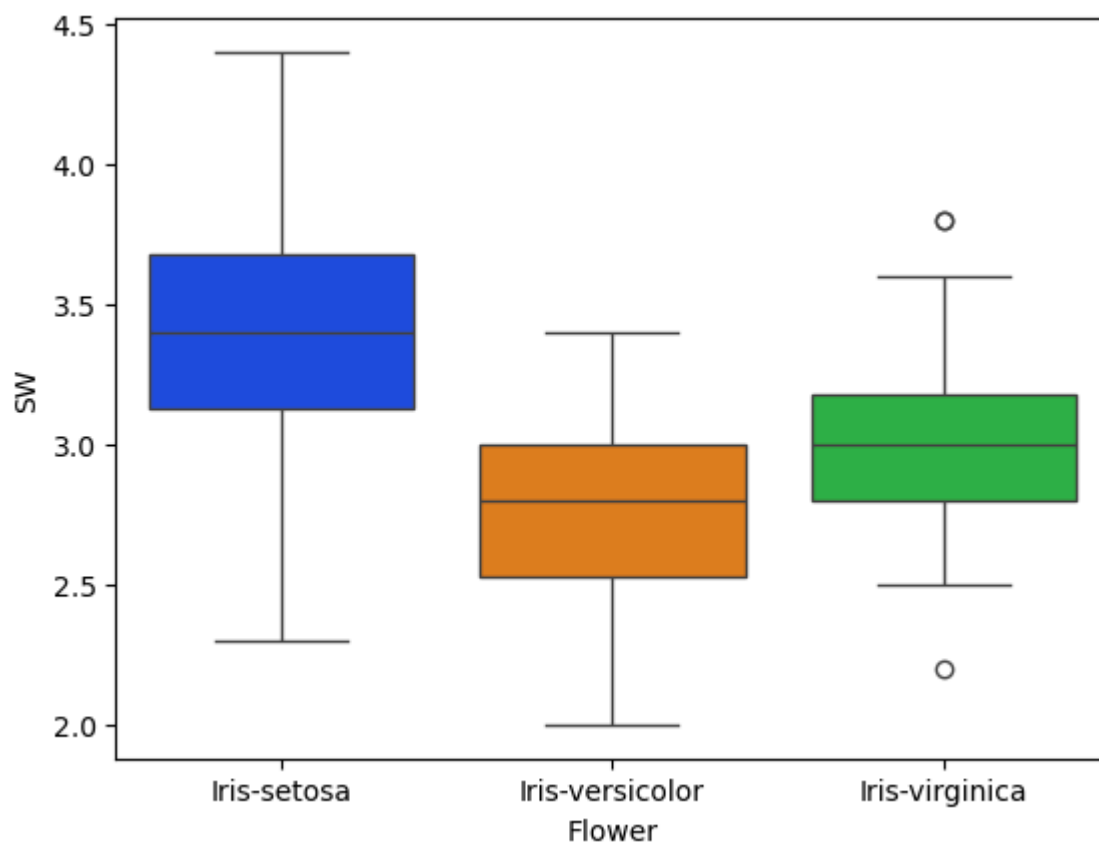
```
In [24]: sns.lineplot(data=iris)
plt.show()
```



Box Plot



```
In [25]: #Box Plot
sns.boxplot(x=iris.Flower,y=iris.SW,palette='bright')
plt.show()
```

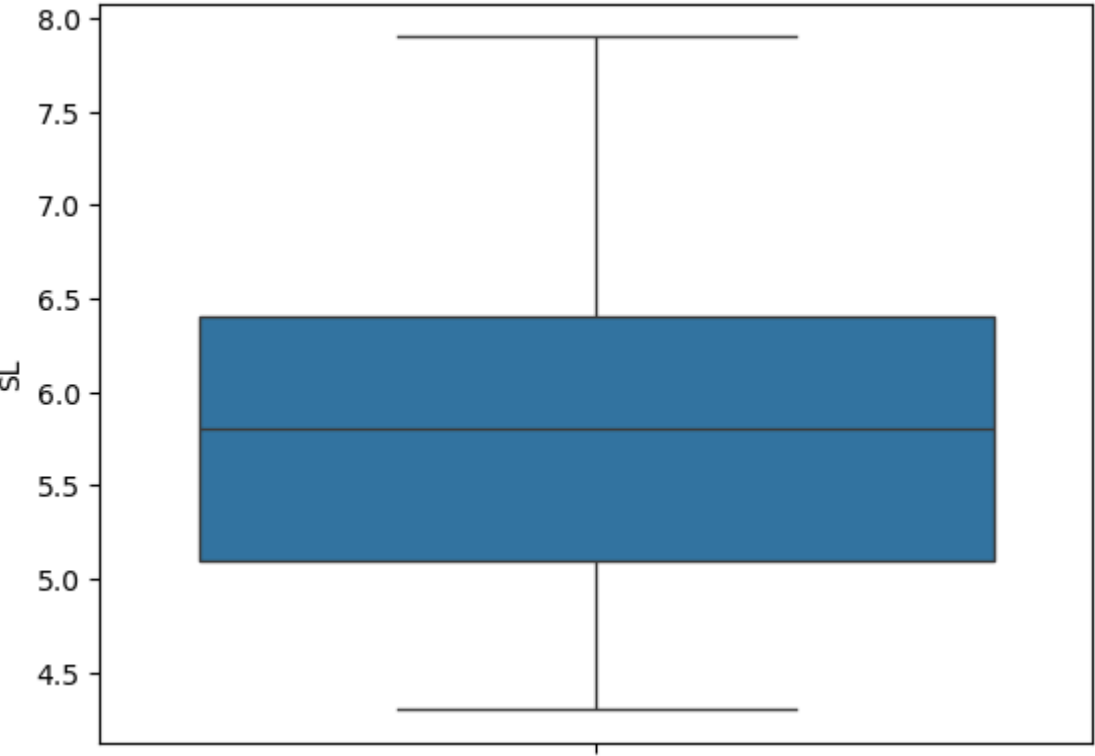


```
In [26]: iris.describe()
```

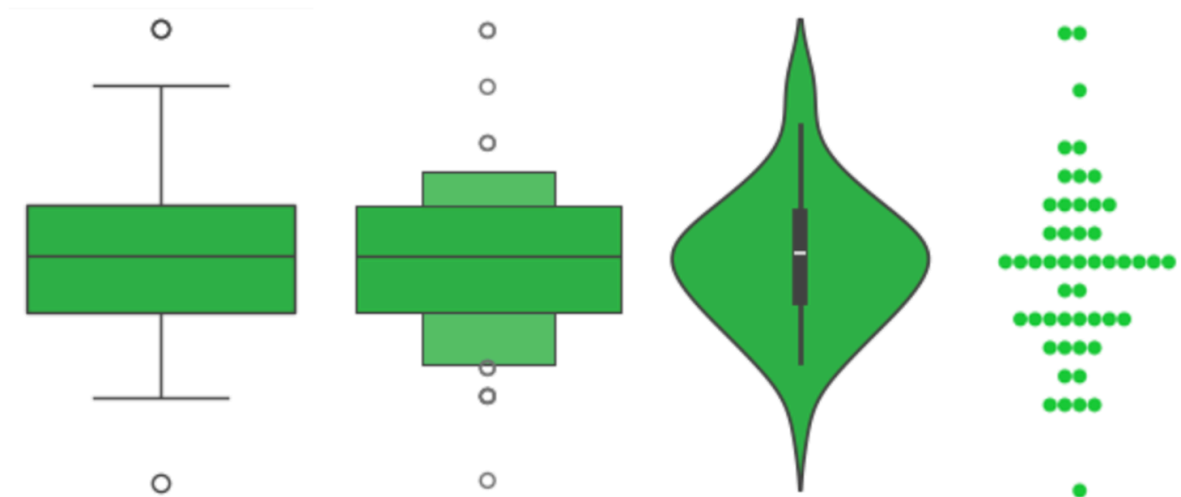
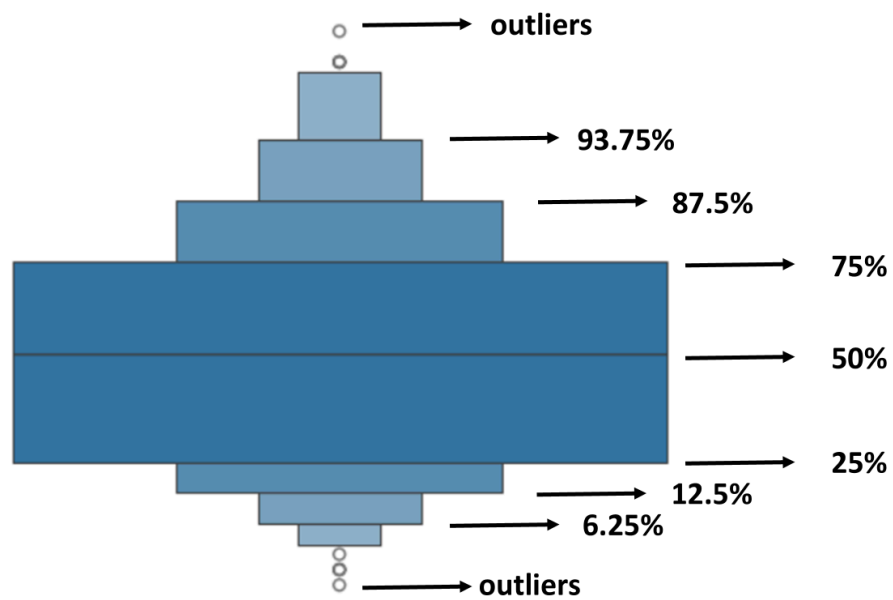
Out[26]:

	SL	SW	PL	PW
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

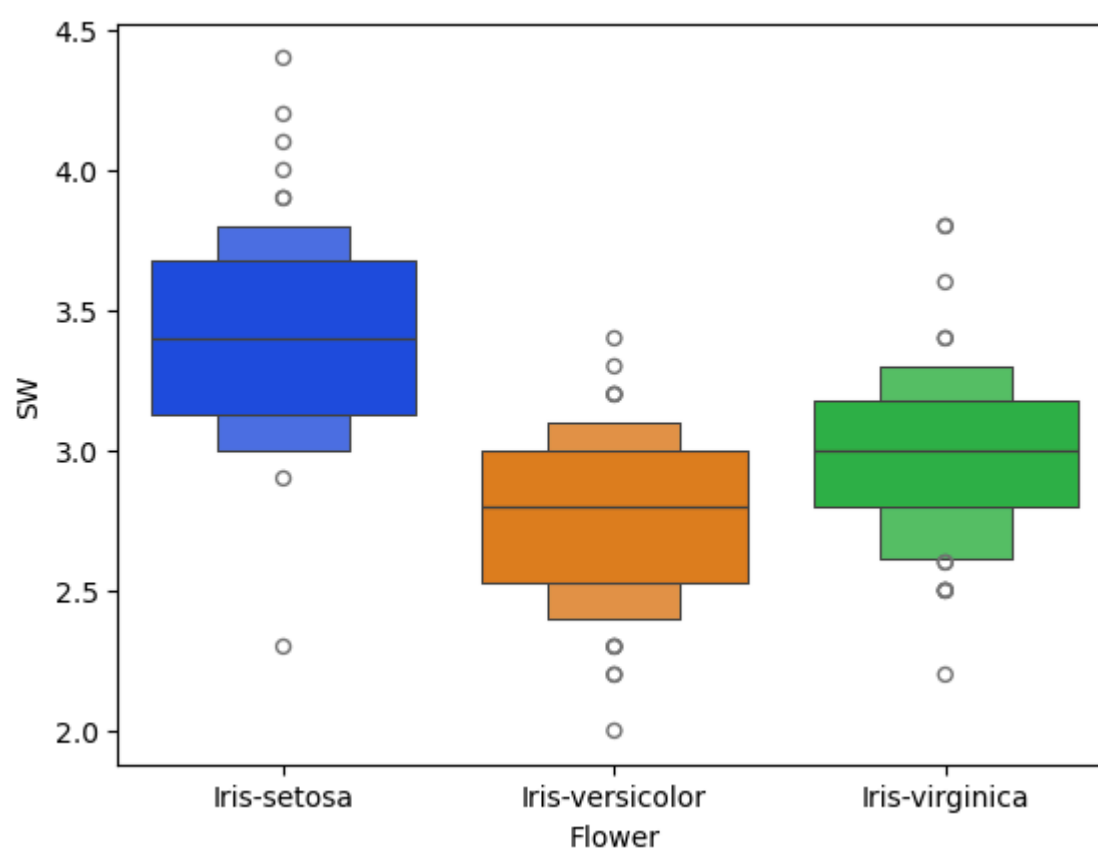
```
In [27]: sns.boxplot(y=iris.SL)
plt.show()
```



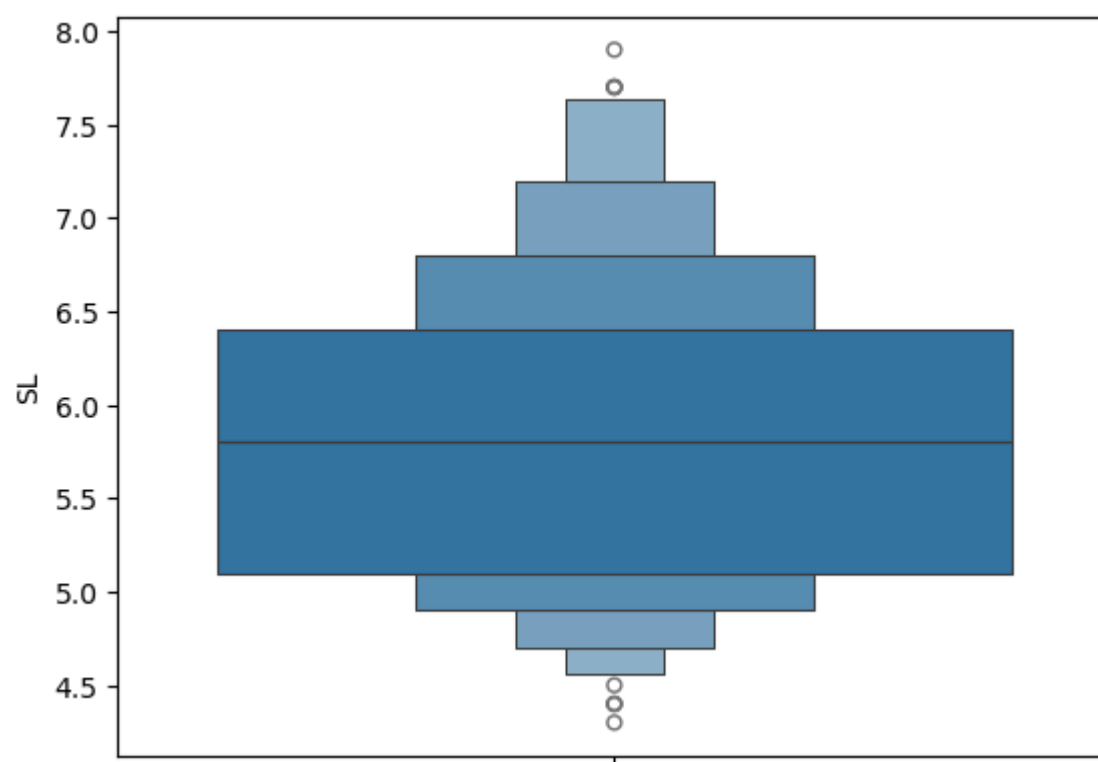
Boxen Plot



```
In [28]: #Boxen Plot
sns.boxenplot(x=iris.Flower,y=iris.SW,palette='bright')
plt.show()
```

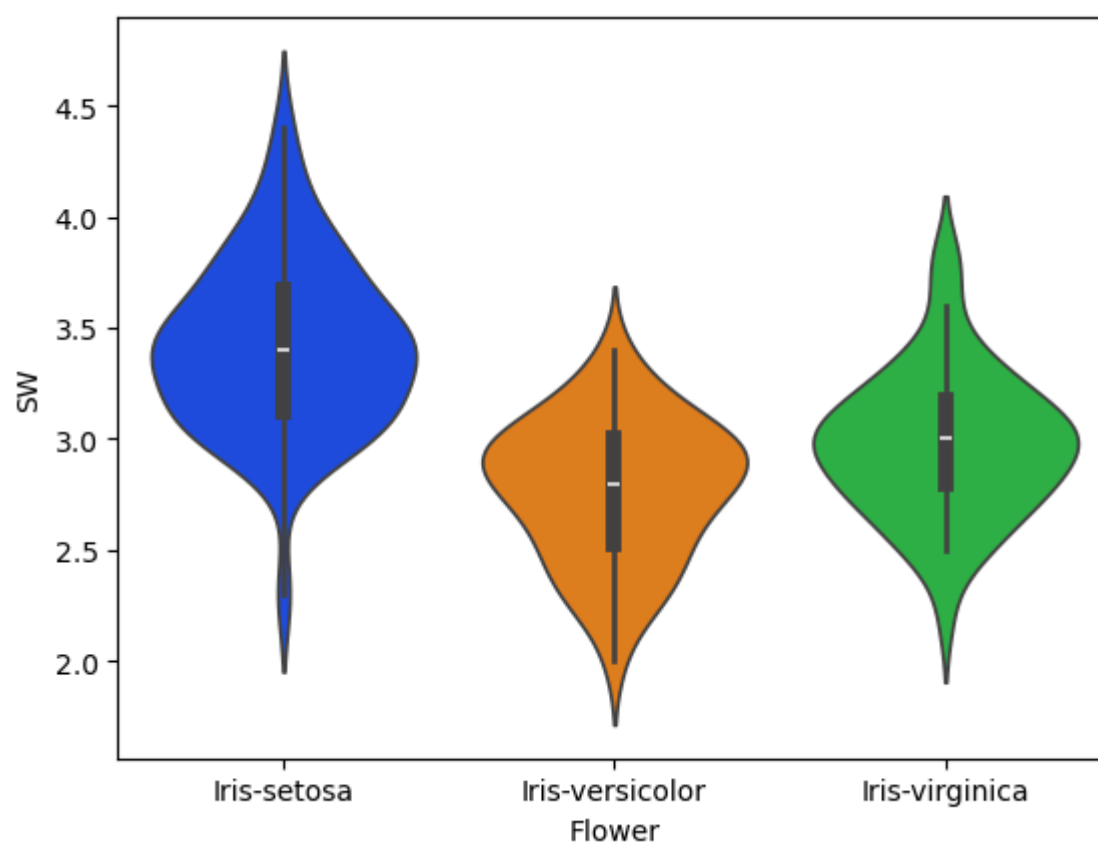
```
In [29]: sns.boxenplot(y=iris.SL)
plt.show()
```



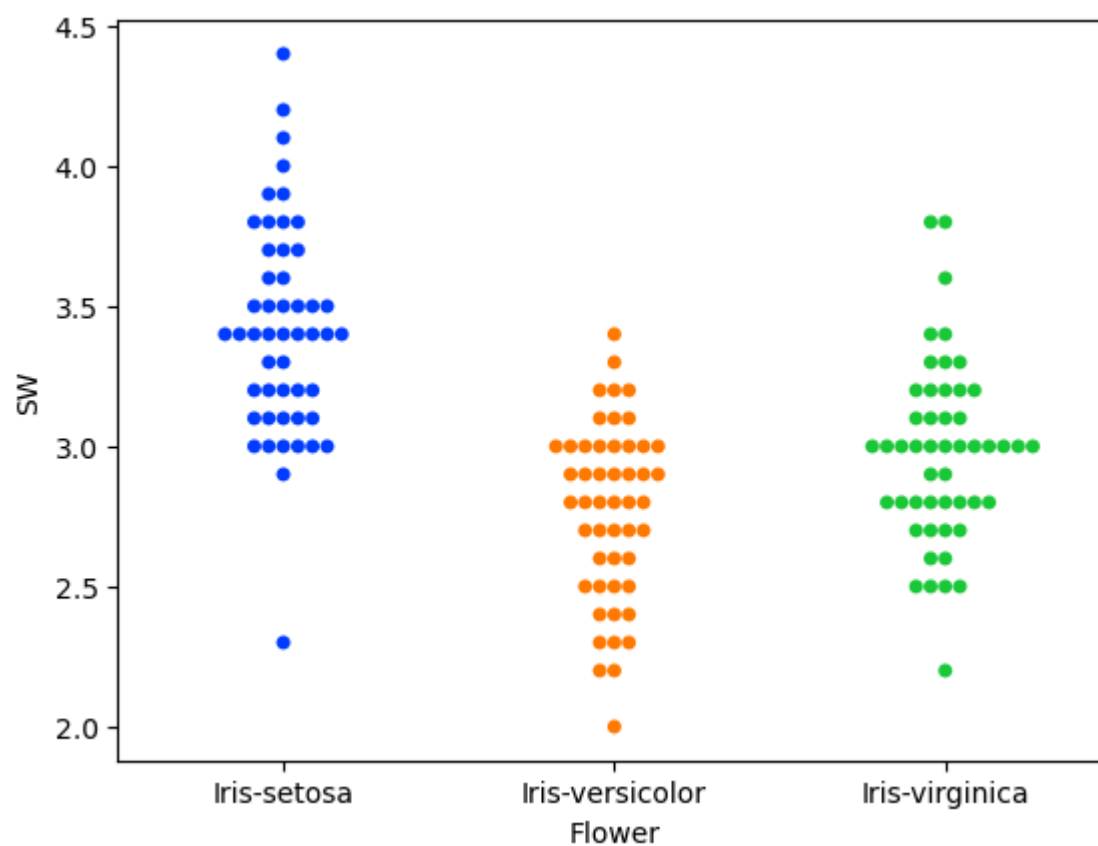
```
In [30]: np.percentile(iris.SL,87.5)
```

```
Out[30]: 6.8
```

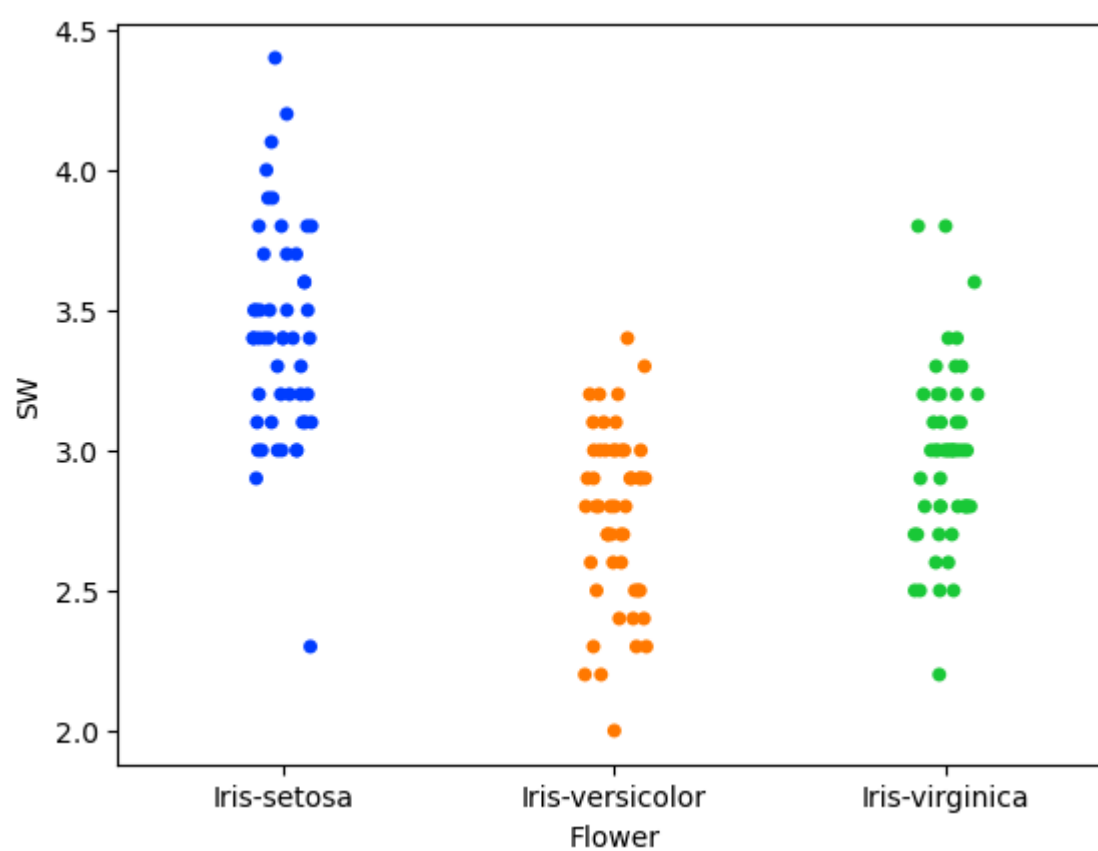
```
In [31]: #Violin Plot
sns.violinplot(x=iris.Flower,y=iris.SW,palette='bright')
plt.show()
```



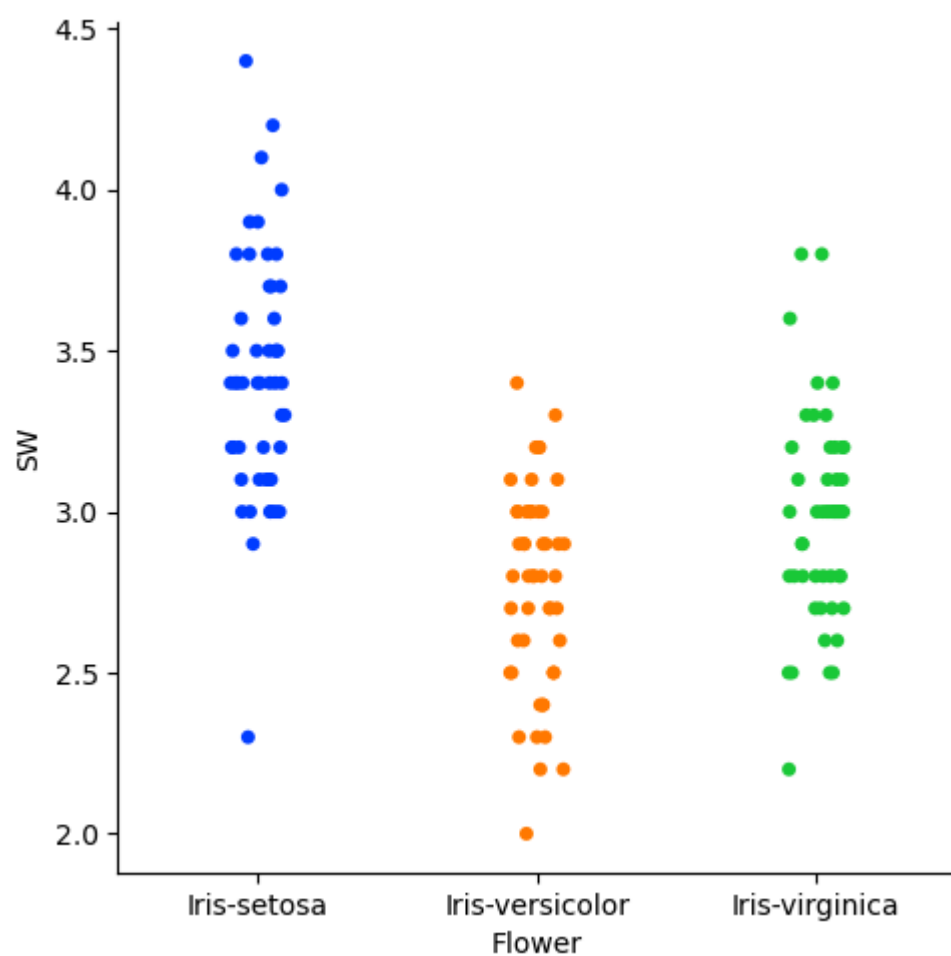
```
In [32]: #Swarm Plot
sns.swarmplot(x=iris.Flower,y=iris.SW,palette='bright')
plt.show()
```



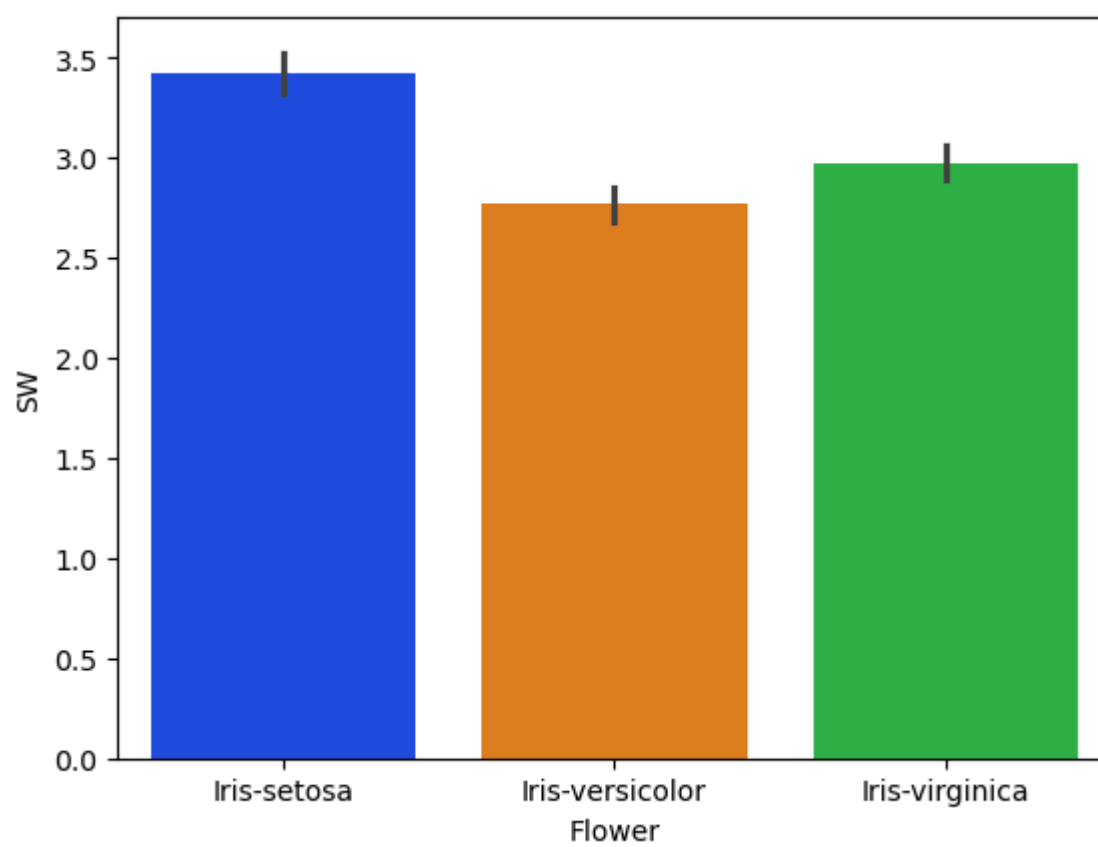
```
In [33]: #Strip Plot
sns.stripplot(x=iris.Flower,y=iris.SW,palette='bright')
plt.show()
```



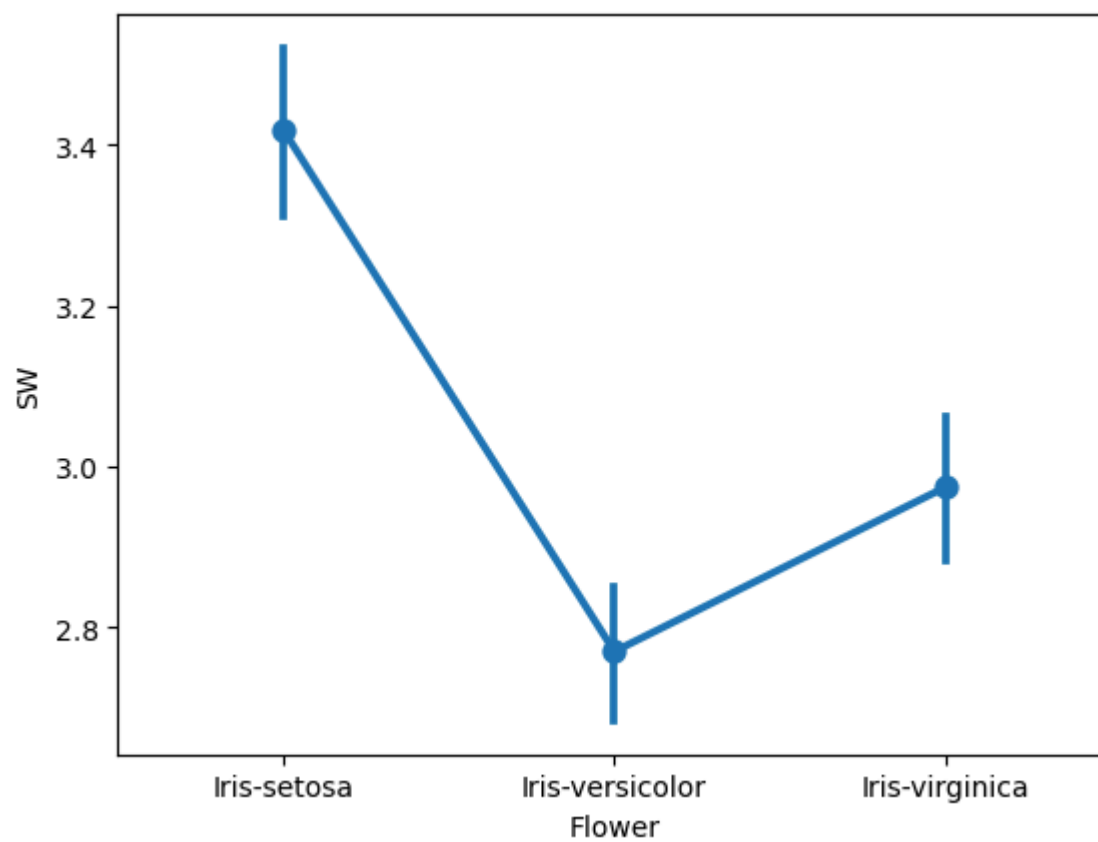
```
In [34]: #Cat Plot
sns.catplot(x=iris.Flower,y=iris.SW,palette='bright')
plt.show()
```



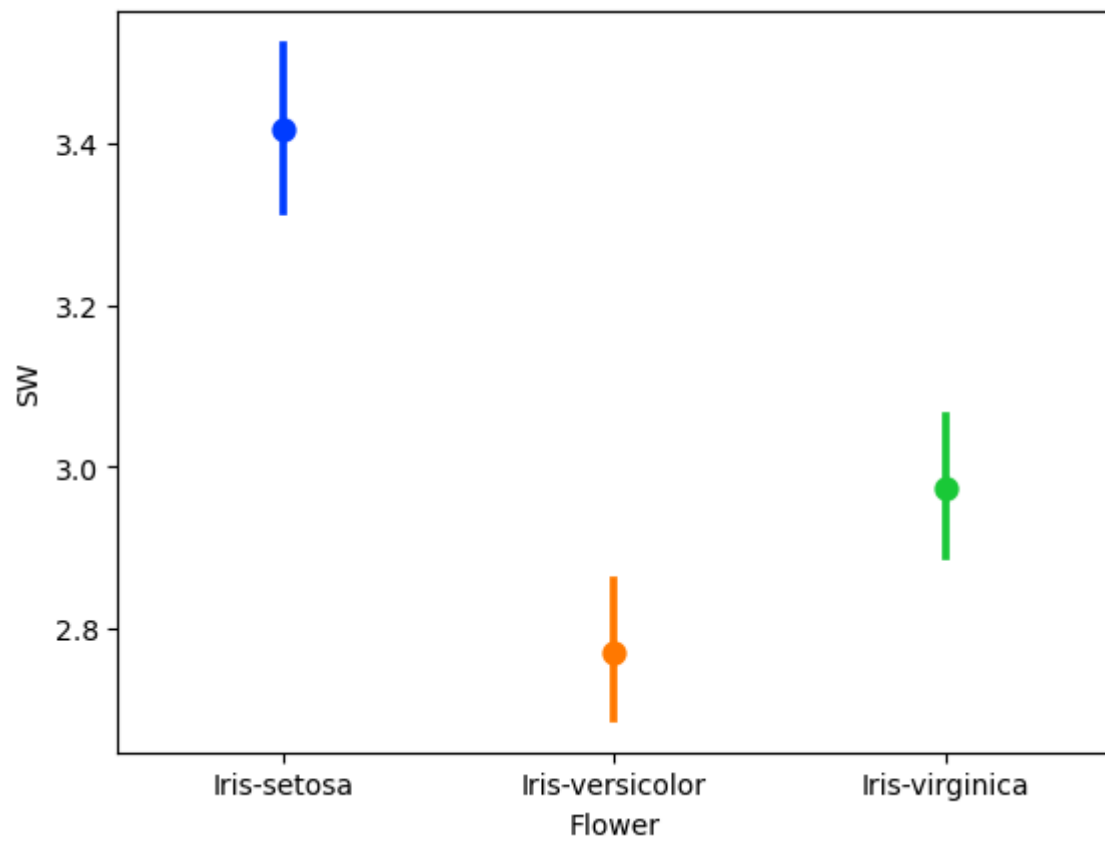
```
In [35]: #Bar Plot
sns.barplot(x=iris.Flower,y=iris.SW,palette='bright')
plt.show()
```



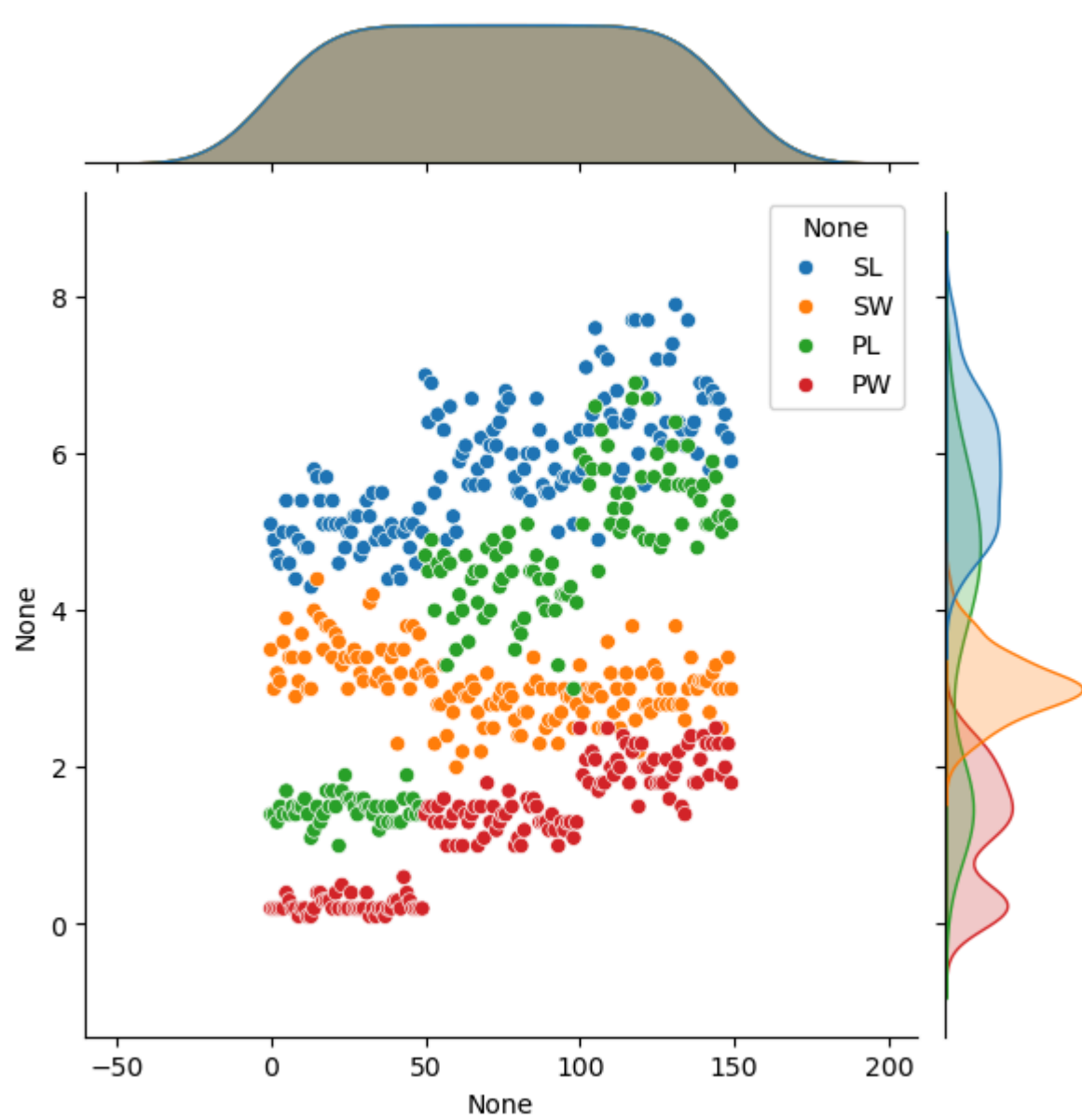
```
In [36]: #Point Plot
sns.pointplot(x=iris.Flower,y=iris.SW)
plt.show()
```



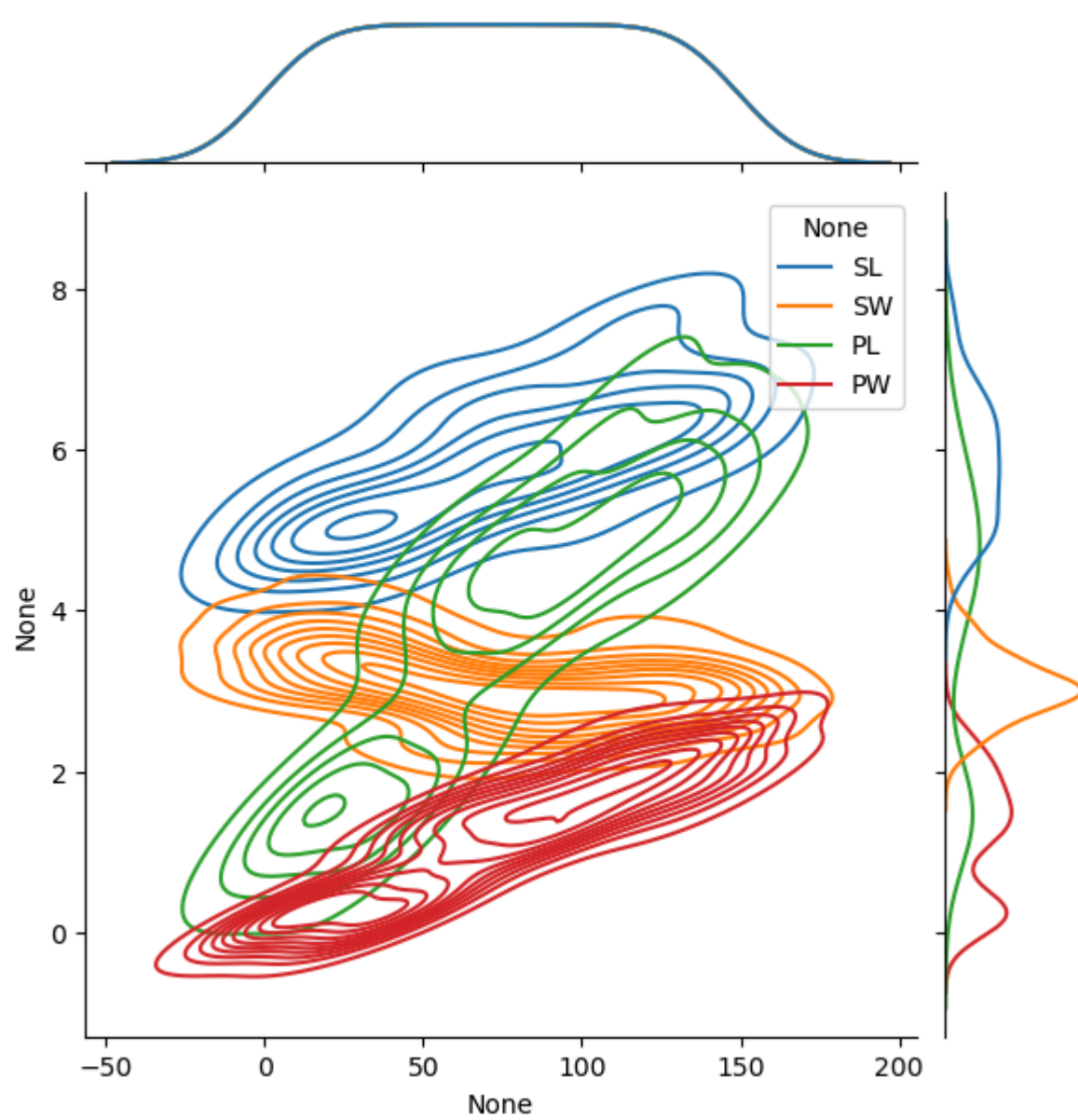
```
In [37]: #Point Plot
sns.pointplot(x=iris.Flower,y=iris.SW,palette='bright')
plt.show()
```



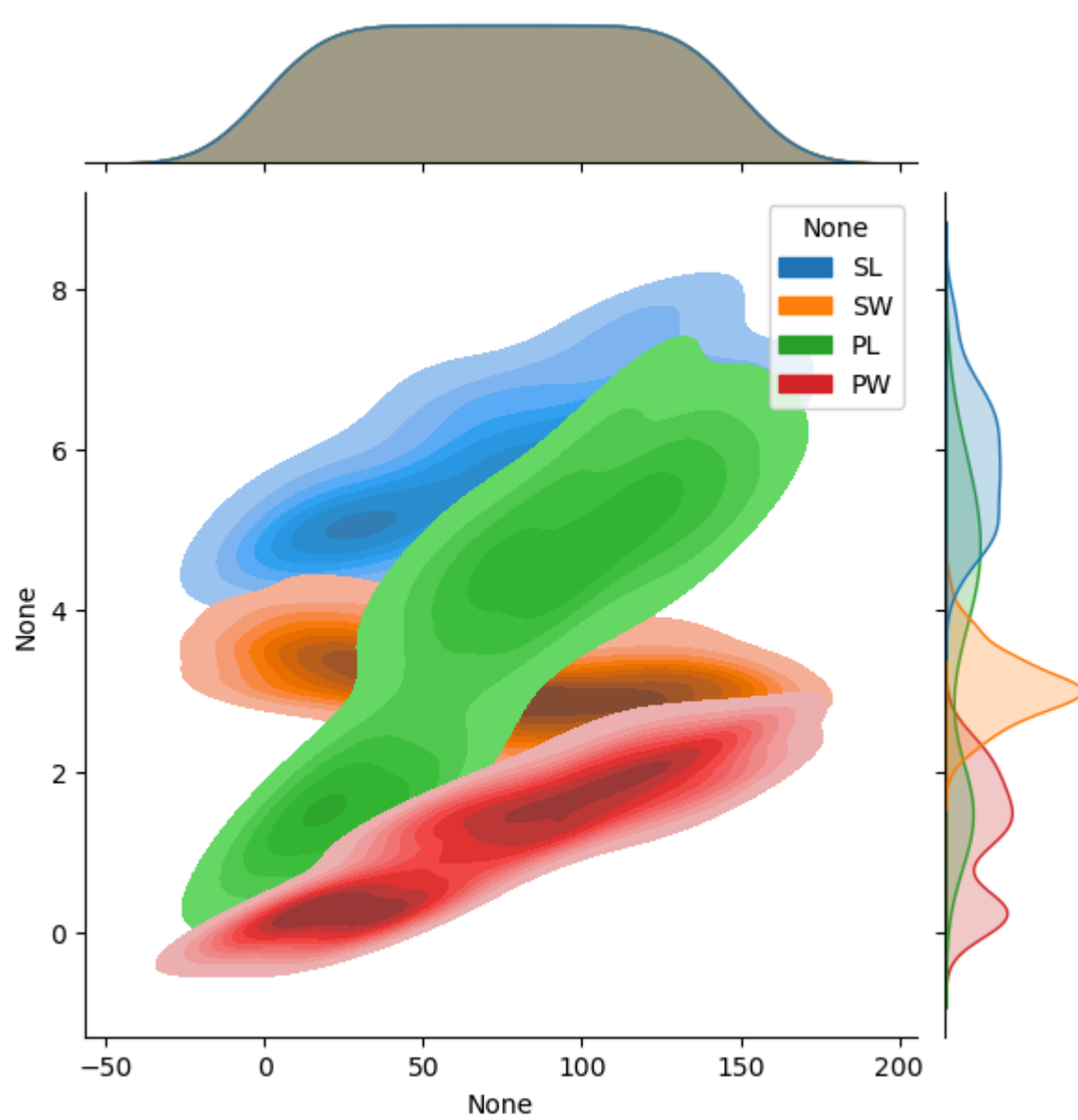
```
In [38]: #Continuous and Continuous  
#Joint Plot  
sns.jointplot(data=iris)  
plt.show()
```



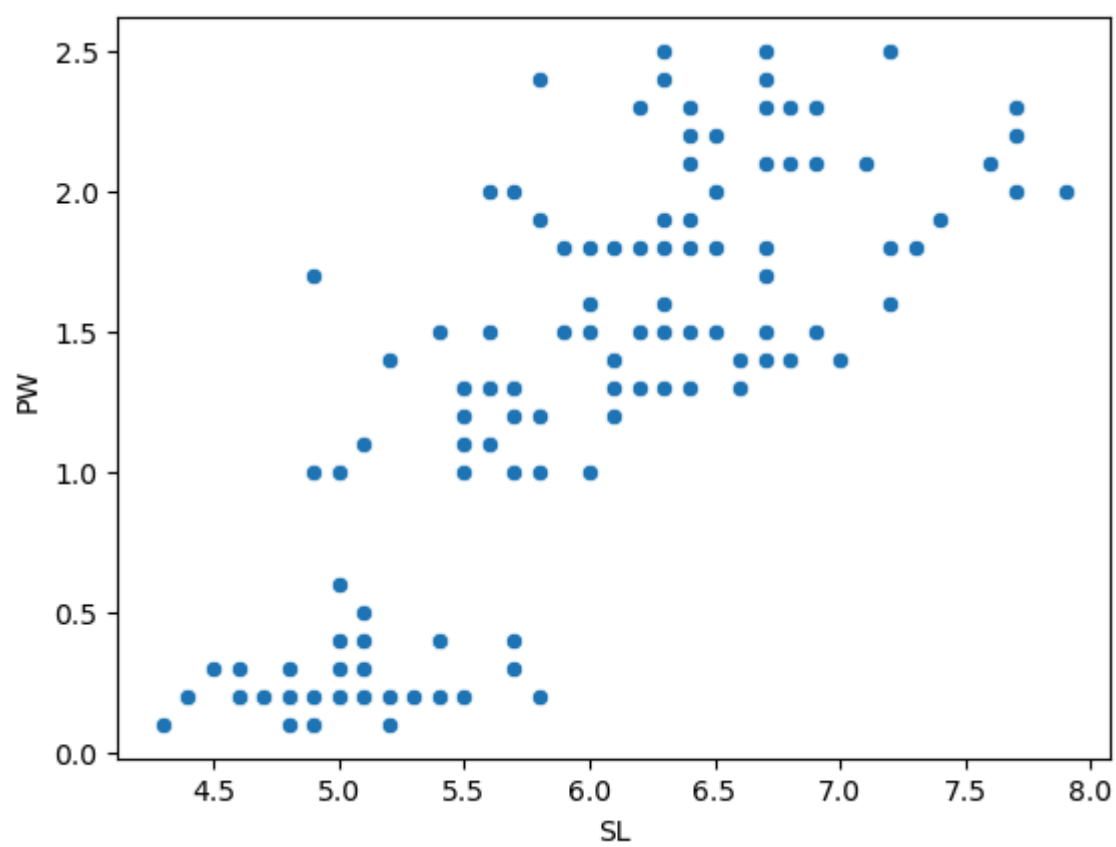
```
In [39]: sns.jointplot(data=iris,kind='kde')  
plt.show()
```



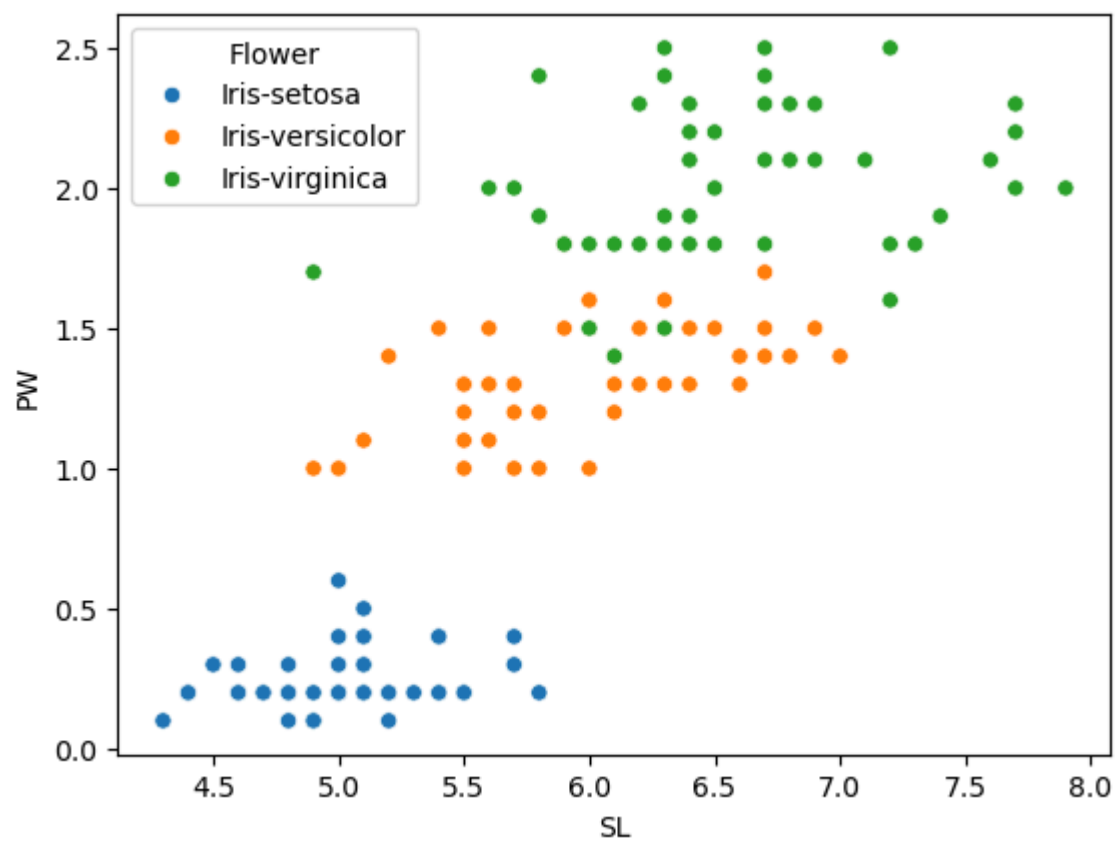
```
In [40]: sns.jointplot(data=iris,kind='kde',fill=True)
plt.show()
```



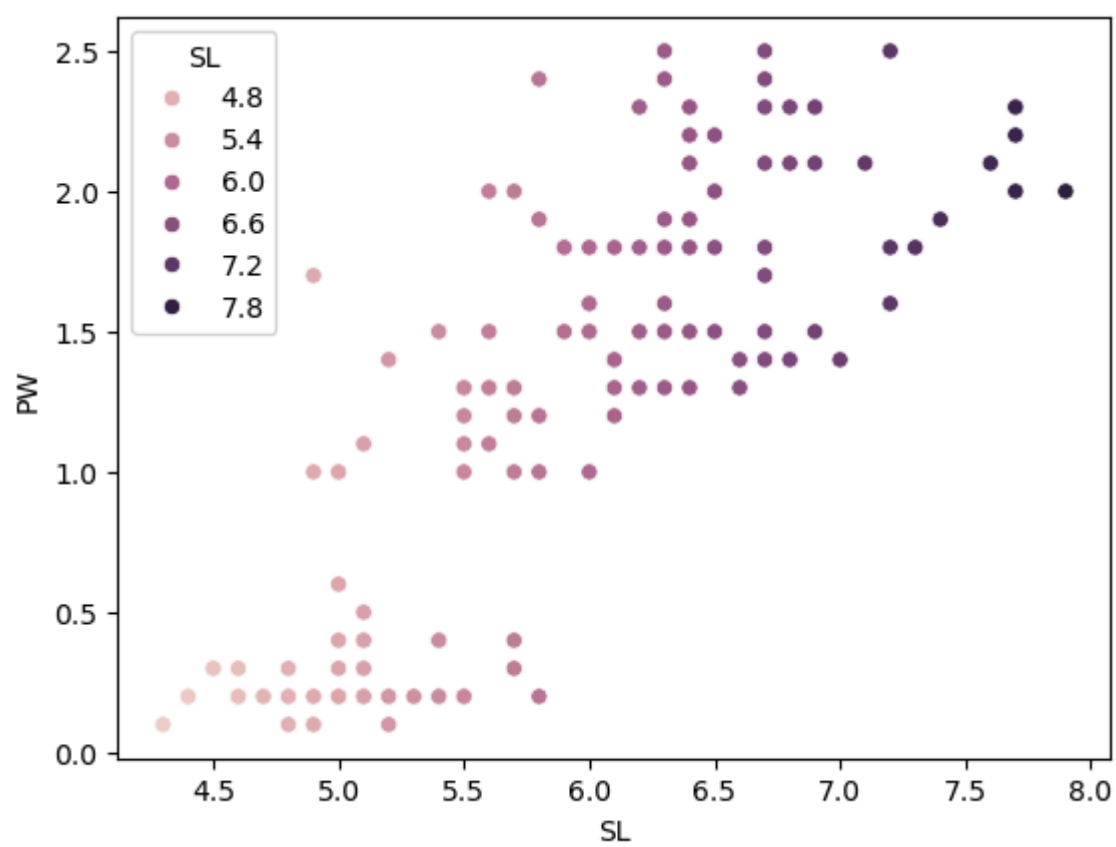
```
In [41]: #Scatter Plot
sns.scatterplot(x=iris.SL,y=iris.PW)
plt.show()
```

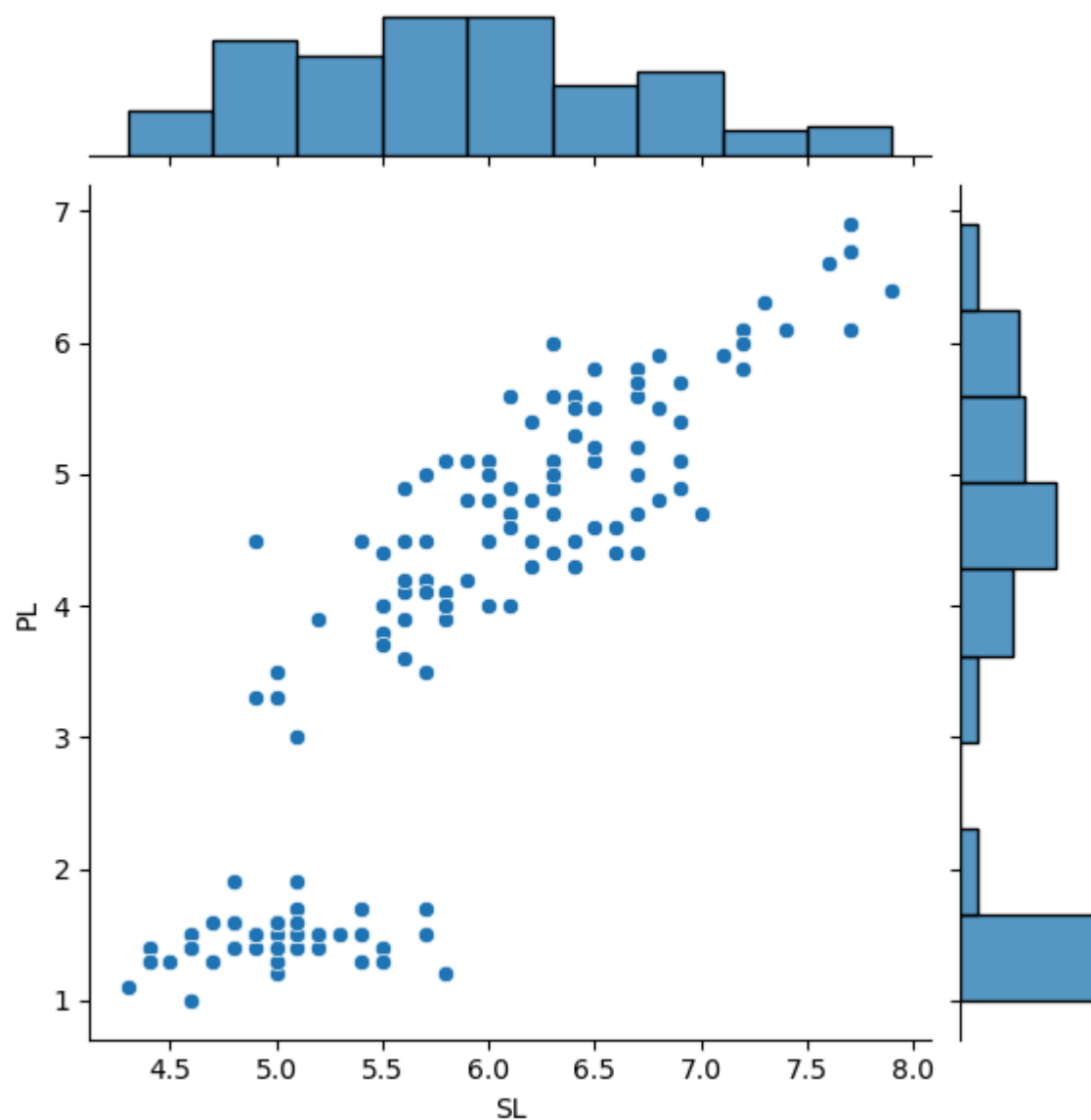
```
In [42]: sns.scatterplot(x=iris.SL,y=iris.PW,hue=iris.Flower)
plt.show()
```



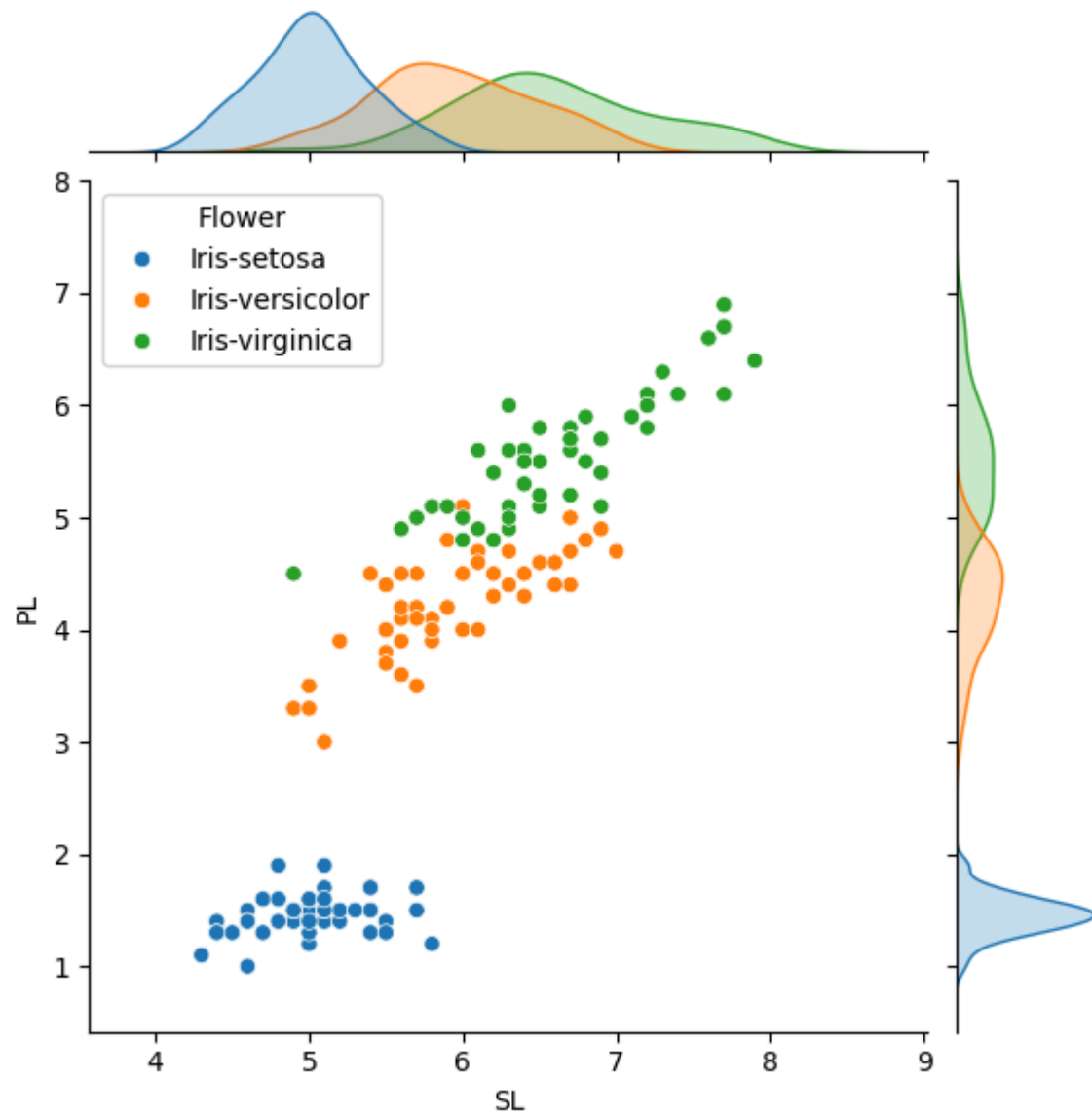
```
In [43]: sns.scatterplot(x=iris.SL,y=iris.PW,hue=iris.SL)
plt.show()
```



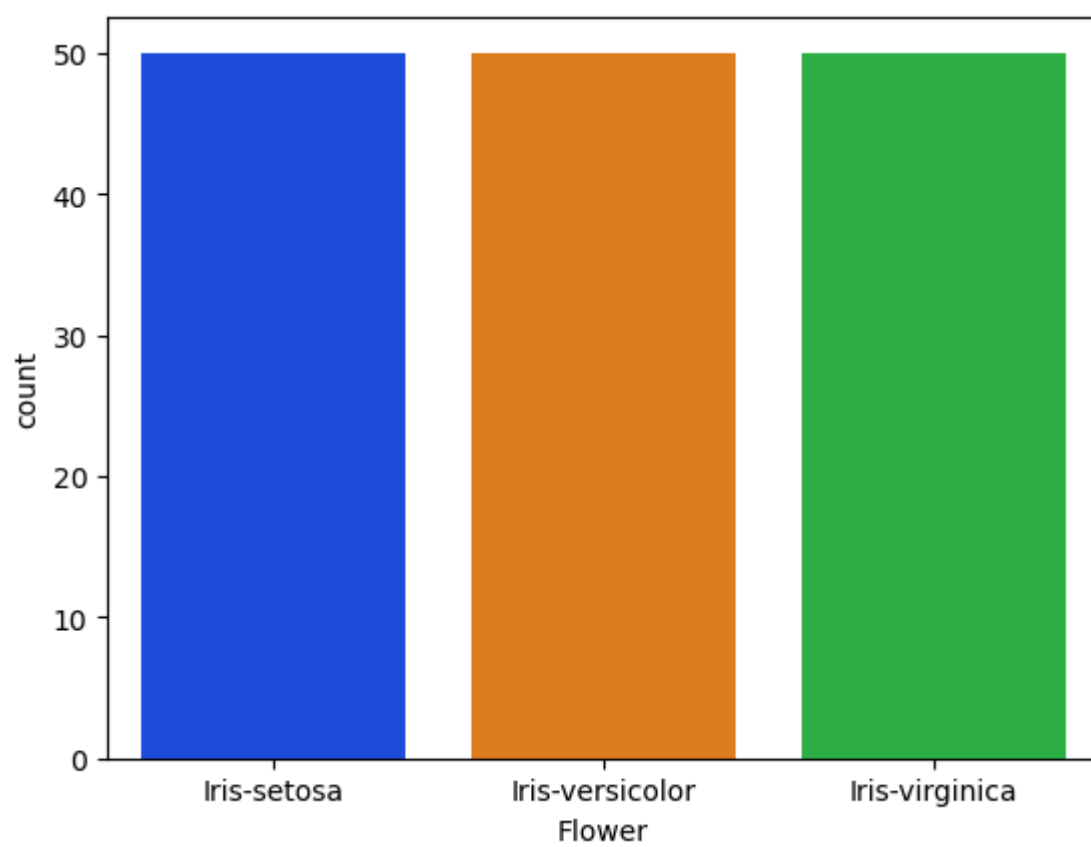
```
In [45]: sns.jointplot(x=iris.SL,y=iris.PL)
plt.show()
```



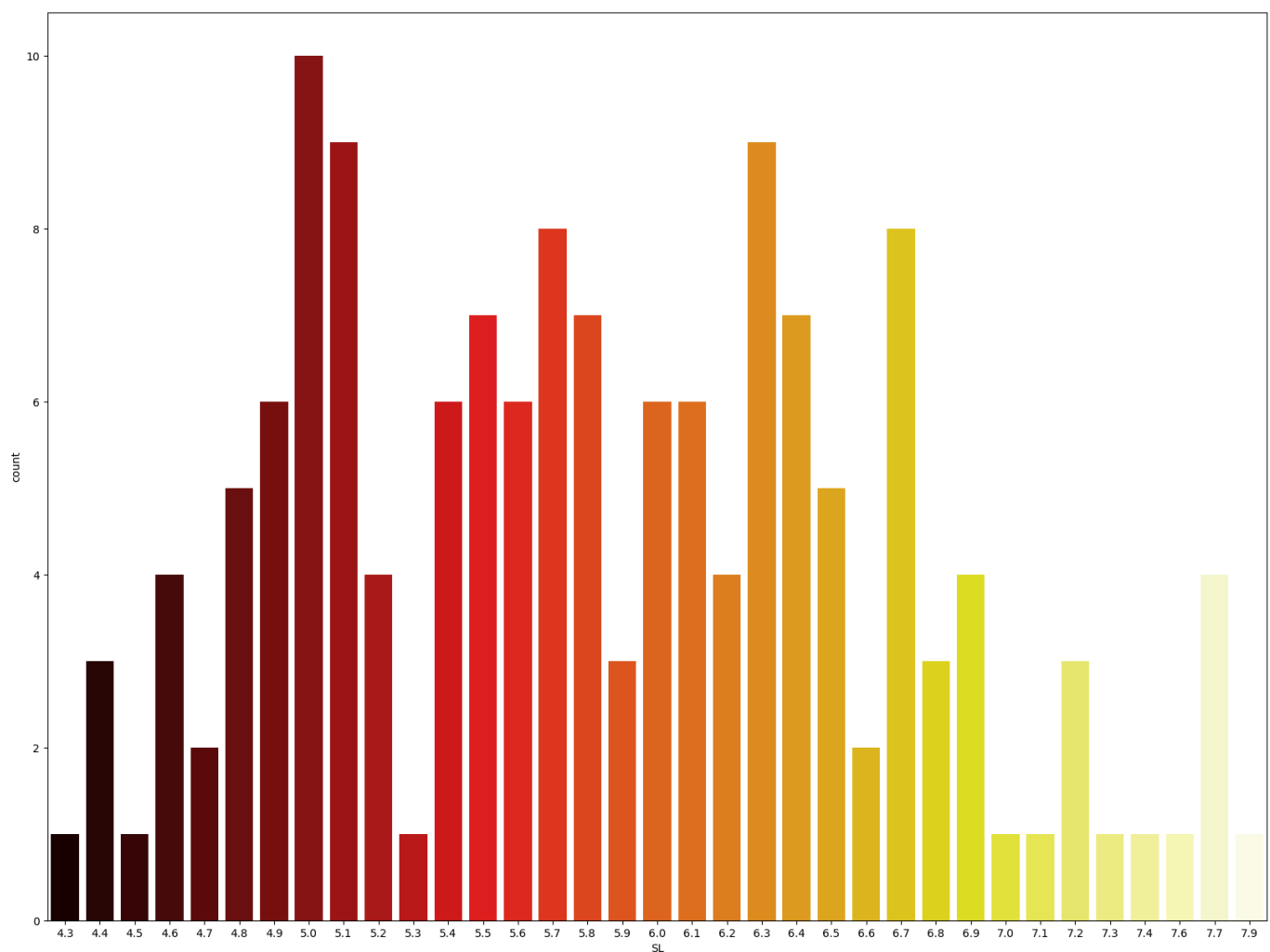
```
In [46]: sns.jointplot(x=iris.SL,y=iris.PL,hue=iris.Flower)
plt.show()
```



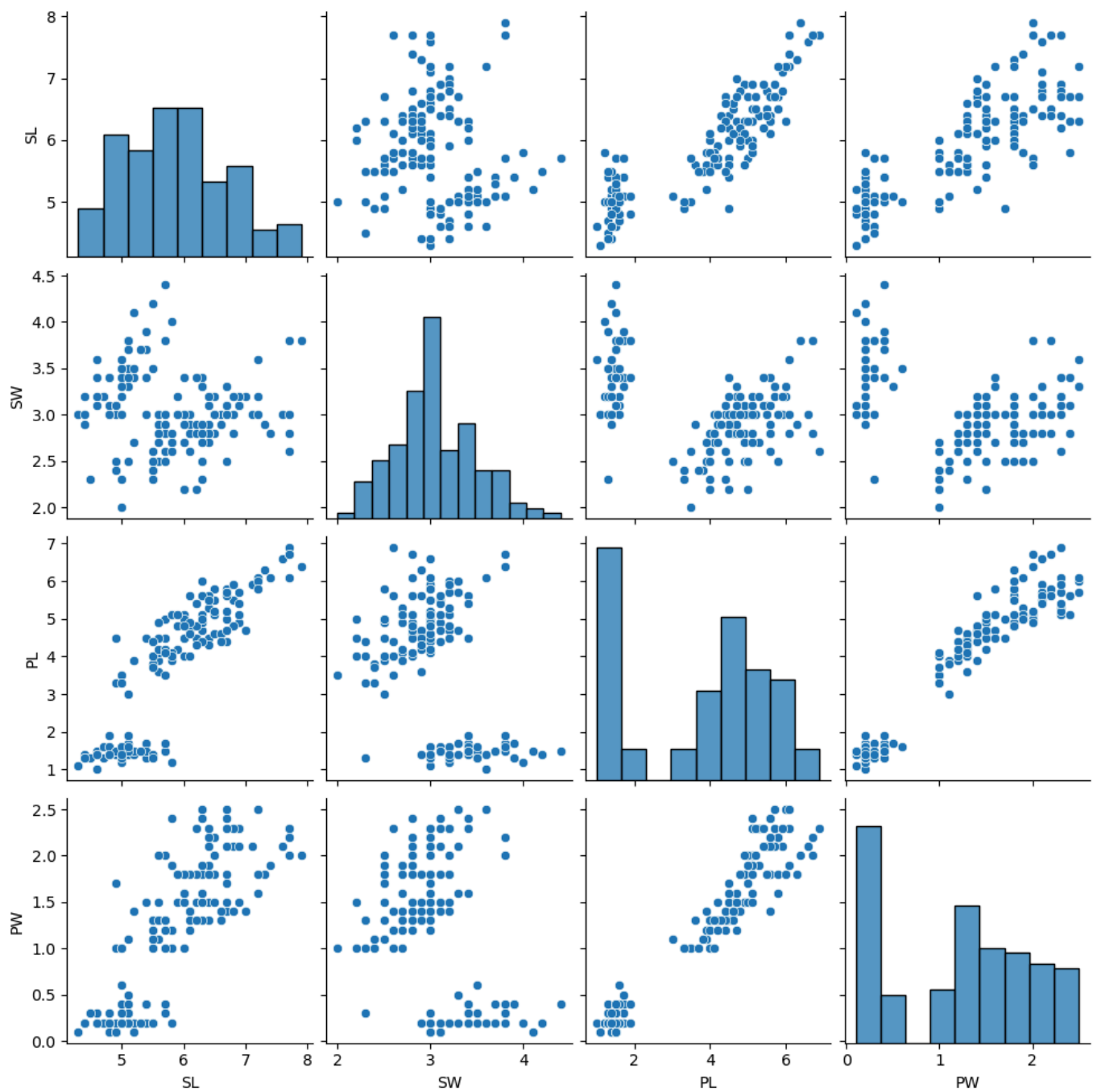
```
In [48]: #Categorical
#Count Plot
sns.countplot(x=iris.Flower,palette='bright')
plt.show()
```



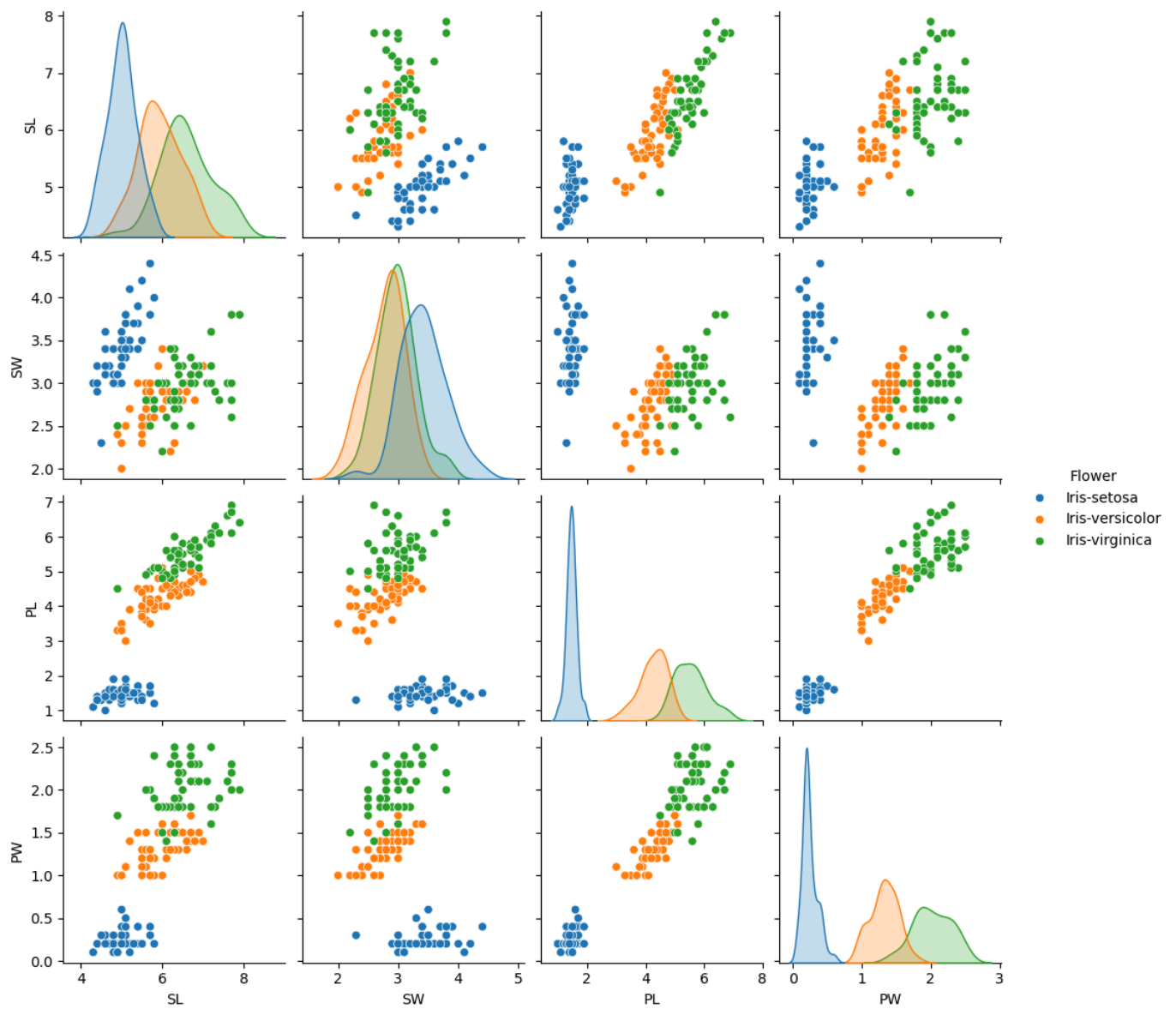
```
In [61]: plt.figure(figsize=(20,15))
sns.countplot(x=iris.SL,palette='hot')
plt.show()
```



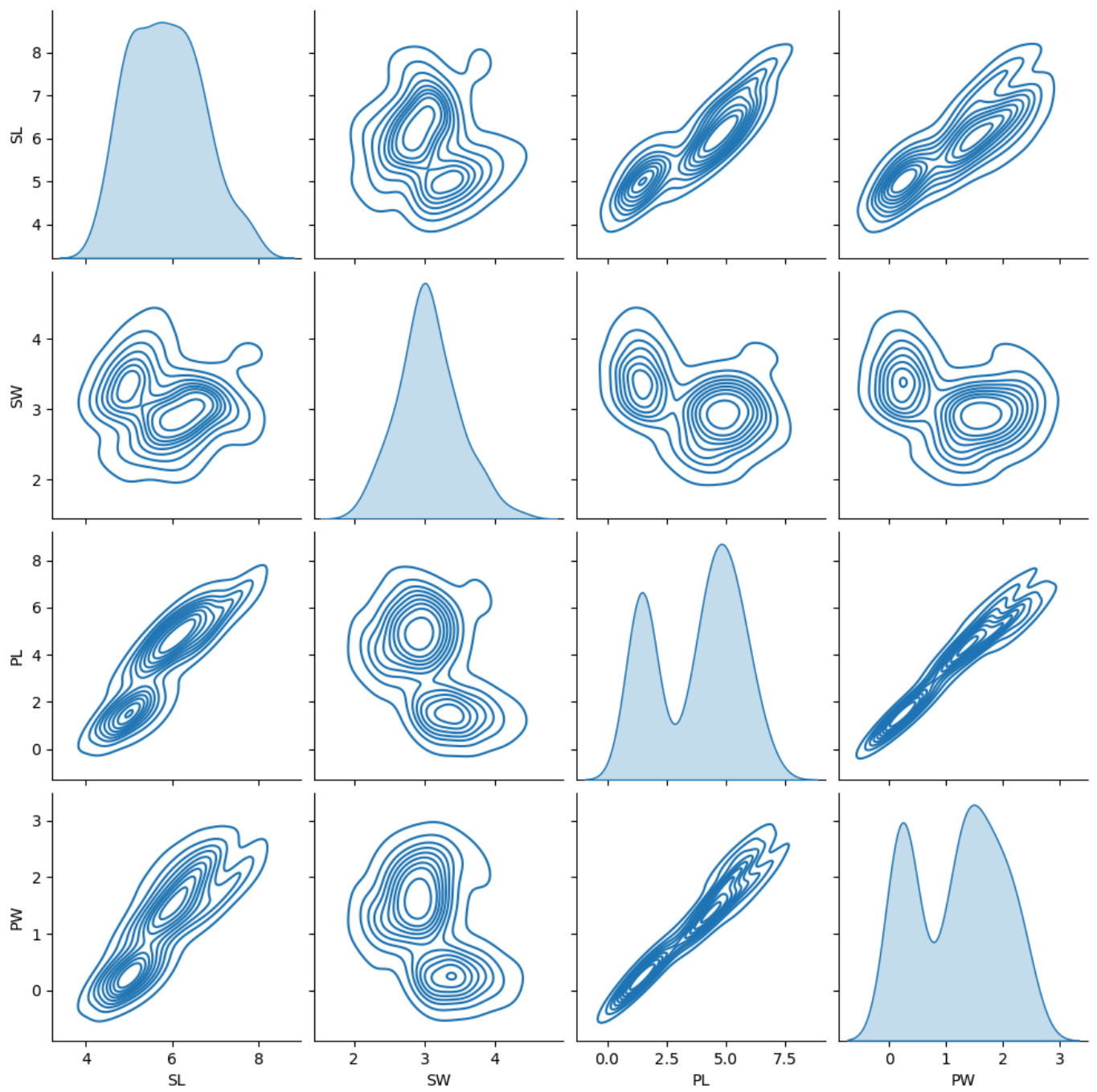
```
In [62]: #Pair Plot
sns.pairplot(iris)
plt.show()
```



```
In [63]: sns.pairplot(iris,hue='Flower')
plt.show()
```



```
In [64]: sns.pairplot(iris, kind='kde')
plt.show()
```



```
In [65]: #HeatMap
c = iris.corr()
c
```

ValueError

Traceback (most recent call last)

Cell In[65], line 2

```
1 #HeatMap
----> 2 c = iris.corr()
3 c
```

File E:\anaconda\Lib\site-packages\pandas\core\frame.py:11049, in DataFrame.corr(self, method, min_periods, numeric_only)

```
11047 cols = data.columns
11048 idx = cols.copy()
> 11049 mat = data.to_numpy(dtype=float, na_value=np.nan, copy=False)
11051 if method == "pearson":
11052     correl = libalgos.nancorr(mat, minp=min_periods)
```

File E:\anaconda\Lib\site-packages\pandas\core\frame.py:1993, in DataFrame.to_numpy(self, dtype, copy, na_value)

```
1991 if dtype is not None:
1992     dtype = np.dtype(dtype)
-> 1993 result = self._mgr.as_array(dtype=dtype, copy=copy, na_value=na_value)
1994 if result.dtype is not dtype:
1995     result = np.asarray(result, dtype=dtype)
```

File E:\anaconda\Lib\site-packages\pandas\core\internals\managers.py:1694, in BlockManager.as_array(self, dtype, copy, na_value)

```
1692     arr.flags.writeable = False
1693 else:
-> 1694     arr = self._interleave(dtype=dtype, na_value=na_value)
1695     # The underlying data was copied within _interleave, so no need
1696     # to further copy if copy=True or setting na_value
1698 if na_value is lib.no_default:
```

File E:\anaconda\Lib\site-packages\pandas\core\internals\managers.py:1753, in BlockManager._interleave(self, dtype, na_value)

```
1751     else:
1752         arr = blk.get_values(dtype)
-> 1753     result[rl.indexer] = arr
1754     itemmask[rl.indexer] = 1
1756 if not itemmask.all():
```

ValueError: could not convert string to float: 'Iris-setosa'

```
In [66]: iris.drop('Flower',axis=1,inplace=True)
iris.head(3)
```

```
Out[66]:
```

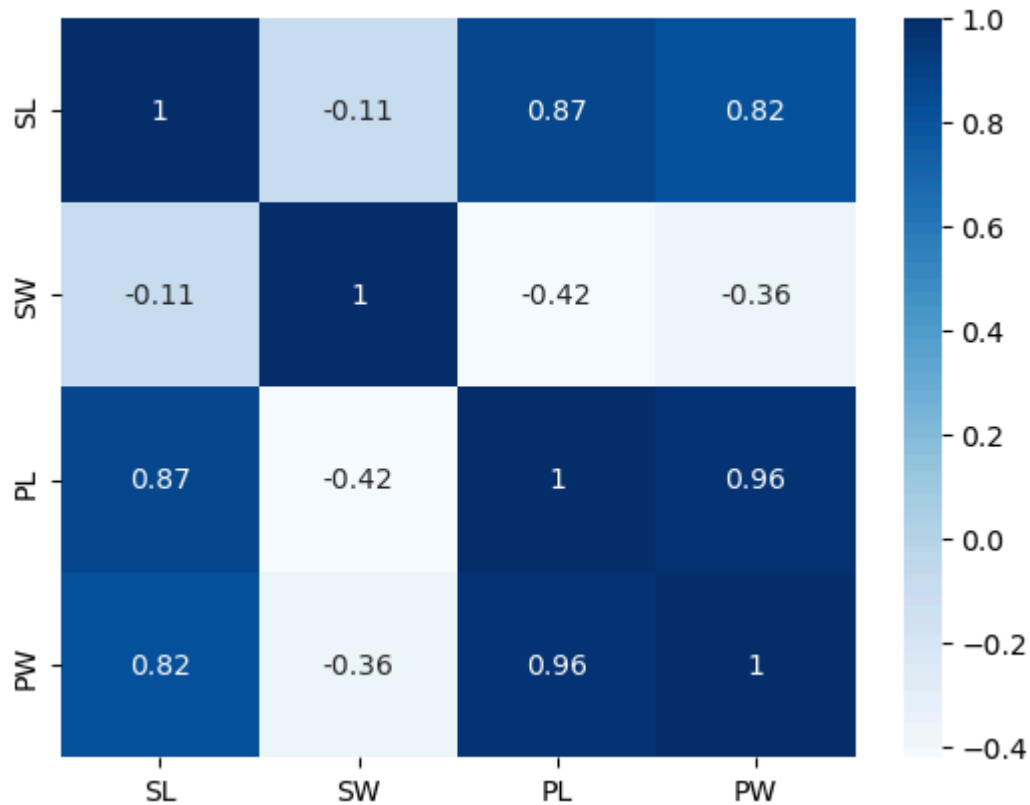
	SL	SW	PL	PW
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2

```
In [67]: c = iris.corr()
c
```


Out[67]:

	SL	SW	PL	PW
SL	1.000000	-0.109369	0.871754	0.817954
SW	-0.109369	1.000000	-0.420516	-0.356544
PL	0.871754	-0.420516	1.000000	0.962757
PW	0.817954	-0.356544	0.962757	1.000000

```
In [72]: sns.heatmap(c,annot=True,cmap='Blues')  
plt.show()
```



In []:

Dataset Practice

```
In [74]: import numpy as np  
import pandas as pd
```

Wine Quality Dataset

```
In [81]: wine = pd.read_csv(r"E:\Downloads data\winequality-red (2).csv",sep=';')  
wine
```

Out[81]:

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol
0	7.4	0.700	0.00	1.9	0.076	11.0	34.0	0.99780	3.51	0.56	9.4
1	7.8	0.880	0.00	2.6	0.098	25.0	67.0	0.99680	3.20	0.68	9.8
2	7.8	0.760	0.04	2.3	0.092	15.0	54.0	0.99700	3.26	0.65	9.8
3	11.2	0.280	0.56	1.9	0.075	17.0	60.0	0.99800	3.16	0.58	9.8
4	7.4	0.700	0.00	1.9	0.076	11.0	34.0	0.99780	3.51	0.56	9.4
...	...	...	...	...	...	...	...	...	...	...	...
1594	6.2	0.600	0.08	2.0	0.090	32.0	44.0	0.99490	3.45	0.58	10.5
1595	5.9	0.550	0.10	2.2	0.062	39.0	51.0	0.99512	3.52	0.76	11.2
1596	6.3	0.510	0.13	2.3	0.076	29.0	40.0	0.99574	3.42	0.75	11.0
1597	5.9	0.645	0.12	2.0	0.075	32.0	44.0	0.99547	3.57	0.71	10.2
1598	6.0	0.310	0.47	3.6	0.067	18.0	42.0	0.99549	3.39	0.66	11.0

1599 rows × 12 columns



In []:

Titanic Dataset

In [82]:

```
titanic = pd.read_csv(r"C:\Users\CTTC\Downloads\titanic dataset.csv")
titanic
```

Out[82]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN
...	...	...	...	...	...	...	...	...	...	...	...
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	NaN
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	B42
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	NaN
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	C148
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	NaN

891 rows × 12 columns



In [83]: `titanic.isnull().sum()`

```
Out[83]: PassengerId      0
Survived      0
Pclass        0
Name          0
Sex           0
Age          177
SibSp         0
Parch         0
Ticket        0
Fare          0
Cabin        687
Embarked      2
dtype: int64
```

```
In [84]: titanic.drop('Cabin',axis=1,inplace=True)
titanic.head(3)
```

Out[84]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S

```
In [85]: d = titanic.describe(include='all')
d
```

Out[85]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket
count	891.000000	891.000000	891.000000	891	891	714.000000	891.000000	891.000000	891
unique	NaN	NaN	NaN	891	2	NaN	NaN	NaN	688
top	NaN	NaN	NaN	Braund, Mr. Owen Harris	male	NaN	NaN	NaN	347081
freq	NaN	NaN	NaN	1	577	NaN	NaN	NaN	1
mean	446.000000	0.383838	2.308642	NaN	NaN	29.699118	0.523008	0.381594	NaN
std	257.353842	0.486592	0.836071	NaN	NaN	14.526497	1.102743	0.806057	NaN
min	1.000000	0.000000	1.000000	NaN	NaN	0.420000	0.000000	0.000000	NaN
25%	223.500000	0.000000	2.000000	NaN	NaN	20.125000	0.000000	0.000000	NaN
50%	446.000000	0.000000	3.000000	NaN	NaN	28.000000	0.000000	0.000000	NaN
75%	668.500000	1.000000	3.000000	NaN	NaN	38.000000	1.000000	0.000000	NaN
max	891.000000	1.000000	3.000000	NaN	NaN	80.000000	8.000000	6.000000	NaN

```
In [86]: titanic.Age.fillna(d['Age'][4],inplace=True)
```

```
In [87]: titanic.Embarked.replace(np.nan,'S',inplace=True)
```

```
In [88]: titanic.isnull().sum()
```

```
Out[88]: PassengerId      0  
Survived      0  
Pclass      0  
Name      0  
Sex      0  
Age      0  
SibSp      0  
Parch      0  
Ticket      0  
Fare      0  
Embarked      0  
dtype: int64
```

```
In [89]: titanic.dtypes
```

```
Out[89]: PassengerId      int64  
Survived      int64  
Pclass      int64  
Name      object  
Sex      object  
Age      float64  
SibSp      int64  
Parch      int64  
Ticket      object  
Fare      float64  
Embarked      object  
dtype: object
```

```
In [90]: for i in titanic.columns:  
    print(i,':','\n',titanic[i].unique(),'\n')
```

```
PassengerId :
[ 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15 16 17 18
 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36
 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54
 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72
 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90
 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108
109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126
127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144
145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162
163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180
181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198
199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216
217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234
235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252
253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270
271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288
289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306
307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324
325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342
343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360
361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378
379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396
397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414
415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432
433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450
451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468
469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486
487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504
505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522
523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540
541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558
559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576
577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594
595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612
613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630
631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648
649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666
667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684
685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702
703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720
721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738
739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756
757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774
775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792
793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810
811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828
829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846
847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864
865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882
883 884 885 886 887 888 889 890 891]
```

```
Survived :
[0 1]
```

```
Pclass :
[3 1 2]
```

```
Name :
['Braund, Mr. Owen Harris'
 'Cumings, Mrs. John Bradley (Florence Briggs Thayer)'
 'Heikkinen, Miss. Laina' 'Futrelle, Mrs. Jacques Heath (Lily May Peel)'
 'Allen, Mr. William Henry' 'Moran, Mr. James' 'McCarthy, Mr. Timothy J'
 'Palsson, Master. Gosta Leonard'
 'Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)'
 'Nasser, Mrs. Nicholas (Adele Achem)' 'Sandstrom, Miss. Marguerite Rut'
```

'Bonnell, Miss. Elizabeth' 'Sundercock, Mr. William Henry'
'Andersson, Mr. Anders Johan' 'Vestrom, Miss. Hulda Amanda Adolfina'
'Hewlett, Mrs. (Mary D Kingcome)' 'Rice, Master. Eugene'
'Williams, Mr. Charles Eugene'
'Vander Planke, Mrs. Julius (Emelia Maria Vandemoortele)'
'Masselmani, Mrs. Fatima' 'Fynney, Mr. Joseph J' 'Beesley, Mr. Lawrence'
'McGowan, Miss. Anna "Annie"' 'Sloper, Mr. William Thompson'
'Palsson, Miss. Torborg Danira'
'Asplund, Mrs. Carl Oscar (Selma Augusta Emilia Johansson)'
'Emir, Mr. Farred Chehab' 'Fortune, Mr. Charles Alexander'
'O'Dwyer, Miss. Ellen "Nellie"' 'Todoroff, Mr. Lalio'
'Uruchurtu, Don. Manuel E'
'Spencer, Mrs. William Augustus (Marie Eugenie)'
'Glynn, Miss. Mary Agatha' 'Wheadon, Mr. Edward H'
'Meyer, Mr. Edgar Joseph' 'Holverson, Mr. Alexander Oskar'
'Mamee, Mr. Hanna' 'Cann, Mr. Ernest Charles'
'Vander Planke, Miss. Augusta Maria' 'Nicola-Yarred, Miss. Jamila'
'Ahlin, Mrs. Johan (Johanna Persdotter Larsson)'
'Turpin, Mrs. William John Robert (Dorothy Ann Wonnacott)'
'Kraeff, Mr. Theodor' 'Laroche, Miss. Simonne Marie Anne Andree'
'Devaney, Miss. Margaret Delia' 'Rogers, Mr. William John'
'Lennon, Mr. Denis' 'O'Driscoll, Miss. Bridget' 'Samaan, Mr. Youssef'
'Arnold-Franchi, Mrs. Josef (Josefine Franchi)'
'Panula, Master. Juha Niilo' 'Nosworthy, Mr. Richard Cater'
'Harper, Mrs. Henry Sleeper (Myna Haxtun)'
'Faunthorpe, Mrs. Lizzie (Elizabeth Anne Wilkinson)'
'Ostby, Mr. Engelhart Cornelius' 'Woolner, Mr. Hugh' 'Rugg, Miss. Emily'
'Novel, Mr. Mansouer' 'West, Miss. Constance Mirium'
'Goodwin, Master. William Frederick' 'Sirayanian, Mr. Orsen'
'Icard, Miss. Amelie' 'Harris, Mr. Henry Birkhardt'
'Skoog, Master. Harald' 'Stewart, Mr. Albert A'
'Moubarek, Master. Gerios' 'Nye, Mrs. (Elizabeth Ramell)'
'Crease, Mr. Ernest James' 'Andersson, Miss. Erna Alexandra'
'Kink, Mr. Vincenz' 'Jenkin, Mr. Stephen Curnow'
'Goodwin, Miss. Lillian Amy' 'Hood, Mr. Ambrose Jr'
'Chronopoulos, Mr. Apostolos' 'Bing, Mr. Lee' 'Moen, Mr. Sigurd Hansen'
'Staneff, Mr. Ivan' 'Moutal, Mr. Rahamin Haim'
'Caldwell, Master. Alden Gates' 'Dowdell, Miss. Elizabeth'
'Waelens, Mr. Achille' 'Sheerlinck, Mr. Jan Baptist'
'McDermott, Miss. Brigdet Delia' 'Carrau, Mr. Francisco M'
'Ilett, Miss. Bertha'
'Backstrom, Mrs. Karl Alfred (Maria Mathilda Gustafsson)'
'Ford, Mr. William Neal' 'Slocovski, Mr. Selman Francis'
'Fortune, Miss. Mabel Helen' 'Celotti, Mr. Francesco'
'Christmann, Mr. Emil' 'Andreasson, Mr. Paul Edwin'
'Chaffee, Mr. Herbert Fuller' 'Dean, Mr. Bertram Frank'
'Coxon, Mr. Daniel' 'Shorney, Mr. Charles Joseph'
'Goldschmidt, Mr. George B' 'Greenfield, Mr. William Bertram'
'Doling, Mrs. John T (Ada Julia Bone)' 'Kantor, Mr. Sinai'
'Petranec, Miss. Matilda' 'Petroff, Mr. Pastcho ("Pentcho")'
'White, Mr. Richard Frasar' 'Johansson, Mr. Gustaf Joel'
'Gustafsson, Mr. Anders Vilhelm' 'Mionoff, Mr. Stoytcho'
'Salkjelsvik, Miss. Anna Kristine' 'Moss, Mr. Albert Johan'
'Rekic, Mr. Tido' 'Moran, Miss. Bertha' 'Porter, Mr. Walter Chamberlain'
'Zabour, Miss. Hileni' 'Barton, Mr. David John' 'Jussila, Miss. Katriina'
'Attalah, Miss. Malake' 'Pekoniemi, Mr. Edvard' 'Connors, Mr. Patrick'
'Turpin, Mr. William John Robert' 'Baxter, Mr. Quigg Edmond'
'Andersson, Miss. Ellis Anna Maria' 'Hickman, Mr. Stanley George'
'Moore, Mr. Leonard Charles' 'Nasser, Mr. Nicholas' 'Webber, Miss. Susan'
'White, Mr. Percival Wayland' 'Nicola-Yarred, Master. Elias'
'McMahon, Mr. Martin' 'Madsen, Mr. Fridtjof Arne' 'Peter, Miss. Anna'
'Ekstrom, Mr. Johan' 'Drazenoic, Mr. Jozef'
'Coelho, Mr. Domingos Fernando'
'Robins, Mrs. Alexander A (Grace Charity Laury)'
'Weisz, Mrs. Leopold (Mathilde Francoise Pede)'
'Sobey, Mr. Samuel James Hayden' 'Richard, Mr. Emile'

'Newsom, Miss. Helen Monypeny' 'Futrelle, Mr. Jacques Heath'
'Osen, Mr. Olaf Elon' 'Giglio, Mr. Victor'
'Boulos, Mrs. Joseph (Sultana)' 'Nysten, Miss. Anna Sofia'
'Hakkarainen, Mrs. Pekka Pietari (Elin Matilda Dolck)'
'Burke, Mr. Jeremiah' 'Andrew, Mr. Edgardo Samuel'
'Nicholls, Mr. Joseph Charles'
'Andersson, Mr. August Edvard ("Wennerstrom")'
'Ford, Miss. Robina Maggie "Ruby"'
'Navratil, Mr. Michel ("Louis M Hoffman")'
'Byles, Rev. Thomas Roussel Davids' 'Bateman, Rev. Robert James'
'Pears, Mrs. Thomas (Edith Wearne)' 'Meo, Mr. Alfonzo'
'van Billiard, Mr. Austin Blyler' 'Olsen, Mr. Ole Martin'
'Williams, Mr. Charles Duane' 'Gilnagh, Miss. Katherine "Katie"'
'Corn, Mr. Harry' 'Smiljanic, Mr. Mile' 'Sage, Master. Thomas Henry'
'Cribb, Mr. John Hatfield'
'Watt, Mrs. James (Elizabeth "Bessie" Inglis Milne)'
'Bengtsson, Mr. John Viktor' 'Calic, Mr. Jovo'
'Panula, Master. Eino Viljami'
'Goldsmith, Master. Frank John William "Frankie"'
'Chibnall, Mrs. (Edith Martha Bowerman)'
'Skoog, Mrs. William (Anna Bernhardina Karlsson)' 'Baumann, Mr. John D'
'Ling, Mr. Lee' 'Van der hoef, Mr. Wyckoff' 'Rice, Master. Arthur'
'Johnson, Miss. Eleanor Ileen' 'Sivola, Mr. Antti Wilhelm'
'Smith, Mr. James Clinch' 'Klasen, Mr. Klas Albin'
'Lefebvre, Master. Henry Forbes' 'Isham, Miss. Ann Elizabeth'
'Hale, Mr. Reginald' 'Leonard, Mr. Lionel' 'Sage, Miss. Constance Gladys'
'Pernot, Mr. Rene' 'Asplund, Master. Clarence Gustaf Hugo'
'Becker, Master. Richard F' 'Kink-Heilmann, Miss. Luise Gretchen'
'Rood, Mr. Hugh Roscoe'
'O'Brien, Mrs. Thomas (Johanna "Hannah" Godfrey)'
'Romaine, Mr. Charles Hallace ("Mr C Rolmane")' 'Bourke, Mr. John'
'Turcin, Mr. Stjepan' 'Pinsky, Mrs. (Rosa)' 'Carbines, Mr. William'
'Andersen-Jensen, Miss. Carla Christine Nielsine'
'Navratil, Master. Michel M' 'Brown, Mrs. James Joseph (Margaret Tobin)'
'Lurette, Miss. Elise' 'Mernagh, Mr. Robert'
'Olsen, Mr. Karl Siegwart Andreas' 'Madigan, Miss. Margaret "Maggie"'
'Yrois, Miss. Henriette ("Mrs Harbeck")' 'Vande Walle, Mr. Nestor Cyriel'
'Sage, Mr. Frederick' 'Johanson, Mr. Jakob Alfred' 'Youseff, Mr. Gerious'
'Cohen, Mr. Gurshon "Gus"' 'Strom, Miss. Telma Matilda'
'Backstrom, Mr. Karl Alfred' 'Albimona, Mr. Nassef Cassem'
'Carr, Miss. Helen "Ellen"' 'Blank, Mr. Henry' 'Ali, Mr. Ahmed'
'Cameron, Miss. Clear Annie' 'Perkin, Mr. John Henry'
'Givard, Mr. Hans Kristensen' 'Kiernan, Mr. Philip'
'Newell, Miss. Madeleine' 'Honkanen, Miss. Eliina'
'Jacobsohn, Mr. Sidney Samuel' 'Bazzani, Miss. Albina'
'Harris, Mr. Walter' 'Sunderland, Mr. Victor Francis'
'Bracken, Mr. James H' 'Green, Mr. George Henry' 'Nenkoff, Mr. Christo'
'Hoyt, Mr. Frederick Maxfield' 'Berglund, Mr. Karl Ivar Sven'
'Mellors, Mr. William John' 'Lovell, Mr. John Hall ("Henry")'
'Fahlstrom, Mr. Arne Jonas' 'Lefebvre, Miss. Mathilde'
'Harris, Mrs. Henry Birkhardt (Irene Wallach)' 'Larsson, Mr. Bengt Edvin'
'Sjostedt, Mr. Ernst Adolf' 'Asplund, Miss. Lillian Gertrud'
'Leyson, Mr. Robert William Norman' 'Harknett, Miss. Alice Phoebe'
'Hold, Mr. Stephen' 'Collyer, Miss. Marjorie "Lottie"'
'Pengelly, Mr. Frederick William' 'Hunt, Mr. George Henry'
'Zabour, Miss. Thamine' 'Murphy, Miss. Katherine "Kate"'
'Coleridge, Mr. Reginald Charles' 'Maenpaa, Mr. Matti Alexanteri'
'Attalah, Mr. Sleiman' 'Minahan, Dr. William Edward'
'Lindhahl, Miss. Agda Thorilda Viktoria' 'Hamalainen, Mrs. William (Anna)'
'Beckwith, Mr. Richard Leonard' 'Carter, Rev. Ernest Courtenay'
'Reed, Mr. James George' 'Strom, Mrs. Wilhelm (Elna Matilda Persson)'
'Stead, Mr. William Thomas' 'Lobb, Mr. William Arthur'
'Rosblom, Mrs. Viktor (Helena Wilhelmina)'
'Touma, Mrs. Darwis (Hanne Youssef Razi)'
'Thorne, Mrs. Gertrude Maybelle' 'Cherry, Miss. Gladys'
'Ward, Miss. Anna' 'Parrish, Mrs. (Lutie Davis)' 'Smith, Mr. Thomas'

'Asplund, Asplund, Edwin' 'Rojj Felix' 'Taussig, Mr. Emil'
'Harrison, Mr. William' 'Henry, Miss. Delia' 'Reeves, Mr. David'
'Panula, Mr. Ernesti Arvid' 'Persson, Mr. Ernst Ulrik'
'Graham, Mrs. William Thompson (Edith Junkins)' 'Bissette, Miss. Amelia'
'Cairns, Mr. Alexander' 'Tornquist, Mr. William Henry'
'Mellinger, Mrs. (Elizabeth Anne Maidment)' 'Natsch, Mr. Charles H'
'Healy, Miss. Hanora "Nora"' 'Andrews, Miss. Kornelia Theodosia'
'Lindblom, Miss. Augusta Charlotta' 'Parkes, Mr. Francis "Frank"'
'Rice, Master. Eric' 'Abbott, Mrs. Stanton (Rosa Hunt)'
'Duane, Mr. Frank' 'Olsson, Mr. Nils Johan Goransson'
'de Pelsmaeker, Mr. Alfons' 'Dorking, Mr. Edward Arthur'
'Smith, Mr. Richard William' 'Stankovic, Mr. Ivan'
'de Mulder, Mr. Theodore' 'Naidenoff, Mr. Penko' 'Hosono, Mr. Masabumi'
'Connolly, Miss. Kate' 'Barber, Miss. Ellen "Nellie"'
'Bishop, Mrs. Dickinson H (Helen Walton)' 'Levy, Mr. Rene Jacques'
'Haas, Miss. Aloisia' 'Mineff, Mr. Ivan' 'Lewy, Mr. Ervin G'
'Hanna, Mr. Mansour' 'Allison, Miss. Helen Loraine'
'Saalfeld, Mr. Adolphe' 'Baxter, Mrs. James (Helene DeLaudeniére Chaput)'
'Kelly, Miss. Anna Katherine "Annie Kate"' 'McCoy, Mr. Bernard'
'Johnson, Mr. William Cahoon Jr' 'Keane, Miss. Nora A'
'Williams, Mr. Howard Hugh "Harry"' 'Allison, Master. Hudson Trevor'
'Fleming, Miss. Margaret'
'Penasco y Castellana, Mrs. Victor de Satode (Maria Josefa Perez de Soto y Vallejo)'
'Abelson, Mr. Samuel' 'Francatelli, Miss. Laura Mabel'
'Hays, Miss. Margaret Bechstein' 'Ryerson, Miss. Emily Borie'
'Lahtinen, Mrs. William (Anna Sylfven)' 'Hendekovic, Mr. Ignjac'
'Hart, Mr. Benjamin' 'Nilsson, Miss. Helmina Josefina'
'Kantor, Mrs. Sinai (Miriam Sternin)' 'Moraweck, Dr. Ernest'
'Wick, Miss. Mary Natalie'
'Spedden, Mrs. Frederic Oakley (Margaretta Corning Stone)'
'Dennis, Mr. Samuel' 'Danoff, Mr. Yoto' 'Slayter, Miss. Hilda Mary'
'Caldwell, Mrs. Albert Francis (Sylvia Mae Harbaugh)'
'Sage, Mr. George John Jr' 'Young, Miss. Marie Grice'
'Nysveen, Mr. Johan Hansen' 'Ball, Mrs. (Ada E Hall)'
'Goldsmith, Mrs. Frank John (Emily Alice Brown)'
'Hippach, Miss. Jean Gertrude' 'McCoy, Miss. Agnes' 'Partner, Mr. Austen'
'Graham, Mr. George Edward' 'Vander Planke, Mr. Leo Edmondus'
'Frauenthal, Mrs. Henry William (Clara Heinsheimer)' 'Denkoff, Mr. Mitto'
'Pears, Mr. Thomas Clinton' 'Burns, Miss. Elizabeth Margaret'
'Dahl, Mr. Karl Edward' 'Blackwell, Mr. Stephen Weart'
'Navratil, Master. Edmond Roger' 'Fortune, Miss. Alice Elizabeth'
'Collander, Mr. Erik Gustaf' 'Sedgwick, Mr. Charles Frederick Waddington'
'Fox, Mr. Stanley Hubert' 'Brown, Miss. Amelia "Mildred"'
'Smith, Miss. Marion Elsie' 'Davison, Mrs. Thomas Henry (Mary E Finck)'
'Coutts, Master. William Loch "William"' 'Dimic, Mr. Jovan'
'Odahl, Mr. Nils Martin' 'Williams-Lambert, Mr. Fletcher Fellows'
'Elias, Mr. Tannous' 'Arnold-Franchi, Mr. Josef' 'Yousif, Mr. Wazli'
'Vanden Steen, Mr. Leo Peter' 'Bowerman, Miss. Elsie Edith'
'Funk, Miss. Annie Clemmer' 'McGovern, Miss. Mary'
'Mockler, Miss. Helen Mary "Ellie"' 'Skoog, Mr. Wilhelm'
'del Carlo, Mr. Sebastiano' 'Barbara, Mrs. (Catherine David)'
'Asim, Mr. Adola' 'O'Brien, Mr. Thomas' 'Adahl, Mr. Mauritz Nils Martin'
'Warren, Mrs. Frank Manley (Anna Sophia Atkinson)'
'Moussa, Mrs. (Mantoura Boulos)' 'Jermyn, Miss. Annie'
'Aubart, Mme. Leontine Pauline' 'Harder, Mr. George Achilles'
'Wiklund, Mr. Jakob Alfred' 'Beavan, Mr. William Thomas'
'Ringhini, Mr. Sante' 'Palsson, Miss. Stina Viola'
'Meyer, Mrs. Edgar Joseph (Leila Saks)' 'Landergrén, Miss. Aurora Adelia'
'Widener, Mr. Harry Elkins' 'Betros, Mr. Tannous'
'Gustafsson, Mr. Karl Gideon' 'Bidois, Miss. Rosalie'
'Nakid, Miss. Maria ("Mary")' 'Tikkanen, Mr. Juho'
'Holverson, Mrs. Alexander Oskar (Mary Aline Towner)'
'Plotcharsky, Mr. Vasil' 'Davies, Mr. Charles Henry'
'Goodwin, Master. Sidney Leonard' 'Buss, Miss. Kate'
'Sadlier, Mr. Matthew' 'Lehmann, Miss. Bertha'
'Carter, Mr. William Ernest' 'Jansson, Mr. Carl Olof'

'Gustafsson, Mr. Johan Birger' 'Newell, Miss. Marjorie'
'Sandstrom, Mrs. Hjalmar (Agnes Charlotta Bengtsson)'
'Johansson, Mr. Erik' 'Olsson, Miss. Elina' 'McKane, Mr. Peter David'
'Pain, Dr. Alfred' 'Trout, Mrs. William H (Jessie L)'
'Niskanen, Mr. Juha' 'Adams, Mr. John' 'Jussila, Miss. Mari Aina'
'Hakkarainen, Mr. Pekka Pietari' 'Oreskovic, Miss. Marija'
'Gale, Mr. Shadrach' 'Widegren, Mr. Carl/Charles Peter'
'Richards, Master. William Rowe' 'Birkeland, Mr. Hans Martin Monsen'
'Lefebvre, Miss. Ida' 'Sdycoff, Mr. Todor' 'Hart, Mr. Henry'
'Minahan, Miss. Daisy E' 'Cunningham, Mr. Alfred Fleming'
'Sundman, Mr. Johan Julian' 'Meek, Mrs. Thomas (Annie Louise Rowley)'
'Drew, Mrs. James Vivian (Lulu Thorne Christian)'
'Silven, Miss. Lyyli Karoliina' 'Matthews, Mr. William John'
'Van Impe, Miss. Catharina' 'Gheorgheff, Mr. Stanio'
'Charters, Mr. David' 'Zimmerman, Mr. Leo'
'Danbom, Mrs. Ernst Gilbert (Anna Sigrid Maria Brogren)'
'Rosblom, Mr. Viktor Richard' 'Wiseman, Mr. Phillippe'
'Clarke, Mrs. Charles V (Ada Maria Winfield)'
'Phillips, Miss. Kate Florence ("Mrs Kate Louise Phillips Marshall")'
'Flynn, Mr. James' 'Pickard, Mr. Berk (Berk Trembisky)'
'Bjornstrom-Steffansson, Mr. Mauritz Hakan'
'Thorneycroft, Mrs. Percival (Florence Kate White)'
'Louch, Mrs. Charles Alexander (Alice Adelaide Slow)'
'Kallio, Mr. Nikolai Erland' 'Silvey, Mr. William Baird'
'Carter, Miss. Lucile Polk' 'Ford, Miss. Doolina Margaret "Daisy"'
'Richards, Mrs. Sidney (Emily Hocking)' 'Fortune, Mr. Mark'
'Kvillner, Mr. Johan Henrik Johannesson'
'Hart, Mrs. Benjamin (Esther Ada Bloomfield)' 'Hampe, Mr. Leon'
'Pettersson, Mr. Johan Emil' 'Reynaldo, Ms. Encarnacion'
'Johannesen-Bratthammer, Mr. Bernt' 'Dodge, Master. Washington'
'Mellinger, Miss. Madeleine Violet' 'Seward, Mr. Frederic Kimber'
'Baclini, Miss. Marie Catherine' 'Peuchen, Major. Arthur Godfrey'
'West, Mr. Edwy Arthur' 'Hagland, Mr. Ingvald Olai Olsen'
'Foreman, Mr. Benjamin Laventall' 'Goldenberg, Mr. Samuel L'
'Peduzzi, Mr. Joseph' 'Jalsevac, Mr. Ivan' 'Millet, Mr. Francis Davis'
'Kenyon, Mrs. Frederick R (Marion)' 'Toomey, Miss. Ellen'
'O'Connor, Mr. Maurice' 'Anderson, Mr. Harry' 'Morley, Mr. William'
'Gee, Mr. Arthur H' 'Milling, Mr. Jacob Christian' 'Maisner, Mr. Simon'
'Goncalves, Mr. Manuel Estanslas' 'Campbell, Mr. William'
'Smart, Mr. John Montgomery' 'Scanlan, Mr. James'
'Baclini, Miss. Helene Barbara' 'Keefe, Mr. Arthur' 'Cacic, Mr. Luka'
'West, Mrs. Edwy Arthur (Ada Mary Worth)'
'Jerwan, Mrs. Amin S (Marie Marthe Thuillard)'
'Strandberg, Miss. Ida Sofia' 'Clifford, Mr. George Quincy'
'Renouf, Mr. Peter Henry' 'Braund, Mr. Lewis Richard'
'Karlsson, Mr. Nils August' 'Hirvonen, Miss. Hildur E'
'Goodwin, Master. Harold Victor' 'Frost, Mr. Anthony Wood "Archie"'
'Rouse, Mr. Richard Henry' 'Turkula, Mrs. (Hedwig)'
'Bishop, Mr. Dickinson H' 'Lefebvre, Miss. Jeannie'
'Hoyt, Mrs. Frederick Maxfield (Jane Anne Forby)'
'Kent, Mr. Edward Austin' 'Somerton, Mr. Francis William'
'Coutts, Master. Eden Leslie "Neville"'
'Hagland, Mr. Konrad Mathias Reiersen' 'Windelov, Mr. Einar'
'Molson, Mr. Harry Markland' 'Artagaveytia, Mr. Ramon'
'Stanley, Mr. Edward Roland' 'Yousseff, Mr. Gerious'
'Eustis, Miss. Elizabeth Mussey' 'Shellard, Mr. Frederick William'
'Allison, Mrs. Hudson J C (Bessie Waldo Daniels)' 'Svensson, Mr. Olof'
'Calic, Mr. Petar' 'Canavan, Miss. Mary' 'O'Sullivan, Miss. Bridget Mary'
'Laitinen, Miss. Kristina Sofia' 'Maioni, Miss. Roberta'
'Penasco y Castellana, Mr. Victor de Satode'
'Quick, Mrs. Frederick Charles (Jane Richards)'
'Bradley, Mr. George ("George Arthur Brayton")'
'Olsen, Mr. Henry Margido' 'Lang, Mr. Fang' 'Daly, Mr. Eugene Patrick'
'Webber, Mr. James' 'McGough, Mr. James Robert'
'Rothschild, Mrs. Martin (Elizabeth L. Barrett)' 'Coleff, Mr. Satio'
'Walker, Mr. William Anderson' 'Lemore, Mrs. (Amelia Milley)'

'Ryan, Mr. Patrick'
'Angle, Mrs. William A (Florence "Mary" Agnes Hughes)'
'Pavlovic, Mr. Stefo' 'Perreault, Miss. Anne' 'Vovk, Mr. Janko'
'Lahoud, Mr. Sarkis' 'Hippach, Mrs. Louis Albert (Ida Sophia Fischer)'
'Kassem, Mr. Fared' 'Farrell, Mr. James' 'Ridsdale, Miss. Lucy'
'Farthing, Mr. John' 'Salonen, Mr. Johan Werner'
'Hocking, Mr. Richard George' 'Quick, Miss. Phyllis May'
'Toufik, Mr. Nakli' 'Elias, Mr. Joseph Jr'
'Peter, Mrs. Catherine (Catherine Rizk)' 'Cacic, Miss. Marija'
'Hart, Miss. Eva Miriam' 'Butt, Major. Archibald Willingham'
'LeRoy, Miss. Bertha' 'Risien, Mr. Samuel Beard'
'Frolicher, Miss. Hedwig Margaritha' 'Crosby, Miss. Harriet R'
'Andersson, Miss. Ingeborg Constanzia'
'Andersson, Miss. Sigrid Elisabeth' 'Beane, Mr. Edward'
'Douglas, Mr. Walter Donald' 'Nicholson, Mr. Arthur Ernest'
'Beane, Mrs. Edward (Ethel Clarke)' 'Padro y Manent, Mr. Julian'
'Goldsmith, Mr. Frank John' 'Davies, Master. John Morgan Jr'
'Thayer, Mr. John Borland Jr' 'Sharp, Mr. Percival James R'
'O'Brien, Mr. Timothy' 'Leeni, Mr. Fahim ("Philip Zenni")'
'Ohman, Miss. Velin' 'Wright, Mr. George'
'Duff Gordon, Lady. (Lucille Christiana Sutherland) ("Mrs Morgan")'
'Robbins, Mr. Victor' 'Taussig, Mrs. Emil (Tillie Mandelbaum)'
'de Messemaeker, Mrs. Guillaume Joseph (Emma)' 'Morrow, Mr. Thomas Rowan'
'Sivic, Mr. Husein' 'Norman, Mr. Robert Douglas' 'Simmons, Mr. John'
'Meanwell, Miss. (Marion Ogden)' 'Davies, Mr. Alfred J'
'Stoytcheff, Mr. Ilia' 'Palsson, Mrs. Nils (Alma Cornelia Berglund)'
'Doharr, Mr. Tannous' 'Jonsson, Mr. Carl' 'Harris, Mr. George'
'Appleton, Mrs. Edward Dale (Charlotte Lamson)'
'Flynn, Mr. John Irwin ("Irving")' 'Kelly, Miss. Mary'
'Rush, Mr. Alfred George John' 'Patchett, Mr. George'
'Garside, Miss. Ethel' 'Silvey, Mrs. William Baird (Alice Munger)'
'Caram, Mrs. Joseph (Maria Elias)' 'Jussila, Mr. Eiriik'
'Christy, Miss. Julie Rachel'
'Thayer, Mrs. John Borland (Marian Longstreth Morris)'
'Downton, Mr. William James' 'Ross, Mr. John Hugo' 'Paulner, Mr. Uscher'
'Taussig, Miss. Ruth' 'Jarvis, Mr. John Denzil'
'Frolicher-Stehli, Mr. Maxmillian' 'Gilinski, Mr. Eliezer'
'Murdlin, Mr. Joseph' 'Rintamaki, Mr. Matti'
'Stephenson, Mrs. Walter Bertram (Martha Eustis)'
'Elsbury, Mr. William James' 'Bourke, Miss. Mary'
'Chapman, Mr. John Henry' 'Van Impe, Mr. Jean Baptiste'
'Leitch, Miss. Jessie Wills' 'Johnson, Mr. Alfred' 'Boulos, Mr. Hanna'
'Duff Gordon, Sir. Cosmo Edmund ("Mr Morgan")'
'Jacobsohn, Mrs. Sidney Samuel (Amy Frances Christy)'
'Slabenoff, Mr. Petco' 'Harrington, Mr. Charles H'
'Torber, Mr. Ernst William' 'Homer, Mr. Harry ("Mr E Haven")'
'Lindell, Mr. Edvard Bengtsson' 'Karaic, Mr. Milan'
'Daniel, Mr. Robert Williams'
'Laroche, Mrs. Joseph (Juliette Marie Louise Lafargue)'
'Shutes, Miss. Elizabeth W'
'Andersson, Mrs. Anders Johan (Alfrida Konstantia Brogren)'
'Jardin, Mr. Jose Neto' 'Murphy, Miss. Margaret Jane' 'Horgan, Mr. John'
'Brocklebank, Mr. William Alfred' 'Herman, Miss. Alice'
'Danbom, Mr. Ernst Gilbert'
'Lobb, Mrs. William Arthur (Cordelia K Stanlick)'
'Becker, Miss. Marion Louise' 'Gavey, Mr. Lawrence' 'Yasbeck, Mr. Antoni'
'Kimball, Mr. Edwin Nelson Jr' 'Nakid, Mr. Sahid'
'Hansen, Mr. Henry Damsgaard' 'Bowen, Mr. David John "Dai"
'Sutton, Mr. Frederick' 'Kirkland, Rev. Charles Leonard'
'Longley, Miss. Gretchen Fiske' 'Bostandyeff, Mr. Guentcho'
'O'Connell, Mr. Patrick D' 'Barkworth, Mr. Algernon Henry Wilson'
'Lundahl, Mr. Johan Svensson' 'Stahelin-Maeglin, Dr. Max'
'Parr, Mr. William Henry Marsh' 'Skoog, Miss. Mabel' 'Davis, Miss. Mary'
'Leinonen, Mr. Antti Gustaf' 'Collyer, Mr. Harvey'
'Panula, Mrs. Juha (Maria Emilia Ojala)' 'Thorneycroft, Mr. Percival'
'Jensen, Mr. Hans Peder' 'Sagesser, Mlle. Emma'

'Skoog, Margit Elizabeth' 'Foo, Mr. Choong'
'Baclini, Miss. Eugenie' 'Harper, Mr. Henry Sleeper' 'Cor, Mr. Liudevit'
'Simonius-Blumer, Col. Oberst Alfons' 'Willey, Mr. Edward'
'Stanley, Miss. Amy Zillah Elsie' 'Mitkoff, Mr. Mito'
'Doling, Miss. Elsie' 'Kalvik, Mr. Johannes Halvorsen'
'O\Leary, Miss. Hanora "Norah"' 'Hegarty, Miss. Hanora "Nora"'
'Hickman, Mr. Leonard Mark' 'Radeff, Mr. Alexander'
'Bourke, Mrs. John (Catherine)' 'Eitemiller, Mr. George Floyd'
'Newell, Mr. Arthur Webster' 'Frauenthal, Dr. Henry William'
'Badt, Mr. Mohamed' 'Colley, Mr. Edward Pomeroy' 'Coleff, Mr. Peju'
'Lindqvist, Mr. Eino William' 'Hickman, Mr. Lewis'
'Butler, Mr. Reginald Fenton' 'Rommetvedt, Mr. Knud Paust'
'Cook, Mr. Jacob' 'Taylor, Mrs. Elmer Zebley (Juliet Cummins Wright)'
'Brown, Mrs. Thomas William Solomon (Elizabeth Catherine Ford)'
'Davidson, Mr. Thornton' 'Mitchell, Mr. Henry Michael'
'Wilhelms, Mr. Charles' 'Watson, Mr. Ennis Hastings'
'Edvardsson, Mr. Gustaf Hjalmar' 'Sawyer, Mr. Frederick Charles'
'Turja, Miss. Anna Sofia' 'Goodwin, Mrs. Frederick (Augusta Tyler)'
'Cardeza, Mr. Thomas Drake Martinez' 'Peters, Miss. Katie'
'Hassab, Mr. Hammad' 'Olsvigen, Mr. Thor Anderson'
'Goodwin, Mr. Charles Edward' 'Brown, Mr. Thomas William Solomon'
'Laroche, Mr. Joseph Philippe Lemercier' 'Panula, Mr. Jaako Arnold'
'Dakic, Mr. Branko' 'Fischer, Mr. Eberhard Thelander'
'Madill, Miss. Georgette Alexandra' 'Dick, Mr. Albert Adrian'
'Karun, Miss. Manca' 'Lam, Mr. Ali' 'Saad, Mr. Khalil' 'Weir, Col. John'
'Chapman, Mr. Charles Henry' 'Kelly, Mr. James'
'Mullens, Miss. Katherine "Katie"' 'Thayer, Mr. John Borland'
'Humblen, Mr. Adolf Mathias Nicolai Olsen'
'Astor, Mrs. John Jacob (Madeleine Talmadge Force)'
'Silverthorne, Mr. Spencer Victor' 'Barbara, Miss. Saiide'
'Gallagher, Mr. Martin' 'Hansen, Mr. Henrik Juul'
'Morley, Mr. Henry Samuel ("Mr Henry Marshall")'
'Kelly, Mrs. Florence "Fannie"' 'Calderhead, Mr. Edward Pennington'
'Cleaver, Miss. Alice'
'Moubarek, Master. Halim Gonios ("William George")'
'Mayne, Mlle. Berthe Antonine ("Mrs de Villiers")' 'Klaber, Mr. Herman'
'Taylor, Mr. Elmer Zebley' 'Larsson, Mr. August Viktor'
'Greenberg, Mr. Samuel' 'Soholt, Mr. Peter Andreas Lauritz Andersen'
'Endres, Miss. Caroline Louise' 'Troutt, Miss. Edwina Celia "Winnie"'
'McEvoy, Mr. Michael' 'Johnson, Mr. Malkolm Joackim'
'Harper, Miss. Annie Jessie "Nina"' 'Jensen, Mr. Svend Lauritz'
'Gillespie, Mr. William Henry' 'Hodges, Mr. Henry Price'
'Chambers, Mr. Norman Campbell' 'Oreskovic, Mr. Luka'
'Renouf, Mrs. Peter Henry (Lillian Jefferys)' 'Mannion, Miss. Margareth'
'Bryhl, Mr. Kurt Arnold Gottfrid' 'Ilmakangas, Miss. Pieta Sofia'
'Allen, Miss. Elisabeth Walton' 'Hassan, Mr. Houssein G N'
'Knight, Mr. Robert J' 'Berriman, Mr. William John'
'Troupiansky, Mr. Moses Aaron' 'Williams, Mr. Leslie'
'Ford, Mrs. Edward (Margaret Ann Watson)' 'Lesurer, Mr. Gustave J'
'Ivanoff, Mr. Kanio' 'Nankoff, Mr. Minko' 'Hawksford, Mr. Walter James'
'Cavendish, Mr. Tyrell William' 'Ryerson, Miss. Susan Parker "Suzette"'
'McNamee, Mr. Neal' 'Stranden, Mr. Juho' 'Crosby, Capt. Edward Gifford'
'Abbott, Mr. Rossmore Edward' 'Sinkkonen, Miss. Anna'
'Marvin, Mr. Daniel Warner' 'Connaghton, Mr. Michael' 'Wells, Miss. Joan'
'Moor, Master. Meier' 'Vande Velde, Mr. Johannes Joseph'
'Jonkoff, Mr. Lallo' 'Herman, Mrs. Samuel (Jane Laver)'
'Hamalainen, Master. Viljo' 'Carlsson, Mr. August Sigfrid'
'Bailey, Mr. Percy Andrew' 'Theobald, Mr. Thomas Leonard'
'Roths, the Countess. of (Lucy Noel Martha Dyer-Edwards)'
'Garfirth, Mr. John' 'Nirva, Mr. Iisakki Antino Aijo'
'Barah, Mr. Hanna Assi' 'Carter, Mrs. William Ernest (Lucile Polk)'
'Eklund, Mr. Hans Linus' 'Hogeboom, Mrs. John C (Anna Andrews)'
'Brewer, Dr. Arthur Jackson' 'Mangan, Miss. Mary' 'Moran, Mr. Daniel J'
'Gronnestad, Mr. Daniel Danielsen' 'Lievens, Mr. Rene Aime'
'Jensen, Mr. Niels Peder' 'Mack, Mrs. (Mary)' 'Elias, Mr. Dibo'
'Hocking, Mrs. Elizabeth (Eliza Needs)'

'Myhrman, Mr. Pehr Fabian Oliver Malkolm' 'Tobin, Mr. Roger'
 'Emanuel, Miss. Virginia Ethel' 'Kilgannon, Mr. Thomas J'
 'Robert, Mrs. Edward Scott (Elisabeth Walton McMillan)'
 'Ayoub, Miss. Banoura' 'Dick, Mrs. Albert Adrian (Vera Gillespie)'
 'Long, Mr. Milton Clyde' 'Johnston, Mr. Andrew G' 'Ali, Mr. William'
 'Harmer, Mr. Abraham (David Lishin)' 'Sjoblom, Miss. Anna Sofia'
 'Rice, Master. George Hugh' 'Dean, Master. Bertram Vere'
 'Guggenheim, Mr. Benjamin' 'Keane, Mr. Andrew "Andy"'
 'Gaskell, Mr. Alfred' 'Sage, Miss. Stella Anna'
 'Hoyt, Mr. William Fisher' 'Dantcheff, Mr. Ristiu' 'Otter, Mr. Richard'
 'Leader, Dr. Alice (Farnham)' 'Osman, Mrs. Mara'
 'Ibrahim Shawah, Mr. Yousseff'
 'Van Impe, Mrs. Jean Baptiste (Rosalie Paula Govaert)'
 'Ponesell, Mr. Martin' 'Collyer, Mrs. Harvey (Charlotte Annie Tate)'
 'Carter, Master. William Thornton II' 'Thomas, Master. Assad Alexander'
 'Hedman, Mr. Oskar Arvid' 'Johansson, Mr. Karl Johan'
 'Andrews, Mr. Thomas Jr' 'Pettersson, Miss. Ellen Natalia'
 'Meyer, Mr. August' 'Chambers, Mrs. Norman Campbell (Bertha Griggs)'
 'Alexander, Mr. William' 'Lester, Mr. James' 'Slemen, Mr. Richard James'
 'Andersson, Miss. Ebba Iris Alfrida' 'Tomlin, Mr. Ernest Portage'
 'Fry, Mr. Richard' 'Heininen, Miss. Wendla Maria' 'Mallet, Mr. Albert'
 'Holm, Mr. John Fredrik Alexander' 'Skoog, Master. Karl Thorsten'
 'Hays, Mrs. Charles Melville (Clara Jennings Gregg)' 'Lulic, Mr. Nikola'
 'Reuchlin, Jonkheer. John George' 'Moor, Mrs. (Beila)'
 'Panula, Master. Urho Abraham' 'Flynn, Mr. John' 'Lam, Mr. Len'
 'Mallet, Master. Andre' 'McCormack, Mr. Thomas Joseph'
 'Stone, Mrs. George Nelson (Martha Evelyn)'
 'Yasbeck, Mrs. Antoni (Selini Alexander)'
 'Richards, Master. George Sibley' 'Saad, Mr. Amin'
 'Augustsson, Mr. Albert' 'Allum, Mr. Owen George'
 'Compton, Miss. Sara Rebecca' 'Pasic, Mr. Jakob' 'Sirota, Mr. Maurice'
 'Chip, Mr. Chang' 'Marechal, Mr. Pierre' 'Alhomaki, Mr. Ilmari Rudolf'
 'Mudd, Mr. Thomas Charles' 'Serepeca, Miss. Augusta'
 'Lemberopolous, Mr. Peter L' 'Culumovic, Mr. Jeso' 'Abbing, Mr. Anthony'
 'Sage, Mr. Douglas Bullen' 'Markoff, Mr. Marin' 'Harper, Rev. John'
 'Goldenberg, Mrs. Samuel L (Edwiga Grabowska)'
 'Andersson, Master. Sigvard Harald Elias' 'Svensson, Mr. Johan'
 'Boulos, Miss. Nourelain' 'Lines, Miss. Mary Conover'
 'Carter, Mrs. Ernest Courtenay (Lilian Hughes)'
 'Aks, Mrs. Sam (Leah Rosen)' 'Wick, Mrs. George Dennick (Mary Hitchcock)'
 'Daly, Mr. Peter Denis' 'Baclini, Mrs. Solomon (Latifa Qurban)'
 'Razi, Mr. Raihed' 'Hansen, Mr. Claus Peter'
 'Giles, Mr. Frederick Edward'
 'Swift, Mrs. Frederick Joel (Margaret Welles Barron)'
 'Sage, Miss. Dorothy Edith "Dolly"' 'Gill, Mr. John William'
 'Bystrom, Mrs. (Karolina)' 'Duran y More, Miss. Asuncion'
 'Roebbing, Mr. Washington Augustus II' 'van Melkebeke, Mr. Philemon'
 'Johnson, Master. Harold Theodor' 'Balkic, Mr. Cerin'
 'Beckwith, Mrs. Richard Leonard (Sallie Monypeny)'
 'Carlsson, Mr. Frans Olof' 'Vander Cruyssen, Mr. Victor'
 'Abelson, Mrs. Samuel (Hannah Wizosky)'
 'Najib, Miss. Adele Kiamie "Jane"' 'Gustafsson, Mr. Alfred Ossian'
 'Petroff, Mr. Nedelio' 'Laleff, Mr. Kristo'
 'Potter, Mrs. Thomas Jr (Lily Alexenia Wilson)'
 'Shelley, Mrs. William (Imanita Parrish Hall)' 'Markun, Mr. Johann'
 'Dahlberg, Miss. Gerda Ulrika' 'Banfield, Mr. Frederick James'
 'Sutehall, Mr. Henry Jr' 'Rice, Mrs. William (Margaret Norton)'
 'Montvila, Rev. Juozas' 'Graham, Miss. Margaret Edith'
 'Johnston, Miss. Catherine Helen "Carrie"' 'Behr, Mr. Karl Howell'
 'Dooley, Mr. Patrick']

Sex :

['male' 'female']

Age :

[22. 38. 26. 35. 29.69911765 54.]

2.	27.	14.	4.	58.	20.
39.	55.	31.	34.	15.	28.
8.	19.	40.	66.	42.	21.
18.	3.	7.	49.	29.	65.
28.5	5.	11.	45.	17.	32.
16.	25.	0.83	30.	33.	23.
24.	46.	59.	71.	37.	47.
14.5	70.5	32.5	12.	9.	36.5
51.	55.5	40.5	44.	1.	61.
56.	50.	36.	45.5	20.5	62.
41.	52.	63.	23.5	0.92	43.
60.	10.	64.	13.	48.	0.75
53.	57.	80.	70.	24.5	6.
0.67	30.5	0.42	34.5	74.	]

SibSp :

[1 0 3 4 2 5 8]

Parch :

[0 1 2 5 3 4 6]

Ticket :

['A/5 21171' 'PC 17599' 'STON/O2. 3101282' '113803' '373450' '330877' '17463' '349909' '347742' '237736' 'PP 9549' '113783' 'A/5. 2151' '347082' '350406' '248706' '382652' '244373' '345763' '2649' '239865' '248698' '330923' '113788' '347077' '2631' '19950' '330959' '349216' 'PC 17601' 'PC 17569' '335677' 'C.A. 24579' 'PC 17604' '113789' '2677' 'A./5. 2152' '345764' '2651' '7546' '11668' '349253' 'SC/Paris 2123' '330958' 'S.C./A.4. 23567' '370371' '14311' '2662' '349237' '3101295' 'A/4. 39886' 'PC 17572' '2926' '113509' '19947' 'C.A. 31026' '2697' 'C.A. 34651' 'CA 2144' '2669' '113572' '36973' '347088' 'PC 17605' '2661' 'C.A. 29395' 'S.P. 3464' '3101281' '315151' 'C.A. 33111' 'S.O.C. 14879' '2680' '1601' '348123' '349208' '374746' '248738' '364516' '345767' '345779' '330932' '113059' 'SO/C 14885' '3101278' 'W./C. 6608' 'SOTON/OQ 392086' '343275' '343276' '347466' 'W.E.P. 5734' 'C.A. 2315' '364500' '374910' 'PC 17754' 'PC 17759' '231919' '244367' '349245' '349215' '35281' '7540' '3101276' '349207' '343120' '312991' '349249' '371110' '110465' '2665' '324669' '4136' '2627' 'STON/O 2. 3101294' '370369' 'PC 17558' 'A4. 54510' '27267' '370372' 'C 17369' '2668' '347061' '349241' 'SOTON/O.Q. 3101307' 'A/5. 3337' '228414' 'C.A. 29178' 'SC/PARIS 2133' '11752' '7534' 'PC 17593' '2678' '347081' 'STON/O2. 3101279' '365222' '231945' 'C.A. 33112' '350043' '230080' '244310' 'S.O.P. 1166' '113776' 'A.5. 11206' 'A/5. 851' 'Fa 265302' 'PC 17597' '35851' 'SOTON/OQ 392090' '315037' 'CA. 2343' '371362' 'C.A. 33595' '347068' '315093' '363291' '113505' 'PC 17318' '111240' 'STON/O 2. 3101280' '17764' '350404' '4133' 'PC 17595' '250653' 'LINE' 'SC/PARIS 2131' '230136' '315153' '113767' '370365' '111428' '364849' '349247' '234604' '28424' '350046' 'PC 17610' '368703' '4579' '370370' '248747' '345770' '3101264' '2628' 'A/5 3540' '347054' '2699' '367231' '112277' 'SOTON/O.Q. 3101311' 'F.C.C. 13528' 'A/5 21174' '250646' '367229' '35273' 'STON/O2. 3101283' '243847' '11813' 'W/C 14208' 'SOTON/OQ 392089' '220367' '21440' '349234' '19943' 'PP 4348' 'SW/PP 751' 'A/5 21173' '236171' '347067' '237442' 'C.A. 29566' 'W./C. 6609' '26707' 'C.A. 31921' '28665' 'SCO/W 1585' '367230' 'W./C. 14263' 'STON/O 2. 3101275' '2694' '19928' '347071' '250649' '11751' '244252' '362316' '113514' 'A/5. 3336' '370129' '2650' 'PC 17585' '110152' 'PC 17755' '230433' '384461' '110413' '112059' '382649' 'C.A. 17248' '347083' 'PC 17582' 'PC 17760' '113798' '250644' 'PC 17596' '370375' '13502' '347073' '239853' 'C.A. 2673' '336439' '347464' '345778' 'A/5. 10482' '113056' '349239' '345774' '349206' '237798' '370373' '19877' '11967' 'SC/Paris 2163' '349236' '349233' 'PC 17612' '2693' '113781' '19988' '9234' '367226' '226593' 'A/5 2466' '17421' 'PC 17758' 'P/PP 3381' 'PC 17485' '11767' 'PC 17608' '250651' '349243' 'F.C.C. 13529' '347470' '29011' '36928' '16966' 'A/5 21172' '349219' '234818' '345364' '28551' '111361' '113043' 'PC 17611' '349225' '7598' '113784' '248740' '244361' '229236' '248733' '31418' '386525']

'C.A. 37671' '315088' '7267' '113510' '2695' '2647' '345783' '237671'
 '330931' '330980' 'SC/PARIS 2167' '2691' 'SOTON/O.Q. 3101310' 'C 7076'
 '110813' '2626' '14313' 'PC 17477' '11765' '3101267' '323951' 'C 7077'
 '113503' '2648' '347069' 'PC 17757' '2653' 'STON/O 2. 3101293' '349227'
 '27849' '367655' 'SC 1748' '113760' '350034' '3101277' '350052' '350407'
 '28403' '244278' '240929' 'STON/O 2. 3101289' '341826' '4137' '315096'
 '28664' '347064' '29106' '312992' '349222' '394140' 'STON/O 2. 3101269'
 '343095' '28220' '250652' '28228' '345773' '349254' 'A/5. 13032' '315082'
 '347080' 'A/4. 34244' '2003' '250655' '364851' 'SOTON/O.Q. 392078'
 '110564' '376564' 'SC/AH 3085' 'STON/O 2. 3101274' '13507' 'C.A. 18723'
 '345769' '347076' '230434' '65306' '33638' '113794' '2666' '113786'
 '65303' '113051' '17453' 'A/5 2817' '349240' '13509' '17464'
 'F.C.C. 13531' '371060' '19952' '364506' '111320' '234360' 'A/S 2816'
 'SOTON/O.Q. 3101306' '113792' '36209' '323592' '315089' 'SC/AH Basle 541'
 '7553' '31027' '3460' '350060' '3101298' '239854' 'A/5 3594' '4134'
 '11771' 'A.5. 18509' '65304' 'SOTON/OQ 3101317' '113787' 'PC 17609'
 'A/4 45380' '36947' 'C.A. 6212' '350035' '315086' '364846' '330909'
 '4135' '26360' '111427' 'C 4001' '382651' 'SOTON/OQ 3101316' 'PC 17473'
 'PC 17603' '349209' '36967' 'C.A. 34260' '226875' '349242' '12749'
 '349252' '2624' '2700' '367232' 'W./C. 14258' 'PC 17483' '3101296'
 '29104' '2641' '2690' '315084' '113050' 'PC 17761' '364498' '13568'
 'WE/P 5735' '2908' '693' 'SC/PARIS 2146' '244358' '330979' '2620'
 '347085' '113807' '11755' '345572' '372622' '349251' '218629'
 'SOTON/OQ 392082' 'SOTON/O.Q. 392087' 'A/4 48871' '349205' '2686'
 '350417' 'S.W./PP 752' '11769' 'PC 17474' '14312' 'A/4. 20589' '358585'
 '243880' '2689' 'STON/O 2. 3101286' '237789' '13049' '3411' '237565'
 '13567' '14973' 'A./5. 3235' 'STON/O 2. 3101273' 'A/5 3902' '364848'
 'SC/AH 29037' '248727' '2664' '349214' '113796' '364511' '111426'
 '349910' '349246' '113804' 'SOTON/O.Q. 3101305' '370377' '364512'
 '220845' '31028' '2659' '11753' '350029' '54636' '36963' '219533'
 '349224' '334912' '27042' '347743' '13214' '112052' '237668'
 'STON/O 2. 3101292' '350050' '349231' '13213' 'S.O./P.P. 751' 'CA. 2314'
 '349221' '8475' '330919' '365226' '349223' '29751' '2623' '5727' '349210'
 'STON/O 2. 3101285' '234686' '312993' 'A/5 3536' '19996' '29750'
 'F.C. 12750' 'C.A. 24580' '244270' '239856' '349912' '342826' '4138'
 '330935' '6563' '349228' '350036' '24160' '17474' '349256' '2672'
 '113800' '248731' '363592' '35852' '348121' 'PC 17475' '36864' '350025'
 '223596' 'PC 17476' 'PC 17482' '113028' '7545' '250647' '348124' '34218'
 '36568' '347062' '350048' '12233' '250643' '113806' '315094' '36866'
 '236853' 'STON/O2. 3101271' '239855' '28425' '233639' '349201' '349218'
 '16988' '376566' 'STON/O 2. 3101288' '250648' '113773' '335097' '29103'
 '392096' '345780' '349204' '350042' '29108' '363294' 'SOTON/O2 3101272'
 '2663' '347074' '112379' '364850' '8471' '345781' '350047' 'S.O./P.P. 3'
 '2674' '29105' '347078' '383121' '36865' '2687' '113501' 'W./C. 6607'
 'SOTON/O.Q. 3101312' '374887' '3101265' '12460' 'PC 17600' '349203'
 '28213' '17465' '349244' '2685' '2625' '347089' '347063' '112050'
 '347087' '248723' '3474' '28206' '364499' '112058' 'STON/O2. 3101290'
 'S.C./PARIS 2079' 'C 7075' '315098' '19972' '368323' '367228' '2671'
 '347468' '2223' 'PC 17756' '315097' '392092' '11774' 'SOTON/O2 3101287'
 '2683' '315090' 'C.A. 5547' '349213' '347060' 'PC 17592' '392091'
 '113055' '2629' '350026' '28134' '17466' '233866' '236852'
 'SC/PARIS 2149' 'PC 17590' '345777' '349248' '695' '345765' '2667'
 '349212' '349217' '349257' '7552' 'C.A./SOTON 34068' 'SOTON/OQ 392076'
 '211536' '112053' '111369' '370376']

Fare :

[7.25	71.2833	7.925	53.1	8.05	8.4583	51.8625	21.075
11.1333	30.0708	16.7	26.55	31.275	7.8542	16.	29.125
13.	18.	7.225	26.	8.0292	35.5	31.3875	263.
7.8792	7.8958	27.7208	146.5208	7.75	10.5	82.1708	52.
7.2292	11.2417	9.475	21.	41.5792	15.5	21.6792	17.8
39.6875	7.8	76.7292	61.9792	27.75	46.9	80.	83.475
27.9	15.2458	8.1583	8.6625	73.5	14.4542	56.4958	7.65
29.	12.475	9.	9.5	7.7875	47.1	15.85	34.375
61.175	20.575	34.6542	63.3583	23.	77.2875	8.6542	7.775
24.15	9.825	14.4583	247.5208	7.1417	22.3583	6.975	7.05

14.5	15.0458	26.2833	9.2167	79.2	6.75	11.5	36.75
7.7958	12.525	66.6	7.3125	61.3792	7.7333	69.55	16.1
15.75	20.525	55.	25.925	33.5	30.6958	25.4667	28.7125
0.	15.05	39.	22.025	50.	8.4042	6.4958	10.4625
18.7875	31.	113.275	27.	76.2917	90.	9.35	13.5
7.55	26.25	12.275	7.125	52.5542	20.2125	86.5	512.3292
79.65	153.4625	135.6333	19.5	29.7	77.9583	20.25	78.85
91.0792	12.875	8.85	151.55	30.5	23.25	12.35	110.8833
108.9	24.	56.9292	83.1583	262.375	14.	164.8667	134.5
6.2375	57.9792	28.5	133.65	15.9	9.225	35.	75.25
69.3	55.4417	211.5	4.0125	227.525	15.7417	7.7292	12.
120.	12.65	18.75	6.8583	32.5	7.875	14.4	55.9
8.1125	81.8583	19.2583	19.9667	89.1042	38.5	7.725	13.7917
9.8375	7.0458	7.5208	12.2875	9.5875	49.5042	78.2667	15.1
7.6292	22.525	26.2875	59.4	7.4958	34.0208	93.5	221.7792
106.425	49.5	71.	13.8625	7.8292	39.6	17.4	51.4792
26.3875	30.	40.125	8.7125	15.	33.	42.4	15.55
65.	32.3208	7.0542	8.4333	25.5875	9.8417	8.1375	10.1708
211.3375	57.	13.4167	7.7417	9.4833	7.7375	8.3625	23.45
25.9292	8.6833	8.5167	7.8875	37.0042	6.45	6.95	8.3
6.4375	39.4	14.1083	13.8583	50.4958	5.	9.8458	10.5167]

Embarked :
['S' 'C' 'Q']

```
In [91]: titanic.Age = titanic.Age.astype(int)
titanic.dtypes
```

```
Out[91]: PassengerId      int64
Survived      int64
Pclass        int64
Name          object
Sex           object
Age           int32
SibSp         int64
Parch         int64
Ticket        object
Fare          float64
Embarked      object
dtype: object
```

In []:

Adult Dataset

```
In [105... #delimiter=' ', *, ',engine='python' or skipinitialspace=True
adult = pd.read_csv(r"E:\Downloads data\adult (1).data",header=None,skipinitialspace=True)
adult
```


Out[105...

	0	1	2	3	4	5	6	7	8	9	10	11
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	White	Male	2174	0
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	White	Male	0	0
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	White	Male	0	0
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband	Black	Male	0	0
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife	Black	Female	0	0
...	...	...	...	...	...	...	...	...	...	...	...	...
32556	27	Private	257302	Assoc-acdm	12	Married-civ-spouse	Tech-support	Wife	White	Female	0	0
32557	40	Private	154374	HS-grad	9	Married-civ-spouse	Machine-op-inspct	Husband	White	Male	0	0
32558	58	Private	151910	HS-grad	9	Widowed	Adm-clerical	Unmarried	White	Female	0	0
32559	22	Private	201490	HS-grad	9	Never-married	Adm-clerical	Own-child	White	Male	0	0
32560	52	Self-emp-inc	287927	HS-grad	9	Married-civ-spouse	Exec-managerial	Wife	White	Female	15024	0

32561 rows × 15 columns



In [106...

```
adult.columns = ['age', 'workclass', 'fnlwgt', 'education',
                  'education-num', 'marital-status',
                  'occupation', 'relationship', 'race', 'sex',
                  'capital-gain', 'capital-loss', 'hrs-per-week',
                  'native-country', 'income']
adult.head()
```

Out[106...

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship	race	sex
0	39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical	Not-in-family	White	Male
1	50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial	Husband	White	Male
2	38	Private	215646	HS-grad	9	Divorced	Handlers-cleaners	Not-in-family	White	Male
3	53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaners	Husband	Black	Male
4	28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty	Wife	Black	Female

In [107...

```
adult.isnull().sum()
```

Out[107...

```
age          0
workclass    0
fnlwgt       0
education    0
education-num 0
marital-status 0
occupation   0
relationship 0
race         0
sex          0
capital-gain 0
capital-loss 0
hrs-per-week 0
native-country 0
income       0
dtype: int64
```

In [108...

```
adult.dtypes
```

Out[108...

```
age          int64
workclass    object
fnlwgt       int64
education    object
education-num int64
marital-status object
occupation   object
relationship object
race         object
sex          object
capital-gain  int64
capital-loss  int64
hrs-per-week  int64
native-country object
income       object
dtype: object
```

In [110...

```
for i in adult.columns:
    print(i,':','\n',adult[i].unique(),'\n')
```

```

age :
[39 50 38 53 28 37 49 52 31 42 30 23 32 40 34 25 43 54 35 59 56 19 20 45
22 48 21 24 57 44 41 29 18 47 46 36 79 27 67 33 76 17 55 61 70 64 71 68
66 51 58 26 60 90 75 65 77 62 63 80 72 74 69 73 81 78 88 82 83 84 85 86
87]

workclass :
['State-gov' 'Self-emp-not-inc' 'Private' 'Federal-gov' 'Local-gov' '?'
'Self-emp-inc' 'Without-pay' 'Never-worked']

fnlwgt :
[ 77516  83311 215646 ...  34066  84661 257302]

education :
['Bachelors' 'HS-grad' '11th' 'Masters' '9th' 'Some-college' 'Assoc-acdm'
'Assoc-voc' '7th-8th' 'Doctorate' 'Prof-school' '5th-6th' '10th'
'1st-4th' 'Preschool' '12th']

education-num :
[13  9  7 14  5 10 12 11  4 16 15  3  6  2  1  8]

marital-status :
['Never-married' 'Married-civ-spouse' 'Divorced' 'Married-spouse-absent'
'Separated' 'Married-AF-spouse' 'Widowed']

occupation :
['Adm-clerical' 'Exec-managerial' 'Handlers-cleaners' 'Prof-specialty'
'Other-service' 'Sales' 'Craft-repair' 'Transport-moving'
'Farming-fishing' 'Machine-op-inspct' 'Tech-support' '?'
'Protective-serv' 'Armed-Forces' 'Priv-house-serv']

relationship :
['Not-in-family' 'Husband' 'Wife' 'Own-child' 'Unmarried' 'Other-relative']

race :
['White' 'Black' 'Asian-Pac-Islander' 'Amer-Indian-Eskimo' 'Other']

sex :
['Male' 'Female']

capital-gain :
[ 2174      0 14084  5178  5013  2407 14344 15024  7688 34095  4064  4386
 7298  1409  3674  1055  3464  2050  2176   594 20051  6849  4101  1111
 8614  3411  2597 25236  4650  9386  2463  3103 10605  2964  3325  2580
 3471  4865 99999  6514  1471  2329  2105  2885 25124 10520  2202  2961
27828  6767  2228  1506 13550  2635  5556  4787  3781  3137  3818  3942
  914   401  2829  2977  4934  2062  2354  5455 15020  1424  3273 22040
 4416  3908 10566   991  4931  1086  7430  6497   114  7896  2346  3418
 3432  2907  1151  2414  2290 15831 41310  4508  2538  3456  6418  1848
 3887  5721  9562  1455  2036  1831 11678  2936  2993  7443  6360  1797
 1173  4687  6723  2009  6097  2653  1639 18481  7978  2387  5060]

capital-loss :
[   0 2042 1408 1902 1573 1887 1719 1762 1564 2179 1816 1980 1977 1876
1340 2206 1741 1485 2339 2415 1380 1721 2051 2377 1669 2352 1672  653
2392 1504 2001 1590 1651 1628 1848 1740 2002 1579 2258 1602  419 2547
2174 2205 1726 2444 1138 2238  625  213 1539  880 1668 1092 1594 3004
2231 1844  810 2824 2559 2057 1974  974 2149 1825 1735 1258 2129 2603
2282  323 4356 2246 1617 1648 2489 3770 1755 3683 2267 2080 2457  155
3900 2201 1944 2467 2163 2754 2472 1411]

hrs-per-week :
[40 13 16 45 50 80 30 35 60 20 52 44 15 25 38 43 55 48 58 32 70  2 22 56
41 28 36 24 46 42 12 65  1 10 34 75 98 33 54  8  6 64 19 18 72  5  9 47
37 21 26 14  4 59  7 99 53 39 62 57 78 90 66 11 49 84  3 17 68 27 85 31
51 77 63 23 87 88 73 89 97 94 29 96 67 82 86 91 81 76 92 61 74 95]

```

```
native-country :
['United-States' 'Cuba' 'Jamaica' 'India' '?' 'Mexico' 'South'
 'Puerto-Rico' 'Honduras' 'England' 'Canada' 'Germany' 'Iran'
 'Philippines' 'Italy' 'Poland' 'Columbia' 'Cambodia' 'Thailand' 'Ecuador'
 'Laos' 'Taiwan' 'Haiti' 'Portugal' 'Dominican-Republic' 'El-Salvador'
 'France' 'Guatemala' 'China' 'Japan' 'Yugoslavia' 'Peru'
 'Outlying-US(Guam-USVI-etc)' 'Scotland' 'Trinidad&Tobago' 'Greece'
 'Nicaragua' 'Vietnam' 'Hong' 'Ireland' 'Hungary' 'Holand-Netherlands']
```

```
income :
['<=50K' '>50K']
```

```
In [112... for i in adult.columns:
              print(i,':',sum(adult[i]=='?'))
```

```
age : 0
workclass : 1836
fnlwgt : 0
education : 0
education-num : 0
marital-status : 0
occupation : 1843
relationship : 0
race : 0
sex : 0
capital-gain : 0
capital-loss : 0
hrs-per-week : 0
native-country : 583
income : 0
```

```
In [113... d = adult.describe(include='all')
d
```

	age	workclass	fnlwgt	education	education-num	marital-status	occupation	relationship
count	32561.000000	32561	3.256100e+04	32561	32561.000000	32561	32561	32561
unique	NaN	9	NaN	16	NaN	7	15	15
top	NaN	Private	NaN	HS-grad	NaN	Married-civ-spouse	Prof-specialty	Husband
freq	NaN	22696	NaN	10501	NaN	14976	4140	4140
mean	38.581647	NaN	1.897784e+05	NaN	10.080679	NaN	NaN	NaN
std	13.640433	NaN	1.055500e+05	NaN	2.572720	NaN	NaN	NaN
min	17.000000	NaN	1.228500e+04	NaN	1.000000	NaN	NaN	NaN
25%	28.000000	NaN	1.178270e+05	NaN	9.000000	NaN	NaN	NaN
50%	37.000000	NaN	1.783560e+05	NaN	10.000000	NaN	NaN	NaN
75%	48.000000	NaN	2.370510e+05	NaN	12.000000	NaN	NaN	NaN
max	90.000000	NaN	1.484705e+06	NaN	16.000000	NaN	NaN	NaN

```
In [114... for i in ['workclass','occupation','native-country']:
              adult[i].replace('?',d[i][2],inplace=True)
```

In [115...

```
for i in adult.columns:  
    print(i,':',sum(adult[i]=='?'))
```

```
age : 0  
workclass : 0  
fnlwgt : 0  
education : 0  
education-num : 0  
marital-status : 0  
occupation : 0  
relationship : 0  
race : 0  
sex : 0  
capital-gain : 0  
capital-loss : 0  
hrs-per-week : 0  
native-country : 0  
income : 0
```

In []:

Machine Learning

It is a field of AI that gives machines the capability to learn without being explicitly programmed.

It can extract features and patterns from the given data.

It is used for predictive models and can solve complex problems.

Machine Learning is of three types:

- > Supervised ML**
- > Unsupervised ML**
- > Reinforcement ML**

Machine Learning Procedure

- Reading & Studying the Dataset
- Data Cleaning & Analysis
- Data Visualization or EDA (Exploratory Data Analysis)
- Encoding
- ip/op Creation
- Train Test Split
- Standard Scaler Transform
- ML Algorithm
- Prediction
- Accuracy

In []: